

Generators and commutators in finite groups; abstract quotients of compact groups

Nikolay Nikolov and Dan Segal

October 27, 2019

1 Introduction

Let G be a group and $\{g_1, \dots, g_r\}$ a finite subset of G . If G is *nilpotent*, then

- (A) $G = G' \langle g_1, \dots, g_r \rangle$ implies $G = \langle g_1, \dots, g_r \rangle$;
- (B) $G = \langle g_1, \dots, g_r \rangle$ implies $G' = [G, g_1] \dots [G, g_r]$,

where G' denotes the derived group of G and $\langle g_1, \dots, g_r \rangle$ the subgroup generated by $\{g_1, \dots, g_r\}$; for $g \in G$ we write

$$[G, g] = \{[x, g] \mid x \in G\}$$

where $[x, g] = x^{-1}g^{-1}xg$ is the usual commutator.

(A) is an easy folklore result; (B) is also well known, and first appeared in the unpublished 1966 PhD thesis of Peter Stroud; it is a key element in Serre's proof that subgroups of finite index are open in a finitely generated pro- p group. Neither (A) nor (B) is true in general for groups that are not nilpotent. Rather surprisingly, however, similar results hold without assuming nilpotency, as long the group G is assumed to be *finite*. These are very much harder, relying in their most general form on the classification of finite simple groups. The main technical results of the paper [NS], which enabled us to generalize Serre's theorem to all finitely generated profinite groups, imply the following for a finite d -generator group G :

- (C) every element of G' is equal to a product of $f_1(d)$ commutators;
- (D) if $G = G_* \langle g_1, \dots, g_r \rangle$ then $G = \langle g_{ij} \mid i = 1, \dots, r, j = 1, \dots, f_2(d, \alpha) \rangle$ with g_{ij} conjugate to g_i for all i and j ;

here G_* is a certain characteristic subgroup of G with the property that G/G_* is semisimple-by-soluble, and $\alpha = \alpha(G)$ is a certain measure of the complexity of G (the largest n such that G has $\text{Alt}(n)$ as a section). In [NS] we left open the question of whether f_2 can be made independent of $\alpha(G)$; it appears as Problem 4.7.1 in the book [S2], where further background may be found.

The primary purpose of this paper is to answer that question, and more general versions of it, positively. Although at first glance this may seem a mere technical improvement, we shall see that it has diverse applications. These are described in more detail below; among them are the new theorems:

- *If G is any compact Hausdorff topological group, then every finitely generated (abstract) quotient of G is finite.*
- *Let G be a compact Hausdorff group such that G/G^0 is (topologically) finitely generated. Then G has a countably infinite (abstract) quotient if and only if G has an infinite virtually-abelian (continuous) quotient.*

(Here, G^0 denotes the connected component of 1 in G).

Indeed, what motivated the present work was the need to develop machinery powerful enough to establish results of this kind for profinite groups, for which the methods of [NS] are insufficient; the extension to more general compact groups was then a relatively natural step.

Our second purpose is to provide a new and more streamlined route to the results of [NS] and [NS2] – including the solution of Serre’s problem on finite-index subgroups in finitely generated profinite groups – and of [NSP], where it is proved that in those groups the power subgroups are open. In setting out to prove stronger results, we have found an approach that is both more unified and in some respects simpler than the original proofs. Thus in a sense the present paper is a ‘mark 2’ version of [NS] + [NS2] + [NSP].

In the course of the proofs, we shall quote a few self-contained propositions from [NS]. Apart from these, this work is independent of [NS]. In particular, we shall not be needing the difficult structural results about finite simple groups that form the substance of [NS2]; these are replaced by the material of Subsection 4.1. Some discussion of the new ideas that we use instead appears at the end of this introduction.

1.1 Main results on finite groups

In this subsection all groups are assumed to be *finite*. The minimal size of a generating set for G is denoted $d(G)$. To a finite group G we associate the characteristic subgroup

$$\begin{aligned} G_0 &= \bigcap \{T \triangleleft G \mid G/T \text{ is almost-simple}\} \\ &= \bigcap_{M \in \mathcal{S}} C_G(M) \end{aligned} \tag{1}$$

where $\mathcal{S}$ is the set of all non-abelian simple chief factors of G (a group H is *almost-simple* if $S \triangleleft H \leq \text{Aut}(S)$ for some non-abelian simple group S). We remark that G/G_0 is an extension of a semisimple group by a soluble group of derived length at most 3, because the outer automorphism group of any simple group is soluble of derived length at most 3 (strong form of the Schreier conjecture, see Subsection 1.3.2).

1.1.1 Generators

In Section 2 we prove

Theorem 1.1 *Let G be a group and $K \leq G_0$ a normal subgroup of G . Suppose that $G = K \langle y_1, \dots, y_r \rangle = G' \langle y_1, \dots, y_r \rangle$. Then there exist elements $x_{ij} \in K$ such that*

$$G = \langle y_i^{x_{ij}} \mid i = 1, \dots, r, j = 1, \dots, f_0 \rangle$$

where $f_0 = f_0(r, d(G)) = O(rd(G)^2)$.

It is clear that the y_i must be assumed to generate G modulo G' ; the definition of G_0 serves to exclude obvious counterexamples of the form $G = K \times \langle y_1, \dots, y_r \rangle$ where K is simple or $G = \text{Sym}(n)$ with y_1 a transposition.

Observe that $G_0 = G$ if every non-abelian chief factor of G has composition length at least 2, in particular if G is soluble; the result in the soluble case was established in [S1].

1.1.2 Commutators

For a subset X of a group G , we write

$$X^{*f} = \{x_1 x_2 \dots x_f \mid x_1, x_2, \dots, x_f \in X\}.$$

The subset X is *symmetric* if $x \in X$ implies $x^{-1} \in X$.

For subgroups H, K of G ,

$$[H, K] = \langle [x, y] \mid x \in H, y \in K \rangle.$$

Theorem 1.2 *Let G be a group and $\{y_1, \dots, y_r\}$ a symmetric generating set for G . If H is a normal subgroup of G then*

$$[H, G] = \left(\prod_{i=1}^r [H, y_i] \right)^{*f_1}$$

where $f_1 = f_1(r, d(G)) = O(r^2 d(G)) = O(r^3)$.

This is proved in Section 3, together with the following ‘relative’ version, our main result on finite groups:

Theorem 1.3 *Let G be a group, $H \leq G_0$ a normal subgroup of G , and $\{y_1, \dots, y_r\}$ a symmetric subset of G . If $H \langle y_1, \dots, y_r \rangle = G' \langle y_1, \dots, y_r \rangle = G$ then*

$$[H, G] = \left(\prod_{i=1}^r [H, y_i] \right)^{*f_2}$$

where $f_2 = f_2(r, d(G)) = O(r^6 d(G)^6)$.

This is in effect ‘Key Theorem C’ of [NS], with the fundamental improvement that f_2 no longer depends on $\alpha(G)$. In fact Theorem 1.3 simultaneously generalizes all three versions of the said ‘Key Theorem’ (and strengthens them, with our new definition of G_0).

Sharper estimates for the functions f_0 , f_1 , f_2 will appear in the course of the proofs.

1.1.3 Verbal subgroups

A group word w has *width* m in a group G if every product of w -values in G is equal to such a product of length m ; here, by *w-value* we mean an element of the form $w(\mathbf{g})^{\pm 1}$ with $\mathbf{g} \in G^{(k)}$, where w is a word on k variables. In Subsection 5.2 we show how the following theorem, originally established in [NSP], easily follows from the above results:

Theorem 1.4 *Let w be a non-commutator word and G a finite d -generator group. Then w has width $f(w, d)$ in G , where $f(w, d)$ depends only on w and d .*

1.2 Algebraic properties of compact groups

A compact group (which we take to mean a compact Hausdorff topological group) is an extension G of a compact connected group G^0 , its identity component, by a profinite group G/G^0 . The Levi-Mal’cev Theorem shows that the connected component is essentially a product of compact Lie groups; this makes it relatively tractable, and most of our attention will be focused on the profinite case.

1.2.1 Finitely generated profinite groups

The significance of uniform bounds relating to all d -generator finite groups is that they reflect qualitative properties of d -generator profinite groups. Thus (C) implies that the derived group is closed in every finitely generated profinite group; and the main ‘finite’ results of [NS] were used to show that every subgroup of finite index in a finitely generated profinite group G is open. A more roundabout argument, using results from [NS] related to (D), was used in [NSP] to show that the ‘power subgroups’ G^q are open in G . The sharper results now at our disposal yield further dividends when applied in the profinite context.

Routine compactness arguments (recalled in Subsection 5.1) transform Theorems 1.1, 1.2 and 1.3 into the following.

Theorem 1.5 *Let G be a finitely generated profinite group and $K \leq G_0$ a closed normal subgroup of G . Suppose that $G = K\langle y_1, \dots, y_r \rangle = G'\langle y_1, \dots, y_r \rangle$. Then there exist elements $x_{ij} \in K$ such that*

$$G = \overline{\langle y_i^{x_{ij}} \mid i = 1, \dots, r, j = 1, \dots, f_0 \rangle}$$

where $f_0 = f_0(r, d(G))$.

Here, G_0 is defined by (1) with T ranging over *open* normal subgroups; and $\overline{X}$ denotes the closure of a subset X in G . As in the finite case, G/G_0 is an extension of a semisimple group by a soluble group of derived length at most 3 (a semisimple profinite group is a Cartesian product of finite simple groups).

Theorem 1.6 *Let G be a profinite group and $\{y_1, \dots, y_r\}$ a symmetric (topological) generating set for G . If H is a closed normal subgroup of G then*

$$[H, G] = \left(\prod_{i=1}^r [H, y_i] \right)^{*f_1} \quad (\dagger)$$

where $f_1 = f_1(r, d(G))$.

This implies that $[H, G]$ is closed in G , a result already established in [NS].

Theorem 1.7 *Let G be a finitely generated profinite group, $H \leq G_0$ a closed normal subgroup of G , and $\{y_1, \dots, y_r\}$ a symmetric subset of G . If $H \langle y_1, \dots, y_r \rangle = G' \langle y_1, \dots, y_r \rangle = G$ then*

$$[H, G] = \left(\prod_{i=1}^r [H, y_i] \right)^{*f_2} \quad (\ddagger)$$

where $f_2 = f_2(r, d(G))$.

Why is this important? Suppose that N is a proper normal subgroup in a group G . If G is *finite*, then N is contained in some maximal normal subgroup M of G . If G/M is abelian, then $NG' \leq M < G$; if not, then G/M is a simple chief factor of G , so $M \geq G_0$ and $NG_0 \leq M < G$. So far, so trivial. Now suppose that G is a *profinite* group: unless we assume that N is *closed* in G , we have no grounds to assert that N is contained in a maximal open normal subgroup – indeed N could be dense in G . If G is a *finitely generated* profinite group, however, we claim that at least one of NG' , NG_0 is necessarily properly contained in G . For suppose that $NG' = NG_0 = G$. If G is topologically generated by d elements, we can find $2d$ elements $y_1, \dots, y_{2d} \in N$ such that $G_0 \langle y_1, \dots, y_{2d} \rangle = \overline{G' \langle y_1, \dots, y_{2d} \rangle} = G$, and Theorem 1.7 (with $H = G_0$) then implies that

$$[G_0, G] \leq \langle [G_0, y_i], [G_0, y_i^{-1}] \mid 1 \leq i \leq 2d \rangle \leq N.$$

But then

$$G = NG' = N[NG_0, G] = N.$$

Thus we may state

Corollary 1.8 *Let G be a finitely generated profinite group and N a normal subgroup of (the underlying abstract group) G . If $NG' = NG_0 = G$ then $N = G$.*

This is the key to understanding ‘abstract’ normal subgroups. It shows in particular that if G has a *dense* proper normal subgroup, then at least one of G/G' or G/G_0 has a dense proper normal subgroup; the point is that each of these quotients has relatively transparent structure. This is exploited to good effect in Subsections 5.5 and 5.6.

In Subsection 5.2 we discuss the profinite version of Theorem 1.4:

Theorem 1.9 [NSP] *Let G be a finitely generated profinite group and w a non-trivial non-commutator word. Then the verbal subgroup $w(G)$ is open in G .*

As explained in [NSP], the positive solution of the Restricted Burnside Problem (RBP) is an inevitable ingredient in the proof (indeed, it *follows* from Theorem 1.9). While the openness of all finite-index subgroups is an immediate corollary of Theorem 1.9, it was established in [NS] without using RBP; our new proof of the relevant special case of Theorem 1.9 is likewise independent of RBP.

Such results also imply certain *rigidity* properties for profinite groups, that is, conditions under which abstract group homomorphisms are forced to be continuous. Let G be a profinite group, $Q \neq 1$ an abstract group, and $f : G \rightarrow Q$ a surjective homomorphism, with kernel N .

We can restate the main result of [NS] (re-proved in Subsection 5.2, below) as:

(E) *If G is finitely generated (topologically) and Q is finite, then N is open.*

This is also true if G is a *connected* compact group instead of profinite: indeed, such a group is divisible, hence has no nontrivial finite quotients at all ([HM], Theorem 9.35).

An immediate consequence of (E) is

(F) *If G is finitely generated and Q is residually finite, then N is closed, so Q is profinite (with topology inherited from G/N via f); hence Q cannot be countably infinite.*

Rather surprisingly, it is easy to find countably infinite non-(residually finite) images (if using the axiom of choice counts as ‘finding’): if $\phi : \mathbb{Q}_p \rightarrow \mathbb{Q}$ is any $\mathbb{Q}$ -vector space epimorphism then $\mathbb{Z}_p\phi$ is a countably infinite image of $\mathbb{Z}_p$ (in fact it is an exercise, given (F), to show that $\mathbb{Z}_p\phi = \mathbb{Q}$). This suggests the question: can Q be *finitely generated* and infinite? This is answered below.

1.2.2 Compact groups

Many of the above results hold more generally for compact groups G , assuming usually that the profinite quotient G/G^0 is finitely generated (G^0 denotes the connected component of the identity in G). The structure of a connected compact group is relatively straightforward: it is semisimple modulo its centre (where by a *connected compact semisimple group* we mean a Cartesian product of compact connected simple Lie groups). In Subsection 5.4 we prove:

Theorem 1.10 If G is a semisimple profinite or compact connected group and Q is an infinite quotient of G then $|Q| \geq 2^{\aleph_0}$.

In the *profinite* case, we also give a complete classification of the *maximal* normal subgroups of G . Both results depend on associating to each normal subgroup an ultrafilter on the underlying index set of the Cartesian product.

The main results on quotients of compact groups are established in Section 5.5, using Corollary 1.8 and Theorem 1.10:

Theorem 1.11 *Let G be a compact group such that G/G^0 is (topologically) finitely generated. Let N be a normal subgroup of (the underlying abstract group) G . If G/N is countably infinite then G/N has an infinite virtually-abelian quotient.*

Corollary 1.12 *Let G be a compact group such that G/G^0 is (topologically) finitely generated. Then G has a countably infinite (abstract) quotient if and only if G has an infinite virtually-abelian (continuous) quotient.*

Using (F) in conjunction with Theorem 1.11 it is easy to deduce

Theorem 1.13 *Let G be a compact group and N a normal subgroup of (the underlying abstract group) G such that G/N is finitely generated. Then G/N is finite.*

If G/N is a countable quotient of G then the closure of N must be open in G ; in this case we say that N is *virtually dense* in G . More generally, one might ask: *under what conditions is it possible for a normal subgroup of infinite index to be virtually dense?* The answer is ‘always’ in abelian groups – for example, $\mathbb{Z}$ is dense in $\mathbb{Z}_p$; and the results of Subsection 5.4 shows that a semisimple group can have uncountably many dense normal subgroups. When G is finitely generated profinite, Corollary 1.8 shows that these extreme cases essentially account for all possibilities; when G is connected, the proof of Theorem 1.10 enables us to draw a similar conclusion. Let us say that a semisimple compact group is *strictly infinite* if it is the product of an infinite set of simple connected Lie groups or finite simple groups. In Subsection 5.6 we prove

Theorem 1.14 *Let G be a compact group such that G/G^0 is (topologically) finitely generated. Then G has a virtually dense normal subgroup of infinite index if and only if some open normal subgroup of G has an infinite abelian quotient or a strictly infinite semisimple quotient.*

If G is connected, a virtually dense subgroup is the same thing as a dense subgroup; if G is profinite, however, the conditions for the existence of a proper dense normal subgroup are more stringent. Their precise characterization (which depends only on G/G' and G/G_0) is stated in our final theorem, whose proof will appear elsewhere.

1.3 Overview of the paper, conventions, remarks

The basic idea is very simple. Suppose that $G = \langle g_1, \dots, g_r \rangle$ is a finite group. If M is a non-central chief factor of G then at least one of the generators g_i must centralize a relatively small proportion of the points of M , so the set of commutators $[M, g_i]$ must be relatively large. Although we can't predict which value of i is the relevant one, we can in any case infer that the set

$$\prod_{i=1}^r [M, g_i]$$

is relatively large: thus ‘many’ of the elements of M can be expressed as products, of bounded length, of commutators with the original generators g_i .

For this to be of any use, we need to replace ‘many’ with ‘all’. The most difficult parts of [NS] and [NS2] were devoted to that end; we can now replace some of those arguments with the help of a new ‘portmanteau’ result, which we call ‘the Gowers trick’. This is explained below.

For many applications, one needs to have an analogous result for a subset $\{g_1, \dots, g_r\}$ which may not generate the whole group. This was achieved in [NS] (‘Key Theorem C’) only under severe restrictions on the structure of the group G . Somewhat to our surprise, these restrictions turn out to be unnecessary: in Section 2 we show that the g_i have the necessary ‘fixed-point’ property on chief factors provided only that $\{g_1, \dots, g_r\}$ satisfies the hypotheses of Theorem 1.1. The proof is in principle elementary, relying on the O’Nan-Scott Theorem to analyse the action of G on its chief factors.

In Section 3 the main results on products of commutators are reduced to Theorem 4.28: this technical result, the hard core of the paper, concerns a (quasi-)semisimple group N with operators y_{ij} , and shows that every element of N is equal to a certain product of ‘twisted commutators’ with the y_{ij} . The whole of Section 4 is devoted to the proof of this theorem. While the combinatorial reduction arguments are still quite complicated, the proofs in Subsection 4.1 of the necessary results about finite simple groups are relatively short and transparent.

The final Section 5 can be read independently of the rest. Here we derive all the above-stated applications to topological groups, using only the statements of Theorems 1.5 – 1.7 and Corollary 1.8, with some additional material relating to connected compact groups.

The main theorems stated above are not all stated in their sharpest form: sharper, but less succinct, versions are formulated and proved in the body of the paper.

We take as given the classification of finite simple groups. Some of the main results depend on general consequences of CFSG, such as the facts that finite simple groups can be generated by a bounded number of elements, have bounded commutator width, and have soluble outer automorphism groups (the Schreier conjecture). Others depend on specific properties of groups of Lie type, such as

the proportion of regular semisimple elements in these groups, and the detailed structure of their automorphisms. Recent results such as the proof of the Ore Conjecture [LOST], which says that simple groups have commutator width equal to one, lead to sharper estimates for the implied constants in our main theorems, but are not necessary if one is satisfied with qualitative statements as given above.

1.3.1 The ‘Gowers trick’

A key tool in some of the proofs is a remarkable combinatorial result discovered by Tim Gowers. The basic idea is this: to show that a finite group is equal to the product of some of its subsets, it is enough to know merely that the subsets have sufficiently big cardinalities. We will need the following generalization of Gowers’s result.

For a finite group G let $l(G)$ denote *the minimal dimension of any non-trivial $\mathbb{R}$ -linear representation of G .*

Theorem 1.15 ([BNP] Corollary 2.6) *Let $X_1, \dots, X_t$ be subsets of G , where $t \geq 3$. Then*

$$\prod_{i=1}^t |X_i| \geq |G|^t \cdot l(G)^{2-t} \text{ implies } X_1 \cdot \dots \cdot X_t = G.$$

This holds in particular if $|X_i| \geq |G| \cdot l(G)^{-\mu}$ for each i , where $t\mu \leq t - 2$.

1.3.2 Facts about simple groups

Here we list some frequently quoted results, for ready reference. Here S^* will denote a quasisimple group (see below) and $S = S^*/Z(S^*)$ a finite (non-abelian) simple group.

Proposition 1.16 [AG] *S^* can be generated by 2 elements.*

(This is usually stated for simple groups, but of course any generating set for S lifts to a generating set of S^* .)

Proposition 1.17 ([GLS], Sections 7.1, 2.5) *The outer automorphism group $\text{Out}(S)$ is soluble of derived length at most 3.*

Proposition 1.18 (i) ([W], Proposition 2.4) *There exists $\delta \in \mathbb{N}$ such that every element of S is a product of δ commutators.*

(ii) *There exists $\delta^* \in \mathbb{N}$ such that every element of S^* is a product of δ^* commutators.*

((ii) follows from (i) by a theorem of Blau [B], which asserts that every element of $Z(S^*)$ is a commutator unless S^* is one of finitely many exceptions.)

Corollary 1.19 *S^* can be generated by 2δ commutators.*

For the record, we recall the validity of the *Ore Conjecture* (not strictly necessary for our results but yielding better values for the constants):

Proposition 1.20 ([LOST], [LOST2]) $\delta = 1$, $\delta^* = 2$.

Proposition 1.21 [LaS] *Let S^* be a quasisimple group of Lie type, of untwisted Lie rank r over $\mathbb{F}_q$. Then $l(S^*) \geq q^r - 1$ unless $r = 1$, in which case $l(S^*) \geq (q - 1)/2$.*

Proposition 1.22 [LiSh] *There is an absolute constant c' such that: if Y is a normal subset of S then*

$$|Y|^n \geq |S| \implies Y^{*c'n} = S.$$

It is convenient to define the the *rank* of a simple group as follows: if S is of Lie type, $\text{rank}(S)$ is the (untwisted) Lie rank of S ; if $S \cong \text{Alt}(n)$, $\text{rank}(S) = n$; if S is sporadic, $\text{rank}(S) = 0$. The next result is essentially a special case of the main theorem of [BCP]:

Proposition 1.23 *If C is a proper subgroup of S then $|S : C| \geq |S|^{\varepsilon(r)}$ where $\varepsilon(r) > 0$ depends only on $r = \text{rank}(S)$.*

1.3.3 Notation

For a group G , the centre is $Z(G)$ and the derived group is G' . For $n > 1$, $G^{(n)} = (G^{(n-1)})'$ where $G^{(1)} = G'$.

For a subset X and an element y of G , $[X, y]$ denotes the set $\{[x, y] \mid x \in X\}$. When X and Y are both *subgroups* of G , $[X, Y]$ denotes the *subgroup* $\langle [x, y] \mid x \in X, y \in Y \rangle$. In particular, the terms of the lower central series are defined by $\gamma_1(G) = G$, $\gamma_2(G) = G'$, and for $n > 1$

$$\gamma_n(G) = [G, \gamma_{n-1}(G)].$$

$\gamma_\omega(G) = \bigcap_{n=1}^{\infty} \gamma_n(G)$ is the *nilpotent residual* of G . If G is finite, then for some n we have $\gamma_\omega(G) = \gamma_n(G) = [\gamma_\omega(G), G]$.

The notation $G^{(m)}$ is also used for the Cartesian power $G \times \cdots \times G$ with m factors; which meaning is intended should be clear from the context. For $\mathbf{a}, \mathbf{b} \in G^{(m)}$ and $\alpha \in \text{Aut}(G)^{(m)}$,

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= (a_1 b_1, \dots, a_m b_m) \\ [\mathbf{a}, \alpha] &= ([a_1, \alpha_1], \dots, [a_m, \alpha_m]) \\ \mathbf{c}(\mathbf{a}, \alpha) &= \prod_{j=1}^m [a_j, \alpha_j] \end{aligned}$$

where as usual $[a, \beta] = a^{-1}a^\beta$.

In sections 2 - 4, ‘group’ means ‘finite group’, and ‘simple group’ means ‘non-abelian simple group’.

A direct (or Cartesian) product of simple groups is called *semisimple*. A group G is *quasisimple* if G is perfect (i.e. $G = G'$) and $G/Z(G)$ is simple. A central product of quasisimple groups is called *quasi-semisimple*.

For a topological group G , the connected component of the identity is denoted G^0 (not to be confused with G_0 defined above (1)).

For $m \in \mathbb{N}$ we write $[m] = \{1, \dots, m\}$.

When, occasionally, a lemma is stated without proof, it can be verified by a short direct calculation.

2 Generators

2.1 Fixed-point properties

We begin by defining a key technical concept, in three flavours: the *fixed-point* property (fpp), the *fixed-point space* property (fsp), and the *fixed-group* property (fgp):

Definition Let Ω be a finite G -set, V a finite-dimensional kG -module (k some field), and M a G -group (group acted on by G). Let $\varepsilon \in (0, 1]$. An element $y \in G$ has the

- ε -fpp on Ω if y moves at least $\varepsilon|\Omega|$ points of Ω ,
- ε -fsp on V if $\dim V(y - 1) \geq \varepsilon \dim V$.

If Y is a subset of G , we say that Y has the ε -fpp etc. if there exists $y \in Y$ having the given property.

- Y has the ε -fgp on M if (i) $M = S_1 \times \dots \times S_n$ with $n \geq 2$ and the action of G permutes the factors S_i transitively, and (ii) for each such decomposition of M , the set Y has the ε -fpp on the set $\{S_1, \dots, S_n\}$.

Remarks. (i) Each ε property implies the corresponding ε' property for any $\varepsilon' \leq \varepsilon$. If Y acts non-trivially on Ω , respectively V , then y has the $2/|\Omega|$ -fpp on Ω and the $1/\dim(V)$ -fsp on V .

(ii) Suppose that G is imprimitive on Ω and acts transitively on a set $\overline{\Omega}$ of blocks. If y has the ε -fpp on $\overline{\Omega}$ then y has the ε -fpp on Ω .

(iii) If M is a G -group and y has the ε -fgp on M , then $|C_M(y)| \leq |M|^{1-\varepsilon/2}$.

(iv) Suppose that G acts as an imprimitive linear group on V , permuting a system of imprimitivity Ω transitively. If y has the ε -fpp on Ω then y has the $\varepsilon/2$ -fsp on V .

(v) Say $|Y| = r$. If $C_V(\langle Y \rangle) = 0$ then Y has the $1/r$ -fsp on V ; if $\langle Y \rangle$ has no fixed points in Ω then Y has the $1/r$ -fpp on Ω .

These are all easy to see; for (iii) and (iv), suppose that $M = S_1 \times \dots \times S_n$ is a G -group and $y \in G$ permutes the factors S_i , according to a permutation with r cycles, including exactly k cycles of length 1. Choose representatives

$i(1), \dots, i(r)$ for these cycles. Then any fixed point of y in M is determined by its projections to $S_{i(1)}, \dots, S_{i(r)}$, so $|C_M(y)| \leq |S|^r = |M|^{r/n}$ if $S_1 \cong \dots \cong S_n \cong S$. On the other hand, we have

$$r \leq (n+k)/2 \leq n(1 - \varepsilon/2)$$

if y has the ε -fpp on $\{S_1, \dots, S_n\}$. This gives (iii), and (iv) is similar, using $\dim V$ in place of $|M|$.

We recall the

Definition.

$$G_0 = \bigcap_{M \in \mathcal{S}} C_G(M)$$

where $\mathcal{S}$ denotes the set of all (non-abelian) simple chief factors of G .

Recall also that δ is a number such that every quasisimple group can be generated by 2δ commutators. Note that we can take $\delta = 1$ (see Subsection 1.3.2).

Theorem 2.1 *Suppose $G = G' \langle Y \rangle = G_0 \langle Y \rangle$. Then Y has the $\varepsilon/2$ -fsp on every non-central abelian chief factor of G and the ε -fgp on every non-abelian chief factor of G inside G_0 , where*

$$\varepsilon = \min \left\{ \frac{1}{1 + 6\delta}, \frac{1}{|Y|} \right\}.$$

Reductions. Let $M = S_1 \times \dots \times S_n$ be a non-abelian chief factor of G , where $n > 1$ and G permutes $\Omega = \{S_1, \dots, S_n\}$ transitively. Let $\overline{\Omega}$ be a primitive quotient of the G -set Ω . Suppose that $|\overline{\Omega}| = 2$. Then G' acts trivially on $\overline{\Omega}$, so $\langle Y \rangle$ acts transitively on $\overline{\Omega}$, and it follows by Remarks (i) and (ii) that Y has the 1-fpp on Ω . Thus y has the ε -fgp on M .

Let V be a non-central abelian chief factor of G , so G acts as an irreducible $\mathbb{F}_p$ -linear group on V .

(i) Suppose that this action is not primitive, so it induces a primitive permutation action of G on a system of imprimitivity Ω . If $|\Omega| = 2$ then as above we may deduce that Y has the 1-fpp on Ω , hence the $1/2$ -fsp on V , by Remark (iv).

(ii) Suppose that V is not inside G_0 . Then G_0 centralizes V , so V is a non-trivial simple $\mathbb{F}_p \langle Y \rangle$ -module, and then Y has the ε -fsp on V by Remark (v).

(iii) Suppose that $\dim_{\mathbb{F}_p} V = 1$. Then G' centralizes V , whence $V(y-1) = V$ for some $y \in Y$; thus Y has the ε -fsp on V .

Arguing by induction on the number of non-central factors in a chief series of G inside G_0 , it will therefore suffice to prove the following proposition.

Proposition 2.2 *Let G be a group and Y a subset of G of size $r \geq 1$ such that $G = G' \langle Y \rangle = G_0 \langle Y \rangle$. Suppose that $\langle Y \rangle$ does not centralize any non-central abelian chief factor of G , and that if $M = S_1 \times \cdots \times S_n$ is a non-abelian chief factor of G , with each S_i simple and $n \geq 2$, then $\langle Y \rangle$ does not normalize every S_i . Put $\varepsilon = \min\{1/(1 + 6\delta), 1/r\}$. Then Y has the ε -fsp on every primitive irreducible $\mathbb{F}_p G$ -module of dimension at least two, and the ε -fpp on every primitive G -set of size at least 3.*

2.1.1 Primitive modules

Let G be a group and Y a subset of G of size r satisfying the hypotheses of Proposition 2.2. Let V be a primitive irreducible $\mathbb{F}_p G$ -module of dimension at least two; we may assume that G acts faithfully on V . Put $F = \text{Fit}(G)$, the Fitting subgroup of G .

Lemma 2.3 *Let S be a quasisimple subgroup of G and y an element of G such that $[S, S^y] = 1$. Then there exist $a_j, b_j \in S$ such that $S \leq \langle y, y^{a_j}, y^{b_j}, y^{a_j b_j} \mid 1 \leq j \leq 2\delta \rangle$.*

Proof. For $u, v \in S$ we have

$$[u^{-1}, v] = [[u, y], v] = y^{-1} y^u y^{-uv} y^v.$$

The lemma follows since S is generated by 2δ commutators. ■

Lemma 2.4 *If $y \in G$ satisfies $[F, y] \neq \{1\}$ then y has the $\frac{1}{4}$ -fsp on V .*

Proof. (cf. [GSS], proof of Theorem 5.3) For $x \in G$ put $c(x) = \dim C_V(x)$. As $C_V([x_1, x_2]) \geq C_V(x_1) \cap C_V(x_1) \cdot x_2$ we have

$$c([x_1, x_2]) \geq 2c(x_1) - \dim(V). \quad (2)$$

The hypotheses imply that every abelian normal subgroup of G is cyclic and acts freely on V . It follows by a theorem of P. Hall (see [As], 23.9 or [Go], Theorem 5.4.9) that F is metabelian. Thus if $1 \neq t \in F' \cup Z(F)$ then $c(t) = 0$. Now there exists $x \in F$ such that $[x, y] \neq 1$. If $[x, y] \in Z(F)$ we may infer using (2) that $c(y) \leq \frac{1}{2} \dim(V)$. If $[x, y] \notin Z(F)$ then for some $h \in F$ we have $1 \neq [[x, y], h] \in F'$. Then using (2) twice gives

$$c(y) \leq \frac{1}{2} (c[x, y] + \dim(V)) \leq \frac{1}{2} \left(\frac{1}{2} \dim(V) + \dim(V) \right) = \frac{3}{4} \dim(V).$$

The result follows. ■

In view of the preceding lemma, we may suppose for the rest of this subsection that $[F, Y] = \{1\}$. Since Y does not centralize any non-central abelian chief factor of G , this implies that F is contained in the hypercentre of G , and hence that $[F, \gamma_\omega(G)] = 1$. But $G = G' \langle Y \rangle$ implies $G = \gamma_\omega(G) \langle Y \rangle$; therefore $[F, G] = 1$, and so $F = Z(G)$.

Now let $F^* = F^*(G_0)$ denote the generalized Fitting subgroup of G_0 (see [As], Section 31). Then

$$C_{G_0}(F^*) = Z(F^*) = F \cap G_0.$$

Case 1. Suppose that $F^* \leq F$. Then F^* is central in G_0 and it follows that $G_0 = F \cap G_0 \leq Z(G)$. Hence $C_V(\langle Y \rangle)$ is a G -submodule of V ; as V is faithful and irreducible for G and $\langle Y \rangle \neq 1$ it follows that $C_V(\langle Y \rangle) = 0$. Hence Y has the $1/r$ -fsp on V by Remark (v).

Case 2. Suppose that $F^* \not\leq F$. Then $F^* = E \cdot F_0$ where E is a non-empty central product of quasisimple groups, $F_0 = F \cap G_0$, and E is characteristic in G_0 with centre $Z_0 = E \cap F$. Let $\overline{N} = N/Z_0$ be a minimal normal subgroup of G/Z_0 contained in E/Z_0 . Then $\overline{N} = \overline{S}_1 \times \cdots \times \overline{S}_n$, where each $\overline{S}_i = S_i/Z_0$ is a simple group. By hypothesis, there exists $y \in Y$ such that y moves at least one of these factors; say y moves $\overline{S}_1$. Then $[S_1, S_1^y] = 1$, and Lemma 2.3 now shows that $S_1 \leq \langle y_1, \dots, y_t \rangle$ where $t = 1 + 6\delta$ and each y_j is a conjugate of y .

We claim that $C_V(S_1) = 0$. Accepting the claim for now, it follows by Remark (v) that some y_j has the $1/t$ -fsp on V ; as y_j is conjugate to y we may conclude that y has the $1/t$ -fsp on V .

Since V is a primitive irreducible $\mathbb{F}_p G$ -module it is a direct sum of copies of some simple $\mathbb{F}_p N$ -module W . If $C_V(S_1) \neq 0$ then W is a composition factor of the $\mathbb{F}_p N$ -module $C_V(S_1)$, so $W(S_1 - 1) = 0$. But then $V(S_1 - 1) = 0$, a contradiction since V is faithful for G . Thus $C_V(S_1) = 0$ as claimed.

The first claim of Proposition 2.2 clearly follows.

2.1.2 Primitive G -sets

Let G be a group and Y a subset of G of size r satisfying the hypotheses of Proposition 2.2. Let Ω be a primitive G -set of size $n \geq 3$, on which G acts faithfully.

If $\langle Y \rangle$ has no fixed points in Ω then Y has the $1/r$ -fpp on Ω , by Remark (v). We assume henceforth that $\langle Y \rangle$ has at least one fixed point in Ω ; since $G = G_0 \langle Y \rangle$ is transitive this implies also that $G_0 \neq 1$.

According to [DM] Theorem 4.3B (part of the O’Nan-Scott Theorem), one of the following holds:

- (a) G has a unique minimal normal subgroup $N = C_G(N)$ and N acts regularly on Ω ;
- (b) G has exactly two minimal normal subgroups N and $C_G(N)$, and each of them acts regularly on Ω ;
- (c) G has a unique minimal normal subgroup N and $C_G(N) = 1$.

Since $G_0 > 1$, in cases (a) and (c) we have $N \leq G_0$; in case (b) at least one of N and $C_G(N)$ must lie in G_0 , and we choose to call that one N .

Case 1. Suppose that the minimal normal subgroup N of G contained in G_0 acts regularly on Ω . Then $|N| = n$ and N is a non-central chief factor of G ([DM], Theorem 4.3B). Let $\alpha \in \Omega$ be a fixed point for $\langle Y \rangle$. Then for $x \in N$ and $y \in Y$ we have $(\alpha x)y = \alpha x^y$, so y has exactly $|C_N(y)|$ fixed points on $\alpha N = \Omega$. By hypothesis, there exists $y \in Y$ such that $C_N(y) \neq N$. The number of fixed points of y in Ω is then at most

$$|C_N(y)| \leq \frac{1}{2} |N| = \frac{1}{2} n,$$

so y has the $\frac{1}{2}$ -fpp on Ω .

Case 2. The unique minimal normal subgroup N of G is not regular on Ω . Then N is not abelian, so $N = S_1 \times \cdots \times S_m$ where each S_i is simple and $m \geq 2$ since $N \leq G_0$. According to [DM] Theorem 4.6A there are now two possibilities.

Subcase 2.1. G acts as a group of diagonal type on Ω . Fixing an identification of each S_i with a group T , we identify Ω with the right coset space $T^* \backslash T^{(m)}$ where T^* denotes the diagonal subgroup. The action of N is induced by the right regular action, so

$$T^*(t_1, \dots, t_m) \cdot s_1 \dots s_m = T^*(t_1 s_1, \dots, t_m s_m)$$

for $(t_1, \dots, t_m) \in T^{(m)}$ and $s_i \in S_i$. Write $k = |T|$, so that $n = k^{m-1}$.

Let $\alpha = T^*(t_1, \dots, t_m)$ be a fixed point for $y \in G$. The stabilizer of α in N is

$$N_\alpha = \{ (u^{t_1}, \dots, u^{t_m}) \mid u \in T \},$$

so for $x \in N$ we have

$$(\alpha x) \cdot y = \alpha x \iff \alpha x^y = \alpha x \iff x^y x^{-1} = (u^{t_1}, \dots, u^{t_m}), \text{ some } u \in T. \quad (3)$$

Suppose that the conjugation action of y permutes $S_1, S_2, \dots, S_e$ cyclically, and that (3) holds with $x = s_1 s_2 \dots s_m$ ($s_i \in S_i$). Then $s_2, \dots, s_e$ are uniquely determined by u and s_1 . Thus if y has $q = q(y)$ cycles in its action on $\{S_1, S_2, \dots, S_m\}$, then the number of $x \in N$ satisfying (3) is at most $k \cdot k^q$. The mapping $x \mapsto \alpha x$ from N to Ω is surjective and each fibre has size k . It follows that y has at most k^q fixed points in Ω .

Suppose that some $y \in Y$ moves at least 3 of the S_i . Then $q(y) \leq m - 2$, and so the number of fixed points of y in Ω is at most

$$k^{q(y)} \leq k^{m-2} = nk^{-1}.$$

If this holds for no element $y \in Y$, then Y must contain an element y_1 that acts as a transposition (12), say, on $\{S_1, S_2, \dots, S_m\}$.

Assume first that $m \geq 3$. There exists $g \in G$ such that $S_2^g = S_3$; then $y = [y_1, g]$ moves at least 3 of the S_i , and hence fixes at most nk^{-1} points in Ω . It follows that y_1 has at most $(n + nk^{-1})/2$ fixed points.

Suppose now that $m = 2$. Set $y = [y_1, a]$ where $1 \neq a \in S_1$. Suppose that y fixes $\alpha \in \Omega$. Each element of Ω can be put uniquely in the form αx with $x = (s, 1)$, $s \in S_1$, and then (3) gives

$$\begin{aligned} (\alpha x) \cdot y = \alpha x &\iff (s^a s^{-1}, 1) = (u^{t_1}, u^{t_2}) \\ &\iff s \in C_{S_1}(a). \end{aligned}$$

Thus y has at most $|C_{S_1}(a)| \leq \frac{1}{5} |S_1| = \frac{1}{5}n$ fixed points in Ω . It follows that y_1 has at most $\frac{3}{5}n$ fixed points.

Thus in any case, we may conclude (since $k \geq 60$) that y contains an element with the ε -fpp as long as $\varepsilon \leq \frac{2}{5}$.

Subcase 2.2. G is contained in a wreath product $W = H \wr \pi(G)$ where $H \leq \text{Sym}(\Gamma)$, $\pi : G \rightarrow \text{Sym}(d)$ where $d > 1$, and W acts on $\Omega = \Gamma^{(d)}$ by the product action. In this case $N = N_1 \times \dots \times N_d \leq H^{(d)}$, and G permutes the factors N_i via π . Put $k = |\Gamma|$, so $n = k^d$. Note that $k \geq 5$ since N is not soluble.

Suppose that $y = b \cdot \pi(y)$ fixes $(\gamma_1, \dots, \gamma_d) \in \Gamma^{(d)}$, where $b \in H^{(d)}$. If $\pi(y)$ has a cycle $(1, 2, \dots, e)$ then $\gamma_{i+1} = \gamma_i^{b_i}$ for $i = 1, \dots, e-1$. Thus if $\pi(y)$ has $q = q(y)$ cycles then the number of fixed points of y in Ω is at most k^q .

By hypothesis, there exists $y \in Y$ such that $\pi(y) \neq 1$. Then $q(y) \leq d-1$ and so y has at most $k^{d-1} \leq n/5$ fixed points in Ω . Thus y has the $\frac{4}{5}$ -fpp on Ω .

The proof of Proposition 2.2 is now complete.

2.2 Small chief factors

We quote a mild generalization of a well-known result due to Gaschütz [Gch]; the proof given (for example) in [FJ], Lemma 15.30 adapts easily to yield this version:

Lemma 2.5 *Let $Y_1 \subseteq G$ and $D \triangleleft G$. Suppose that*

$$G = D \langle y_1, \dots, y_d, Y_1 \rangle$$

where $d \geq d(G)$. Then there exist $h_1, \dots, h_d \in D$ such that $G = \langle h_1 y_1, \dots, h_d y_d, Y_1 \rangle$.

We have defined δ to be a number such that each element of every simple group is a product of δ commutators, and observed that in fact one can take $\delta = 1$ (Subsection 1.3.2).

Lemma 2.6 Suppose that $M = S_1 \times S_2$ and $\alpha \in \text{Aut}(M)$ satisfies $S_1^\alpha = S_2$, $S_2^\alpha = S_1$. Let $C = \{[x, y] \mid x, y \in S_1\}$. Then

$$C \subseteq [M, \alpha]^{*4}.$$

If S_1 is simple then

$$M = [M, \alpha]^{*8\delta}.$$

Proof. Let $x, y \in S_1$. Then

$$\begin{aligned} [y, x^{-1}] &= [y, [x, \alpha]] = [y, \alpha][y, \alpha^{-x}][[y, \alpha^{-x}], \alpha] \\ &= [y, \alpha] \cdot [y^{x^{-1}}, \alpha^{-1}]^x \cdot [[y, \alpha^{-x}], \alpha] \end{aligned}$$

and the middle factor lies in $[M, \alpha]^{*2}$ because for any $z \in S_1$ we have

$$\begin{aligned} [z, \alpha^{-1}]^x &= [zx, \alpha^{-1}][x^{\alpha^{-1}}, \alpha] \\ &= [(zx)^{-\alpha^{-1}}, \alpha][x^{\alpha^{-1}}, \alpha] \end{aligned}$$

(for the final equality note that $(zx)^\alpha$ commutes with zx). This establishes the first claim.

If S_1 is simple, then $S_1 = C^{*\delta}$, so $M = C^{*\delta} \cdot (C^{*\delta})^\alpha \subseteq [M, \alpha]^{*8\delta}$ since $[M, \alpha] = [M, \alpha]^\alpha$. ■

For technical reasons, we need to introduce a slightly smaller analogue of the subgroup G_0 :

Definition For a group G , let

$$G_1 = \bigcap_{M \in \mathcal{C}(G)} C_G(M) \quad (4)$$

where $\mathcal{C}(G)$ denote the set of all non-abelian chief factors of G that have composition length at most two. We shall call such chief factors ‘bad’.

Remarks. (vi) A non-abelian chief factor belongs to $\mathcal{C}(G)$ if and only if it is either simple or a product of two simple groups. Hence such a factor that occurs inside G_1 is a product of at least 3 simple groups.

(vii) $(G^2)^{(3)}G_1/G_1$ is semisimple: for if $M \in \mathcal{C}(G)$ then $G/C_G(M)$ is an extension of M by $\text{Out}(M)$, $\text{Out}(M)$ is isomorphic to $\text{Out}(S)$ or $\text{Out}(S) \wr C_2$ where S is simple, and $\text{Out}(S)^{(3)} = 1$ (Proposition 1.17).

(viii) If $G > 1$ then $G_1 < G$ or $G' < G$.

Proposition 2.7 Let G be a group and $W = \{w_1, \dots, w_s\}$ a subset such that $G = D \langle W \rangle$ where $D \leq G_0 \cap G^{(4)}G_1$. Then there exist elements $b_{ij} \in D$ such that

$$G = \left\langle w_i^{b_{ij}} \mid i = 1, \dots, s, j = 1, \dots, m \right\rangle (D \cap G_1)$$

where $m = 1 + 8\delta d(G)$.

Proof. Note that $G_1 = \bigcap_{M \in \mathcal{C} \setminus \mathcal{S}} C_{G_0}(M)$, where $\mathcal{S}$ denotes the set of all simple chief factors of G . The section $G^{(4)}G_1/G_1$ is semisimple, and is a product of minimal normal subgroups of G/G_1 belonging to $\mathcal{C}$. We may suppose that $D \cap G_1 = 1$. In that case, D is a product of minimal normal subgroups of G belonging to $\mathcal{C} \setminus \mathcal{S}$.

Let $M = S_1 \times S_2$ be one of these. Then D normalizes S_1 and S_2 , so there exists $y \in W$ such that $S_1^y = S_2$ and $S_2^y = S_1$. Now Lemma 2.6 shows that

$$M = [M, y]^{*8\delta}.$$

As D is the direct product of such normal subgroups M of G , it follows that

$$D = [D, y]^{*8\delta}.$$

If $r < d = d(G)$ put $w_{r+1} = \dots = w_d = w_r$. Now applying Lemma 2.5 we find elements $h_j \in D$ such that $G = \langle h_1 w_1, \dots, h_d w_d, w_{d+1}, \dots, w_r \rangle$. Each h_j lies in the subgroup generated by W and 8δ D -conjugates of the w_i . The result follows. ■

2.3 Lifting generators

Recall that a chief factor of G is *bad* if it is either simple or the product of two simple groups.

Proposition 2.8 *Let $G = N \langle y_1, \dots, y_m \rangle$ be a d -generator group where N is a non-central minimal normal subgroup of G . If N is non-abelian, assume that N is not bad. Let*

$$\mathcal{X} = \left\{ \mathbf{a} \in N^{(m)} \mid \langle y_1^{a_1}, \dots, y_m^{a_m} \rangle = G \right\}.$$

(i) *Suppose that N is abelian and that y_j has the ε -fsp on N for at least k values of j . Then*

$$|\mathcal{X}| \geq |N|^m (1 - |N|^{d-k\varepsilon}).$$

(ii) *Suppose that N is non-abelian and that y_j has the ε -fgp on N for at least k values of j , where $k\varepsilon \geq \max\{2d+4, C\}$ for a certain absolute constant C . Then*

$$|\mathcal{X}| \geq |N|^m (1 - 2^{2-k\varepsilon}).$$

Proof. Part (i) is [NS], Proposition 5.1(i). In the situation of (ii), the proof of [NS], Proposition 5.1(ii) shows that $|\mathcal{X}| \geq |N|^m (1 - z)$ where $z \leq \zeta(k\varepsilon) - 1$ (Riemann zeta function). A crude estimate gives $\zeta(t) - 1 \leq 2^{2-t}$ for $t > 2$. ■

The main result is now

Theorem 2.9 *Let G be a group and $K \leq G_0$ a normal subgroup of G . Let $Y = \{y_1, \dots, y_r\}$ be a subset of G such that $G = G' \langle Y \rangle = K \langle Y \rangle$. Then there exist elements $x_{ij} \in K$ such that*

$$G = \langle y_i^{x_{ij}} \mid i = 1, \dots, r, j = 1, \dots, k \rangle$$

where

$$\begin{aligned} k &= \max\{(1 + 2d(G)\widehat{r})(1 + 8\delta d(G)), \widehat{r}C\} \\ &= f_0(r, d(G)) \leq C_0 r d(G)^2, \end{aligned} \quad (5)$$

$\widehat{r} = \max\{r, 1 + 6\delta\}$, and C and C_0 are absolute constants.

Corollary 2.10 *If G has no simple chief factors and $G = G' \langle y_1, \dots, y_r \rangle$ then $G = \langle y_i^{c_{ij}} \mid i = 1, \dots, r, j = 1, \dots, k \rangle$.*

Proof of Theorem 2.9. Write $d = d(G)$ and set $\varepsilon = \widehat{r}^{-1}$. Let N be a non-central chief factor of G . We will say that a subset W of G has the (k, ε) -property w.r.t. N if N is abelian and at least k elements of W have the $\varepsilon/2$ -fsp on N , or if N is a product of at least 3 simple groups and at least k elements of W have the ε -fpp on the set of simple factors of N . According to Theorem 2.1, the set $\{y_1, \dots, y_r\}$ has the $(1, \varepsilon)$ -property w.r.t. N .

Put $D = K \cap G^{(4)}G_1$. We begin by proving

(*) there exists elements $a_{ij} \in K$ such that

$$G = D \langle y_i^{a_{ij}} \mid i = 1, \dots, r, j = 1, \dots, k_1 \rangle \quad (6)$$

where $k_1 = 1 + 2d\widehat{r}$.

Replacing G by G/D for the moment, we may assume that K is soluble. If $K = 1$ we can take all $a_{ij} = 1$ and there is nothing to prove.

Suppose that $K > 1$ and let N be a minimal normal subgroup of G contained in K ; then N is abelian. Arguing by induction on $|K|$, we may suppose that $G = N \langle W \rangle$ where

$$W = \{y_i^{a_{ij}} \mid i = 1, \dots, r, j = 1, \dots, k_1\}$$

and each $a_{ij} \in K$. If $N \leq Z(G)$ then $G' \leq \langle W \rangle$, so $\langle W \rangle \geq G' \langle Y \rangle = G$ and we are done.

If N is non-central, the set W has the (k_1, ε) -property w.r.t. N . As $d - k_1\varepsilon/2 < 0$, Proposition 2.8(i) shows that there exist elements $b_{ij} \in N$ such that

$$G = \langle y_i^{a_{ij}b_{ij}} \mid i = 1, \dots, r, j = 1, \dots, k_1 \rangle,$$

and (6) follows on replacing a_{ij} by $a_{ij}b_{ij}$. This completes the proof of (*).

Now we apply Proposition 2.7 to find elements $c_{ijl} \in D$ such that

$$G = G_1 \langle y_i^{a_{ij}c_{ijl}} \mid i = 1, \dots, r, j = 1, \dots, k_1, l = 1, \dots, m \rangle$$

where $m = 1 + 8\delta d$.

If $G_1 = 1$ we are done. Otherwise, let N be a minimal normal subgroup of G contained in G_1 , and suppose inductively that $G = N \langle W \rangle$ where $W = \{y_i^{x_{ij}} \mid i = 1, \dots, r, j = 1, \dots, k_1 m\}$. If N is abelian we deduce as above that

G is generated by a set of the form $\{y_i^{x_{ij}b_{ij}} \mid i = 1, \dots, r, j = 1, \dots, k_1m\}$, and the result follows since $k = \max\{k_1m, \widehat{r}C\} \geq k_1m$.

Suppose that N is non-abelian; then N is not bad. If $k_1m < \widehat{r}C$, enlarge the family W by repeating some of its elements to obtain a family containing k conjugates of each y_i ($i = 1, \dots, r$). Then in any case, W has the (k, ε) -property w.r.t. N ; Proposition 2.8(ii) now shows that G is generated by a set of the form $\{y_i^{x_{ij}c_{ij}} \mid i = 1, \dots, r, j = 1, \dots, k\}$ with $c_{ij} \in N$, as required. (Note that $k\varepsilon \geq \max\{2d + 4, C\}$ since $k_1m > 18d$.) ■

Remarks. (ix) Recall that $\delta = 1$ if we accept the validity of the Ore Conjecture (Subsection 1.3.2).

(x) If we assume that $K \leq G_1$ we can take $k = \widehat{r} \cdot \max\{2d + 4, C\} = O(rd)$. In particular, if G has no bad chief factors then the Corollary holds with this smaller value of k .

3 Commutators

In this section we begin the proof of the two main ‘commutator’ results.

Theorem 3.1 *Let $G = \langle g_1, \dots, g_r \rangle$ be a group and H a normal subgroup of G . Then*

$$[H, G] = \left(\prod_{i=1}^r [H, g_i]^H \right)^{*f_3} = \left(\prod_{i=1}^r [H, g_i][H, g_i^{-1}] \right)^{*f_3}$$

where $f_3 = O(rd) = O(r^2)$ depends only on r and $d = d(G)$.

Theorem 3.2 *Let $G = G' \langle g_1, \dots, g_r \rangle$ be a group and H a normal subgroup of G such that $H \langle g_1, \dots, g_r \rangle = G$.*

(i) *If $H \leq G_0$ then*

$$[H, G] = \left(\prod_{i=1}^r [H, g_i]^H \right)^{*f_4} = \left(\prod_{i=1}^r [H, g_i][H, g_i^{-1}] \right)^{*f_4},$$

(ii) *if $H \leq G_1$ then*

$$[H, G] = \left(\prod_{i=1}^r [H, g_i] \right)^{*f_5},$$

where $f_4 = O(r^5d^6)$ and $f_5 = O(rd)$ depend only on r and $d = d(G)$.

These are not quite the same as Theorems 1.2 and 1.3, which refer to a symmetric set $Y = \{y_1, \dots, y_r\}$, and omit the factors $[H, g_i^{-1}]$. To deduce the stated results, note that if Y is symmetric then

$$\left(\prod_{j=1}^r [H, y_j] \right)^{*2r} \supseteq \prod_{j=1}^r [H, y_i][H, y_i^{-1}],$$

and we may take $f_1 = 2rf_3$, $f_2 = 2rf_4$; of course if we are allowed to order Y so that $y_{2i} = y_{2i-1}^{-1}$ for $i = 1, \dots, r/2$ then we can take $f_1 = f_3$ and $f_2 = f_4$.

3.1 Acceptable normal subgroups

Suppose that $A < B$ are normal subgroups of a group G . Recall that B/A is a *bad chief factor* of G if B/A is a minimal normal subgroup of G/A and B/A is either simple or the direct product of two simple groups. Thus G_1 (defined in Subsection 2.1) is precisely the intersection of the centralizers of all bad chief factors of G .

A normal subgroup H of G is said to be *acceptable* in G if

- (a) $H = [H, G]$ and
- (b) if $A < B \leq H$ are normal subgroups of G then B/A is not a bad chief factor of G .

Here we show how the main results may be reduced to the consideration of acceptable normal subgroups.

Lemma 3.3 *$H \triangleleft G$ is acceptable if and only if $H = [H, G] \leq G' \cap G_1$.*

Proof. If $H \geq B > A$ and B/A is a bad chief factor then H does not centralize B/A , so $H \not\leq G_1$. Conversely, if $H \not\leq G_1$ then H does not centralize some bad chief factor B/A ; then $(B \cap H)A = A$ so $(B \cap H)/(A \cap H) \cong B/A$ and $A \cap H < B \cap H \leq H$ contradicts (b), showing that H is not acceptable. ■

The next result is elementary; it is the general form of facts (A) and (B) mentioned in the introduction:

Lemma 3.4 *Let $H \triangleleft G = G' \langle g_1, \dots, g_r \rangle$ and let $n \geq 1$. Then*

$$[H, G] = [H, {}_n G] \prod_{i=1}^r [H, g_i].$$

If in addition we have $G = H \langle g_1, \dots, g_r \rangle$ then

$$G = [H, {}_n G] \langle g_1, \dots, g_r \rangle.$$

Proof. The first claim is [NS], Lemma 2.4 or [S2], Prop. 1.2.5. For the second, we argue by induction on n and reduce to the case where $[H, G] = 1$. Then $G' \leq \langle g_1, \dots, g_r \rangle$ and the claim is evident. ■

Lemma 3.5 *Let G be a group and $\alpha, \beta \in \text{Aut}(G)$. Then*

$$\begin{aligned} [G, \alpha]^G &\subseteq [G, \alpha][G, \alpha^{-1}], \\ [G, \alpha\beta]^G &\subseteq [G, \beta][G, \beta^{-1}][G, \alpha][G, \alpha^{-1}][G, \beta][G, \beta^{-1}] \\ [G, \alpha^{-1}\beta\alpha]^G &\subseteq [G, \beta][G, \beta^{-1}][G, \alpha][G, \alpha^{-1}] \end{aligned}$$

Proof.

$$\begin{aligned} [x, \alpha]^y &= [xy, \alpha][y^\alpha, \alpha^{-1}], \\ [x, \alpha\beta] &= [x, \beta][x, \alpha][[x, \alpha], \beta] \\ [x, \alpha^{-1}\beta\alpha] &= [x^{\alpha^{-1}}, \beta][[x^{\alpha^{-1}}, \beta], \alpha]. \end{aligned}$$

■

Lemma 3.6 *Let G be a quasisimple group and $\alpha \in \text{Aut}(G)$. Put $\overline{G} = G/Z(G)$. If $|\overline{G}, \alpha|^s \geq |\overline{G}|$ then*

$$G = ([G, \alpha]^G)^{*cs}$$

where $c \in \mathbb{N}$ is an absolute constant.

Proof. Proposition 1.22 shows that if Y is a normal subset of $\overline{G}$ with $|Y|^s \geq |\overline{G}|$ then $\overline{G} = Y^{*c's}$, where c' is an absolute constant. Applying this with $Y = XZ(G)/Z(G)$ where $X = [G, \alpha]^G$ we get

$$G = X^{*c's}Z(G).$$

Now for $g, h, k \in G$ we have

$$[[g, \alpha]^k, h] = [g, \alpha]^{-k}[g, \alpha]^{kh} = [g^{-1}, \alpha]^{gk}[g, \alpha]^{kh} \in X^{*2},$$

so if $w \in X^{*c's}$ then

$$[w, h] \in X^{*2c's}.$$

According to Proposition 1.18, there exists an absolute constant δ^* such that every element of G is a product of δ^* commutators (In fact $\delta^* = 2$). It follows that

$$G = X^{*2c's\delta^*}.$$

■

Lemma 3.7 *Let $G = \langle g_1, \dots, g_r \rangle$ and suppose that $T \triangleleft G$ is quasisemisimple with one or two simple composition factors. Then*

$$T = \left(\prod_{i=1}^r [T, g_i]^T \right)^{*k_0 r},$$

where k_0 is an absolute constant.

Proof. Suppose that T is quasisimple, with centre Z . Put $\overline{T} = T/Z$. Then $C_{\overline{T}}(G) = 1$ so $|\overline{T}| \leq \prod_{i=1}^r |\overline{T}, g_i|$ and so $|\overline{T}, g_i| \geq |\overline{T}|^{1/r}$ for some i . Now Lemma 3.6 implies that $T = ([T, g_i]^T)^{*cr}$.

If T is not quasisimple, then $T = S_1 S_2$ with each S_i quasisimple and $[S_1, S_2] = 1$. If G normalizes the factors S_i , we apply the preceding paragraph to each factor and obtain the same result as before. Otherwise, G permutes them

transitively by conjugation. The action of G lifts to an action on the universal cover $\widetilde{T} = \widetilde{S}_1 \times \widetilde{S}_2$, and for some i we have $\widetilde{S}_1^{g_i} = \widetilde{S}_2$, $\widetilde{S}_2^{g_i} = \widetilde{S}_1$. Let C_j denote the set of commutators in $\widetilde{S}_j$; then Lemma 2.6 shows that

$$C_j \subseteq [\widetilde{T}, g_i]^{*4}$$

for $j = 1, 2$. Since $\widetilde{S}_j = C_j^{*4\delta^*}$ (Proposition 1.18), it follows that

$$\widetilde{T} = [\widetilde{T}, g_i]^{*4\delta^*},$$

which implies $T = [T, g_i]^{*4\delta^*}$.

The result follows on setting $k_0 = \max\{c, 4\delta^*\}$. ■

Let us say that $N \triangleleft G$ is *narrow* if

$$\bigcap_{T \in \mathcal{M}} T \leq Z(N)$$

where $\mathcal{M}$ is the set of normal subgroups T of G contained in N such that N/T is semisimple with composition length at most two. This is equivalent to saying that $N/Z(N)$ is a direct product of bad chief factors of G (occurring as minimal normal subgroups of $G/Z(N)$).

Lemma 3.8 *Let $G = \langle g_1, \dots, g_r \rangle$ and let N be a perfect narrow normal subgroup of G . Then*

$$N = \left(\prod_{i=1}^r [N, g_i]^N \right)^{*k_0 r}$$

where k_0 is given in Lemma 3.7.

Proof. The hypotheses imply that N is a central product $N = T_1 \dots T_n$ where each T_i is a quasisemisimple normal subgroup of G having one or two simple composition factors. As the T_i commute elementwise the claim follows from Lemma 3.7. ■

Proposition 3.9 *Let $G = \langle g_1, \dots, g_r \rangle$ and let $H \triangleleft G$. Then G has normal subgroups $H_3 \leq H_2 \leq H_1 \leq [H, G]$ such that*

$$[H, G] = \prod_{i=1}^r [H, g_i] \cdot H_1, \tag{7}$$

$$H_2 = \left(\prod_{i=1}^r [H, g_i]^H \right)^{*k_0 r} \cdot H_3, \tag{8}$$

H_1/H_2 is acceptable in G/H_2 and H_3 is acceptable in G .

Proof. Let G_s be the soluble residual of G and set $H_1 = [H, {}_\omega G]$, $H_2 = [H_1, G_s]$. Let D be the intersection of all $M \triangleleft G$ such that $M < H_2$ and H_2/M is either simple or a product of two simple groups, and put $H_3 = [D, G_s]$.

Then (7) follows from Lemma 3.4. Also $H_1 = [H_1, G]$ and H_1/H_2 is soluble, so H_1/H_2 is acceptable in G/H_2 .

Now

$$[H_3, G] \geq [H_3, G_s] = [D, G_s, G_s] = [D, G_s] = H_3$$

since G_s is perfect. To complete the proof that H_3 is acceptable, suppose that $K \leq H_3$ is a minimal normal subgroup of G and that K is either simple or a product of two simple groups. Then $G/KC_G(K)$ is soluble by the Schreier conjecture (Proposition 1.17), so $G_s \leq KC_G(K)$ and as $K \leq H_2 \leq G_s$ it follows that $H_2 = K \times C_{H_2}(K)$. This implies that $C_{H_2}(K) \geq D \geq K$, a contradiction. Applying this argument to an arbitrary quotient of G we infer that H_3 is acceptable in G .

Finally, H_2/H_3 is narrow in G/H_3 so Lemma 3.8 gives (8). ■

3.2 The ‘Key Theorem’

The ‘Key Theorem’ of [NS] described certain product decompositions of an acceptable normal subgroup in a d -generator group. As one of us wrote in [S2], ‘each part has an undesirable feature in either its hypothesis or its conclusion’. These are now swept away in our core technical result. To state this we need some notation:

Definition For $\mathbf{g}, \mathbf{v} \in G^{(m)}$ and $1 \leq j \leq m$,

$$\tau_j(\mathbf{g}, \mathbf{v}) = v_j[g_{j-1}, v_{j-1}] \dots [g_1, v_1].$$

Theorem 3.10 *There exists a function $k : \mathbb{N}^{(2)} \rightarrow \mathbb{N}$ with the following property. Let G be a d -generator group and H an acceptable normal subgroup of G . Suppose that $G = H \langle g_1, \dots, g_r \rangle$. Put $m = r \cdot k(d, r)$, and for $1 \leq j < k(d, r)$ and $1 \leq i \leq r$ set*

$$g_{i+jr} = g_i.$$

Then for each $h \in H$ there exist $\mathbf{v}(i) \in H^{(m)}$ ($i = 1, \dots, 10$) such that

$$h = \prod_{i=1}^{10} \prod_{j=1}^m [v(i)_j, g_j] \tag{9}$$

and

$$\langle g_1^{\tau_1(\mathbf{g}, \mathbf{v}(i))}, \dots, g_m^{\tau_m(\mathbf{g}, \mathbf{v}(i))} \rangle = G \quad \text{for } i = 1, \dots, 10. \tag{10}$$

In fact we can take

$$k(d, r) = 1 + \max\{r, 1 + 6\delta\} \cdot \max\{4d + 4, \widehat{C}\} \leq C_1 dr,$$

where $\widehat{C}$ and C_1 are absolute constants.

The proof will occupy the next three subsections. Accepting the theorem for now, we deduce the main results stated above.

Proof of Theorem 3.1. We are given $H \triangleleft G = \langle g_1, \dots, g_r \rangle$. Let $H_3 \leq H_2 \leq H_1 \leq [H, G]$ be the normal subgroups given by Proposition 3.9. Thus H_1/H_2 is acceptable in G/H_2 and H_3 is acceptable in G . Theorem 3.10 shows that

$$H_1 = \left(\prod_{j=1}^r [H_1, g_j] \right)^{*10k(d,r)} \cdot H_2$$

and that

$$H_3 = \left(\prod_{j=1}^r [H_3, g_j] \right)^{*10k(d,r)}$$

where $d = d(G)$. Combining these with (7) and (8) from Proposition 3.9 we deduce that

$$[H, G] = \left(\prod_{j=1}^r [H, g_j]^H \right)^{*f_3}$$

where $f_3 = 1 + k_0 r + 20k(d, r)$; here k_0 is the absolute constant introduced in Lemma 3.7. Finally, Lemma 3.5 shows that $[H, g_j]^H$ can be replaced by $[H, g_j][H, g_j^{-1}]$ for each j .

We observe that $f_3 = O(r + k(d, r)) = O(dr) = O(r^2)$.

Proof of Theorem 3.2 (i). Now $H \triangleleft G$ satisfies $H \leq G_0$, and $G = G' \langle g_1, \dots, g_r \rangle = H \langle g_1, \dots, g_r \rangle$. According to Theorem 2.9, there exist element $x_{ij} \in H$ such that

$$G = \langle g_i^{x_{ij}} \mid i = 1, \dots, r, j = 1, \dots, k \rangle$$

where $k = f_0(r, d(G))$. Using this generating set in Theorem 3.1 gives

$$\begin{aligned} [H, G] &= \left(\prod_{j=1}^k \prod_{i=1}^r [H, g_i^{x_{ij}}]^H \right)^{*f_3(kr)} \\ &= \left(\prod_{j=1}^k \prod_{i=1}^r [H, g_i]^H \right)^{*f_3(kr)} = \left(\prod_{i=1}^r [H, g_i]^H \right)^{*f_4} \end{aligned}$$

where $f_4 = k f_3(kr)$. Again, we may replace $[H, g_j]^H$ by $[H, g_j][H, g_j^{-1}]$, by Lemma 3.5.

Since $k = f_0(r, d) \leq C_0 r d^2$ where $d = d(G)$, we have $f_4 = O(k^3 r^2) = O(r^5 d^6)$.

We remark that this bound for $f_4(r, d)$ is very crude; a much better bound emerges if, instead of quoting Theorem 2.9, one uses the method of proof of that theorem to reduce Theorem 3.2 (i) to Theorem 3.2 (ii).

Proof of Theorem 3.2 (ii). Now we assume that H , as above, satisfies $H \leq G_1$. Put $H_1 = [H, {}_\omega G]$. Then H_1 is acceptable in G , by Lemma 3.3, and $G = H_1 \langle g_1, \dots, g_r \rangle$ by Lemma 3.4. Thus Theorem 3.10 and Lemma 3.4 together yield

$$\begin{aligned} [H, G] &= \prod_{j=1}^r [H, g_j] \cdot H_1 \\ &= \prod_{j=1}^r [H, g_j] \cdot \left(\prod_{j=1}^r [H_1, g_j] \right)^{*10k(d, r)} = \left(\prod_{i=1}^r [H, g_i] \right)^{*f_5} \end{aligned}$$

where $f_5 = 1 + 10k(d, r) = O(rd)$.

3.3 Proof of the Key Theorem: reductions

We follow the strategy of [NS], Section 4.

Notation For $\mathbf{u}, \mathbf{g} \in G^{(m)}$,

$$\mathbf{u} \cdot \mathbf{g} = (u_1 g_1, \dots, u_m g_m), \quad \mathbf{c}(\mathbf{u}, \mathbf{g}) = \prod_{j=1}^m [u_j, g_j].$$

Lemma 3.11

$$\left(\prod_{i=1}^s \mathbf{c}(\mathbf{a}(i) \cdot \mathbf{u}(i), \mathbf{g}) \right) \left(\prod_{i=1}^s \mathbf{c}(\mathbf{u}(i), \mathbf{g}) \right)^{-1} = \prod_{i=1}^s \left(\prod_{j=1}^m [a(i)_j, g_j]^{\tau_j(\mathbf{g}, \mathbf{u}(i))} \right)^{w(i)}$$

where $w(i) = \mathbf{c}(\mathbf{u}(i-1), \mathbf{g})^{-1} \dots \mathbf{c}(\mathbf{u}(1), \mathbf{g})^{-1}$.

This is a direct calculation. The next lemma is easily verified by induction on m (see [NS], Lemma 4.5):

Lemma 3.12

$$\left\langle g_j^{\tau_j(\mathbf{g}, \mathbf{u})} \mid j = 1, \dots, m \right\rangle = \left\langle g_j^{u_j h_j} \mid j = 1, \dots, m \right\rangle$$

where $h_j = g_{j-1}^{-1} \dots g_1^{-1}$.

Now let $H \triangleleft G = H \langle g_1, \dots, g_r \rangle$ be as in Theorem 3.10. If $H = 1$ there is nothing to prove, so we suppose that $H > 1$ and argue by induction on $|H|$. Since H is acceptable, we have $H = [H, G]$. Choose $N \triangleleft G$ with $N \leq H$ minimal subject to $1 < N = [N, G]$ (in [NS] such an N was called a *quasi-minimal normal subgroup* of G). Let Z be a normal subgroup of G maximal subject to $Z < N$. Then $[Z, {}_n G] = 1$ for some n , which implies (i) that $Z = N \cap \zeta_\omega(G)$ is uniquely determined, and (ii) that $[Z, N] \leq [Z, H] \leq [Z, G_\omega] = 1$. By definition, $\overline{N} = N/Z$ is a chief factor of G ; it is not bad because H is acceptable.

Applying Lemma 3.4 to Z we note that Z is contained in the Frattini subgroup $\Phi(G)$ of G .

We fix a natural number k , the candidate for $k(d, r)$, and define g_j for $j = 1, \dots, kr$ as in Theorem 3.10.

Depending on the nature of N , we shall choose a certain normal subgroup K of G with $1 \neq K \leq N$.

Suppose now that $h \in H$. We have to find elements $\mathbf{v}(i) \in H^{(m)}$ ($i = 1, \dots, 10$) such that (9) and (10) hold. By inductive hypothesis, we can do this ‘modulo K ’: thus there exist $\mathbf{u}(i) \in H^{(m)}$ and $\kappa \in K$ such that

$$h = \kappa \prod_{i=1}^{10} \mathbf{c}(\mathbf{u}(i), \mathbf{g})$$

and

$$\begin{aligned} G &= K \left\langle g_j^{\tau_j(\mathbf{g}, \mathbf{u}(i))} \mid j = 1, \dots, m \right\rangle \\ &= K \left\langle g_j^{u(i)_j h_j} \mid j = 1, \dots, m \right\rangle \quad \text{for } i = 1, \dots, 10, \end{aligned} \quad (11)$$

the second equality thanks to Lemma 3.12.

The idea now is to find elements $\mathbf{a}(i) \in N^{(m)}$ such that (9) and (10) are satisfied on setting

$$\mathbf{v}(i) = \mathbf{a}(i) \cdot \mathbf{u}(i).$$

Lemma 3.11 shows that (9) is then equivalent to

$$\prod_{i=1}^{10} \left(\prod_{j=1}^m [a(i)_j, g_j]^{\tau_j(\mathbf{g}, \mathbf{u}(i))} \right)^{w(i)} = \kappa. \quad (12)$$

This can be further simplified by setting

$$\begin{aligned} y(i)_j &= g_j^{\tau_j(\mathbf{g}, \mathbf{u}(i))w(i)}, \quad t(i)_j = g_j^{u(i)_j h_j} \\ b(i)_j &= a(i)_j^{\tau_j(\mathbf{g}, \mathbf{u}(i))w(i)}, \quad c(i)_j = a(i)_j^{u(i)_j h_j}. \end{aligned} \quad (13)$$

Define $\phi(i) : N^{(m)} \rightarrow N$ by

$$\mathbf{b}\phi(i) = \mathbf{c}(\mathbf{b}, \mathbf{y}(i)).$$

Then (12) becomes

$$\prod_{i=1}^{10} \mathbf{b}(i) \phi(i) = \kappa, \quad (14)$$

and (11) is equivalent to

$$\begin{aligned} G &= K \langle y(i)_1, \dots, y(i)_m \rangle \\ &= K \langle t(i)_1, \dots, t(i)_m \rangle \quad \text{for } i = 1, \dots, 10. \end{aligned} \quad (15)$$

Similarly, (10) holds if and only if for $i = 1, \dots, 10$ we have

$$G = \langle t(i)_j^{c(i)_j} \mid j = 1, \dots, m \rangle Z \quad (16)$$

(where Z is added harmlessly since $Z \leq \Phi(G)$). Let $\mathcal{X}(i)$ denote the set of all $\mathbf{c}(i) \in N^{(m)}$ such that (16) holds, and write $W(i)$ for the image of $\mathcal{X}(i)$ under the bijection $N^{(m)} \rightarrow N^{(m)}$ defined in (13) sending $\mathbf{c}(i) \mapsto \mathbf{b}(i)$.

To sum up: to establish the existence of $\mathbf{a}(1), \dots, \mathbf{a}(10) \in N^{(m)}$ such that the $\mathbf{v}(i) = \mathbf{a}(i) \cdot \mathbf{u}(i)$ satisfy (9) and (10), it suffices to find $(\mathbf{b}(1), \dots, \mathbf{b}(10)) \in W(1) \times \dots \times W(10)$ such that (14) holds.

Set $\varepsilon = \min\{\frac{1}{1+6\delta}, \frac{1}{r}\}$, and write $- : G \rightarrow G/Z$ for the quotient map. Now we separate four cases.

3.3.1 The easy case

If $[Z, G] > 1$ we define $K = [Z, G]$. Since $[Z, H] = 1$ and $G = H \langle g_1, \dots, g_r \rangle$, we have $K = \prod_{j=1}^r [Z, g_j]$. Thus $\kappa = \prod_{j=1}^r [z_j, g_j]$ with $z_1, \dots, z_r \in Z$. In this case, (14) is satisfied if we set

$$\begin{aligned} b(1)_j &= z_j & (1 \leq j \leq r) \\ b(1)_j &= 1 & (r < j \leq m) \\ b(i)_j &= 1 & (2 \leq i \leq 10, 1 \leq j \leq m), \end{aligned}$$

because $y(i)_j$ is conjugate to g_j under the action of H and $[Z, H] = 1$.

For each i we have $W(i) \supseteq Z^{(m)}$, since in this case (15) implies (16) if $c(i)_j \in Z$ for all j . So $\mathbf{b}(i) \in W(i)$ for each i , as required.

3.3.2 The abelian case

If $[Z, G] = 1$ and N is *abelian* we set $K = N$. We use additive notation for N and consider it as a G -module. Then (15) implies that

$$\phi(1) : \mathbf{b} \mapsto \sum_{j=1}^m b_j(y(1)_j - 1)$$

is a surjective ($\mathbb{Z}$ -module) homomorphism $N^{(m)} \rightarrow N$. It follows that

$$|\phi(1)^{-1}(c)| = |\ker \phi(1)| = |N|^{m-1}$$

for each $c \in N$.

Now fix $i \in \{1, \dots, 10\}$. According to Theorem 2.1, at least one of the elements g_j has the $\varepsilon/2$ -fsp on $\overline{N}$; therefore at least k of the elements $\overline{t(i)_j}$ have this property. Now we apply Proposition 2.8(i): this shows that (16) holds for at least $|\overline{N}|^m (1 - |\overline{N}|^{d-k\varepsilon/2})$ values of $\overline{\mathbf{c}(i)}$ in $|\overline{N}|^m$. It follows that

$$|W(i)| = |\mathcal{X}(i)| \geq |Z|^m \cdot |\overline{N}|^m (1 - |\overline{N}|^{d-k\varepsilon/2}) = |N|^m (1 - |\overline{N}|^{d-k\varepsilon/2}). \quad (17)$$

We need to compare $|\overline{N}|$ with $|N|$. Let $\{x_1, \dots, x_d\}$ be a generating set for G . Then $\mathbf{b} \mapsto \sum_{j=1}^d b_j(x_j - 1)$ induces an epimorphism from $\overline{N}^{(d)}$ onto N ; consequently $|N| \leq |\overline{N}|^d$. Thus provided $k\varepsilon/2d > 1$ we have

$$|W(i)| \geq |N|^m (1 - |N|^{1-k\varepsilon/2d}).$$

Assume now that $k\varepsilon > 4d$. Then $W(i)$ is non-empty for each i . For $i = 2, \dots, 10$ choose $\mathbf{b}(i) \in W(i)$ and put

$$c = \kappa \left(\prod_{i=2}^{10} \mathbf{b}(i) \phi(i) \right)^{-1}.$$

Then

$$|\phi(1)^{-1}(c)| + |W(1)| \geq |N|^m (|N|^{-1} + 1 - |N|^{1-k\varepsilon/2d}) > |N|^m.$$

It follows that $\phi(1)^{-1}(c) \cap W(1)$ is non-empty. Thus we may choose $\mathbf{b}(1) \in \phi(1)^{-1}(c) \cap W(1)$ and ensure that (14) is satisfied.

3.3.3 The soluble case

Suppose next that $[Z, G] = 1$ and $N > N' > 1$. In this case we take $K = N'$. Since $N' \leq Z$, the argument above again gives (17).

The maps $\phi(i)$ are no longer homomorphisms, however, and it is quite a major undertaking to obtain a good estimate for the fibres. The outcome is Proposition 6.2 of [NS]; translated into the present notation it is

Proposition 3.13 *Assume that $G = Z \langle y(i)_1, \dots, y(i)_m \rangle$ for $i = 1, 2, 3$. Then for each $c \in N'$ there exist $c_1, c_2, c_3 \in N$ such that $c = c_1 c_2 c_3$ and*

$$|\phi(i)^{-1}(c_i)| \geq |N|^m \cdot |\overline{N}|^{-d-1} \quad (i = 1, 2, 3), \quad (18)$$

provided in case N is a 2-group that $N^2 = N'$.

The final condition is easily verified in the present situation, and the initial hypothesis follows from (15) since now $K \leq Z$.

Assume now that $k\varepsilon > 4d + 2$. Then (17) and (18) together imply that $\phi(i)^{-1}(c_i) \cap W(i)$ is non-empty for $i = 1, 2, 3$, while (18) implies that $W(i)$ is non-empty for every i .

Choose $\mathbf{b}(i) \in W(i)$ for $i = 4, \dots, 10$. Put

$$c = \kappa \left(\prod_{i=4}^{10} \mathbf{b}(i) \phi(i) \right)^{-1},$$

and choose c_1, c_2, c_3 as in Proposition 3.13. Then for $i = 1, 2, 3$ we can find $\mathbf{b}(i) \in \phi(i)^{-1}(c_i) \cap W(i)$, and so ensure that (14) is satisfied.

3.3.4 The semisimple case

If $[Z, G] = 1$ and $N = N'$, define $K = N$. Now $\overline{N}$ is semisimple with at least 3 simple factors, and N is quasi-semisimple. In this case, Theorem 2.1 shows that at least one the elements g_j has the ε -fgp on $\overline{N}$; therefore for each i , at least k of the elements $\overline{t(i)}_j$ and at least k of the elements $y(i)_j$ have this property. Proposition 2.8(ii) now shows that

$$|\overline{\mathcal{X}(i)}| \geq |\overline{N}|^m (1 - 2^{2-k\varepsilon}),$$

provided we assume that $k\varepsilon \geq \max\{2d + 4, C\}$ for a certain absolute constant C . This implies

$$|W(i)| = |\mathcal{X}(i)| \geq |N|^m (1 - 2^{2-k\varepsilon}).$$

Now Theorem 4.28, proved below in Subsection 4.2, gives the following: there are absolute constants D, ε_0 such that if for each $i = 1, \dots, 10$

- (a) the group $\langle y(i)_1, \dots, y(i)_m \rangle$ permutes the quasisimple factors of N transitively,
- (b) at least k of the $y(i)_j$ have the ε -fgp on $\overline{N}$, where $k\varepsilon \geq 4 + 2D$,
- (c) the subset $W(i) \subseteq N^{(m)}$ satisfies $|W(i)| \geq (1 - \varepsilon_0/6) |N|^m$,

then

$$\prod_{i=1}^{10} W(i) \phi(i) = N.$$

Condition (a) follows from (15). Thus we can find $\mathbf{b}(i) \in W(i)$ ($i = 1, \dots, 10$) such that (14) is satisfied provided we assume that

$$\begin{aligned} k\varepsilon &> \max\{2d + 4, C, 4 + 2D, 2 + \log_2(6/\varepsilon_0)\} \\ &= \max\{2d + 4, C^*\} \end{aligned}$$

where C^* is an absolute constant.

3.3.5 Conclusion of the proof

Recall that we defined $\varepsilon = \min\{\frac{1}{1+6\delta}, \frac{1}{r}\}$. So if we now define

$$k(d, r) = 1 + \max\{r, 1 + 6\delta\} \cdot \max\{4d + 4, \lceil C^* \rceil\},$$

then $k = k(d, r)$ fulfils the requirements of all the preceding steps. This concludes the proof of Theorem 3.10 modulo Proposition 2.8, Theorem 4.28, and [NS], Proposition 6.2.

4 Semisimple groups

This section is devoted to the proof of Theorem 4.28. This will be stated in Subsection 4.2. Like Proposition 9.2 of [NS], which it in effect generalizes, its proof has two components: (1) a result about products of commutators in quasisimple groups, and (2) a complicated combinatorial reduction argument. These will occupy the next two subsections.

As remarked in the Introduction, the proof of (1) given here is significantly simpler (and shorter) than [NS2], which played the analogous role in our earlier work. The reduction argument (2) is essentially the same as in [NS], though we are now using it to prove something different (specifically, we have to control the image of a certain mapping rather than its fibres). We have re-cast the argument from scratch, in an attempt to make it more transparent (the reader will judge whether we have succeeded!) However, we shall quote one combinatorial result from Section 8 of [NS].

4.1 Twisted commutators in quasisimple groups

For automorphisms α, β of a group S and $x, y \in S$ we write

$$T_{\alpha, \beta}(x, y) = x^{-1}y^{-1}x^{\alpha}y^{\beta}.$$

For $\alpha = (\alpha_1, \dots, \alpha_D)$ and $\beta = (\beta_1, \dots, \beta_D)$ in $\text{Aut}(S)^{(D)}$ the mapping $\mathbf{T}_{\alpha, \beta} : S^{(D)} \times S^{(D)} \rightarrow S$ is defined by

$$\mathbf{T}_{\alpha, \beta}(\mathbf{x}, \mathbf{y}) = \prod_{i=1}^D T_{\alpha_i, \beta_i}(x_i, y_i).$$

Theorem 4.1 *There exist $\varepsilon > 0$ and $D \in \mathbb{N}$ such that if S is a finite quasisimple group, $\alpha, \beta \in \text{Aut}(S)^{(D)}$, and $X \subseteq S^{(2D)}$ has size at least $(1 - \varepsilon) |S^{(2D)}|$, then $|\mathbf{T}_{\alpha, \beta}(X)| \geq \lambda |S|$, where*

$$\lambda = \begin{cases} l(S)^{-3/5} & \text{if } l(S) \geq 3 \\ 1 & \text{if } l(S) = 2 \end{cases}. \quad (19)$$

The following corollary is Theorem 1.1 of [NS2]:

Corollary 4.2 *There exists $D_1 \in \mathbb{N}$ such that if S is a finite quasisimple group and $\alpha, \beta \in \text{Aut}(S)^{(D_1)}$ then*

$$\prod_{i=1}^{D_1} T_{\alpha_i, \beta_i}(S, S) = S.$$

Proof. Set $D_1 = 5D$, and divide α and β into 5 D -tuples $\alpha(j), \beta(j)$. Taking $X = S^{(2D)}$ in the theorem gives $|T_{\alpha(j), \beta(j)}(X)| \geq l(S)^{-3/5} |S|$ for $j = 1, \dots, 5$. The result now follows by the ‘Gowers trick’, since $5 \times \frac{3}{5} = 5 - 2$. ■

4.1.1 Reductions for Theorem 4.1

In this subsection, we fix a finite group S and $\lambda \in (0, 1]$. For any $\alpha, \beta \in \text{Aut}(S)^{(D)}$ we consider the statement

$\mathcal{P}(\alpha, \beta; D, \varepsilon) : \text{For } X \subseteq S^{(2D)},$

$$|X| \geq (1 - \varepsilon) |S^{(2D)}| \implies |\mathbf{T}_{\alpha, \beta}(X)| \geq \lambda |S|.$$

If Γ is a subgroup of $\text{Aut}(S)$, we write

$$\mathcal{P}(\Gamma; D, \varepsilon) \Leftrightarrow \mathcal{P}(\alpha, \beta; D, \varepsilon) \quad \forall \alpha, \beta \in \Gamma^{(D)}.$$

Thus Theorem 4.1 asserts the existence of D and ε such that $\mathcal{P}(\text{Aut}(S); D, \varepsilon)$ holds with λ defined by (19) for every quasisimple group S .

Our aim in the rest of this subsection is to establish the reduction steps Propositions 4.3, 4.4 and 4.12.

Proposition 4.3 *If $D_1 \leq D$ and $\varepsilon_1 \geq \varepsilon$ then $\mathcal{P}(\Gamma; D_1, \varepsilon_1)$ implies $\mathcal{P}(\Gamma; D, \varepsilon)$.*

Proof. If $D_1 = D$ the claim is obvious. Suppose that $D > D_1$. We write

$$\mathbf{T}_{\alpha, \beta}(\mathbf{x}, \mathbf{y}) = \mathbf{T}_{\alpha', \beta'}(\mathbf{x}', \mathbf{y}') \mathbf{T}_{\alpha'', \beta''}(\mathbf{x}'', \mathbf{y}'')$$

where $\mathbf{x}' = (x_1, \dots, x_{D_1})$, $\mathbf{x}'' = (x_{D_1+1}, \dots, x_D)$ etc. Now if $X \subseteq S^{(2D)}$ satisfies $|X| \geq (1 - \varepsilon) |S^{(2D)}|$ then there exist $(\mathbf{x}'', \mathbf{y}'') \in S^{2(D-D_1)}$ and $X_1 \subseteq S^{(2D_1)}$ such that $X_1 \times \{(\mathbf{x}'', \mathbf{y}'')\} \subseteq X$ and $|X_1| \geq (1 - \varepsilon) |S^{(2D_1)}|$. Then

$$T_{\alpha, \beta}(X) \supseteq T_{\alpha', \beta'}(X_1) \cdot \mathbf{T}_{\alpha'', \beta''}(\mathbf{x}'', \mathbf{y}''),$$

a set of size at least $\lambda |S|$ since $1 - \varepsilon \geq 1 - \varepsilon_1$. ■

Proposition 4.4 *If $\Delta \triangleleft \Gamma$ and $|\Gamma : \Delta| \leq n$ then $\mathcal{P}(\Delta; D, \varepsilon)$ implies $\mathcal{P}(\Gamma; n^2 D, \varepsilon)$.*

This is a little more complicated. It will follow from

Proposition 4.5 *Let Δ be a normal subgroup of index n in Γ , and let $\alpha, \beta \in \Gamma^{(n^2 D)}$. Then there exist $\bar{\alpha}, \bar{\beta} \in \Delta^{(D)}$ and a bijection $\pi : S^{(n^2 D)} \rightarrow S^{((2n^2 - 2)D)} \times S^{(2D)}$ such that, for each $\mathbf{x} \in S^{(n^2 D)}$,*

$$\prod_{i=1}^{n^2 D} T_{\alpha_i, \beta_i}(x_{2i-1}, x_{2i}) = \prod_{i=1}^D T_{\bar{\alpha}_i, \bar{\beta}_i}(\hat{x}_{2i-1}, \hat{x}_{2i}) \cdot R(\tilde{x}),$$

where $(\tilde{x}, \hat{x}) = \mathbf{x}\pi$ and $R(\tilde{x})$ depends only on $\tilde{x}$.

Accepting this for now we deduce Proposition 4.4. Let $\varepsilon \in (0, 1)$ and suppose that $W \subseteq S^{(2n^2 D)}$ satisfies $|W| \geq (1 - \varepsilon) |S^{(2n^2 D)}|$. Then $|W\pi| = |W|$; so for at least one element $u \in S^{((2n^2 - 2)D)}$ the set

$$Y_u := \left\{ y \in S^{(2D)} \mid (u, y) \in W\pi \right\}$$

satisfies $|Y_u| \geq (1 - \varepsilon) |S^{(2D)}|$. Then

$$\mathbf{T}_{\alpha,\beta}(W) \supseteq \mathbf{T}_{\bar{\alpha},\bar{\beta}}(Y_u) \cdot R(u).$$

If $\mathcal{P}(\Delta; D, \varepsilon)$ holds then $|\mathbf{T}_{\bar{\alpha},\bar{\beta}}(Y_u)| \geq \lambda |S|$, and so $|\mathbf{T}_{\alpha,\beta}(W)| \geq \lambda |S|$. Thus $\mathcal{P}(\Gamma; n^2 D, \varepsilon)$ holds as claimed.

Now we embark on the proof of Proposition 4.5.

Lemma 4.6

$$\begin{aligned} T_{a_1,\beta_1}(x_1, x_2) T_{a_2,\beta_2}(x_3, x_4) &= A(x\nu) \cdot T_{a_1\alpha_2,\beta_2}(x\mu) \cdot B(x\nu) \\ &= C(x\tau) \cdot T_{a_1,\beta_1\beta_2}(x\sigma) \end{aligned}$$

where $x \mapsto (x\mu, x\nu)$ and $x \mapsto (x\sigma, x\tau)$ are bijections from $S^{(4)}$ to $S^{(2)} \times S^{(2)}$.

Proof. Take $z = x_3(x_1^{\alpha_1}x_2^{\beta_1})^{-1}$, $w = z^{-\alpha_2}x_4zx_2$ and $t = x_1x_2^{\beta_1\alpha_1^{-1}}z^{\alpha_2\beta_2\alpha_2^{-1}\alpha_1^{-1}}$ and set

$$\begin{aligned} x\mu &= (t, w), \quad x\nu = (x_2, z) \\ A(y, z) &= (y^{\beta_1}z^{\alpha_2\beta_2\alpha_2^{-1}})^{\alpha_1^{-1}}, \quad B(y, z) = (yz)^{-\beta_2}. \end{aligned}$$

Take $u = x_2^{\beta_1}x_3^{-1}x_4^{-1}$, $= x_1u^{\alpha_1^{-1}}x_3^{\alpha_2\alpha_1^{-1}}u^{-\beta_2\alpha_1^{-1}}$ and set

$$\begin{aligned} x\sigma &= (v, x_2), \quad x\tau = (u, x_3) \\ C(u, y) &= (y^{\alpha_2}u^{-\beta_2})^{\alpha_1^{-1}}. \end{aligned}$$

■

Lemma 4.7

$$zT_{\alpha,\beta}(x, y) = T_{\alpha,\beta}(x', y')z^{-\gamma}$$

where $x' = z^{\alpha\beta^{-1}\alpha^{-1}}xz^{-1}$, $y' = z^{\alpha\beta^{-1}}yz^{-\alpha\beta^{-1}\alpha^{-1}}$ and $\gamma = [\alpha^{-1}, \beta]$.

Lemma 4.8 Suppose that $|\Gamma : \Delta| = 2$. Given $\alpha_i, \beta_i \in \Gamma$ ($i = 1, \dots, 4$), there exist $\gamma, \delta \in \Delta$, a bijection $x \mapsto (x^*, \tilde{x})$ from $S^{(8)}$ to $S^{(2)} \times S^{(6)}$ and maps $P, Q : S^{(6)} \rightarrow S$ such that

$$\prod_{i=1}^4 T_{\alpha_i, \beta_i}(x_{2i-1}, x_{2i}) = P(\tilde{x})T_{\gamma, \delta}(x^*)Q(\tilde{x}).$$

Proof. Define

$$(\hat{x}_1, \hat{x}_2, \tilde{x}_1, \tilde{x}_2; \gamma_1, \eta_1) = \begin{cases} (x_1, x_2, x_3, x_4; \alpha_1, \beta_1) & \text{if } \alpha_1 \in \Delta \\ (x_3, x_4, x_1, x_2; \alpha_2, \beta_2) & \text{if } \alpha_2 \in \Delta \\ (x\mu, x\nu; \alpha_1\alpha_2, \beta_2) & \text{if } \alpha_1\alpha_2 \in \Delta \end{cases}$$

(assuming in the 2nd and 3d lines that $\alpha_1 \notin \Delta$). Then (using Lemma 4.6 in the 3d case) we see that

$$T_{a_1, \beta_1}(x_1, x_2)T_{a_2, \beta_2}(x_3, x_4) = P_1(\tilde{x}_1, \tilde{x}_2)T_{\gamma_1, \eta_1}(\hat{x}_1, \hat{x}_2)Q_1(\tilde{x}_1, \tilde{x}_2)$$

for suitable maps P_1, Q_1 . Note that $\gamma_1 \in \Delta$ and $(x_1, x_2, x_3, x_4) \mapsto (\hat{x}_1, \hat{x}_2, \tilde{x}_1, \tilde{x}_2)$ is bijective. Similarly

$$T_{a_3, \beta_3}(x_5, x_6)T_{a_4, \beta_4}(x_7, x_8) = P_2(\tilde{x}_3, \tilde{x}_4)T_{\gamma_2, \eta_2}(\hat{x}_3, \hat{x}_4)Q_2(\tilde{x}_3, \tilde{x}_4)$$

where $\gamma_2 \in \Delta$ and $(x_3, x_4, x_5, x_6) \mapsto (\hat{x}_3, \hat{x}_4, \tilde{x}_3, \tilde{x}_4)$ is bijective.

Put $z = Q_1(\tilde{x}_1, \tilde{x}_2)P_2(\tilde{x}_3, \tilde{x}_4)$ and set

$$\bar{x}_3 = z^{\gamma_2 \eta_2^{-1} \gamma_2^{-1}} \hat{x}_3 z^{-1}, \quad \bar{x}_4 = z^{\gamma_2 \eta_2^{-1}} \hat{x}_3 z^{-\gamma_2 \eta_2^{-1} \gamma_2^{-1}}.$$

Lemma 4.7 gives

$$T_{\gamma_1, \eta_1}(\hat{x}_1, \hat{x}_2) \cdot z \cdot T_{\gamma_2, \eta_2}(\hat{x}_3, \hat{x}_4) = T_{\gamma_1, \eta_1}(\hat{x}_1, \hat{x}_2)T_{\gamma_2, \eta_2}(\bar{x}_3, \bar{x}_4)R(\tilde{x})$$

where $\tilde{x} = (\tilde{x}_1, \tilde{x}_2, \tilde{x}_3, \tilde{x}_4)$. Now we repeat the first procedure, applied to the second pair of automorphisms η_1, η_2 . This gives $\delta \in \{\eta_1, \eta_2, \eta_1 \eta_2\} \cap \Delta$, $\gamma \in \{\gamma_1, \gamma_2\}$ and a bijection $(\hat{x}_1, \hat{x}_2, \bar{x}_3, \bar{x}_4) \mapsto (x_1^*, x_2^*, \tilde{x}_5, \tilde{x}_6)$ such that

$$T_{\gamma_1, \eta_1}(\hat{x}_1, \hat{x}_2)T_{\gamma_2, \eta_2}(\bar{x}_3, \bar{x}_4) = P_3(\tilde{x}_5, \tilde{x}_6)T_{\gamma, \delta}(x_1^*, x_2^*)Q_3(\tilde{x}_5, \tilde{x}_6).$$

Then

$$\prod_{i=1}^4 T_{\alpha_i, \beta_i}(x_{2i-1}, x_{2i}) = PT_{\gamma, \delta}(x_1^*, x_2^*)Q$$

where $P = P_1(\tilde{x}_1, \tilde{x}_2)P_3(\tilde{x}_5, \tilde{x}_6)$ and $Q = Q_3(\tilde{x}_5, \tilde{x}_6)R(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3, \tilde{x}_4)$. The result follows. ■

Proof of Proposition 4.5. Suppose first that $n = 2$. Write $T_i = T_{\alpha_i, \beta_i}(x_{2i-1}, x_{2i})$. Grouping these four at a time and applying the preceding lemma we see that

$$\prod_{i=1}^{4D} T_i = \prod_{j=1}^D P_j(y_j)T_{\gamma_j, \delta_j}(u_j)Q_j(y_j)$$

where $\gamma_j, \delta_j \in D, y_j \in S^{(6)}, u_j \in S^{(2)}$ and $(x_1, \dots, x_{4D}) \mapsto (y_1, \dots, y_D; u_1, \dots, u_D)$ is a bijection. Using Lemma 4.7 we now conjugate the factors $T_{\gamma_j, \delta_j}(u_j)$ by $z_j = (P_1^{\lambda_{1j}} Q_1^{\mu_{1j}} \dots Q_{j-1}^{\mu_{j-1,j}} P_j^{\lambda_{jj}})^{-1}$, for suitable automorphisms $\lambda_{ij}, \mu_{ij} \in \Delta$, to obtain

$$\prod_{i=1}^{4D} T_i = \prod_{j=1}^D T_{\gamma_j, \delta_j}(\hat{x}_{2j-1}, \hat{x}_{2j}) \cdot R(\tilde{x})$$

where $\tilde{x} = (y_1, \dots, y_D)$, $R = \prod_{j=1}^D P_j(y_j)^{\lambda_j} Q_j(y_j)^{\mu_j}$ for certain automorphisms $\lambda_j, \mu_j \in \Delta$, and $\hat{x}_{2j-1}, \hat{x}_{2j}$ are obtained from y_j by multiplying on the left and right by expressions depending only on $z_j = z_j(y_1, \dots, y_D)$. The result follows.

Now we consider the general case where $|\Gamma : \Delta| = n > 2$. This follows the same pattern. Suppose first that $D = 1$. There exist i, j with $1 \leq i \leq j \leq n$ such that $\gamma = \alpha_i \alpha_{i+1} \dots \alpha_j \in \Delta$. Using Lemmas 4.6 and 4.7 repeatedly we get

$$\prod_{i=1}^n T_i = P(\tilde{x}) T_{\gamma, \beta_j}(x^*) Q(\tilde{x})$$

where $x \mapsto (x^*, \tilde{x})$ is a bijection $S^{(2n)} \rightarrow S^{(2)} \times S^{(2n-2)}$. Grouping the factors together n at a time and applying this to each group of n factors we get

$$\begin{aligned} \prod_{i=1}^{n^2} T_i &= \prod_{i=1}^n P_i(\tilde{x}_i) T_{\gamma_i, \beta_{j(i)}}(x_i^*) Q_i(\tilde{x}_i) \\ &= \prod_{i=1}^n T_{\gamma_i, \beta_{j(i)}}(\bar{x}_i) \cdot R_1(x^\dagger), \end{aligned}$$

using Lemma 4.7 for the second step; here $x \mapsto (\bar{x}, x^\dagger)$ is a bijection $S^{(2n^2)} \rightarrow S^{(2n)} \times S^{(2n^2-2n)}$ and each $\gamma_i \in \Delta$.

There exist k, l with $1 \leq k \leq l \leq n$ such that $\beta_{j(k)} \dots \beta_{j(l)} = \delta \in \Delta$, and repeating the procedure we get

$$\prod_{i=1}^n T_{\gamma_i, \beta_{j(i)}}(\bar{x}_i) = T_{\gamma_k, \delta}(x^\ddagger) \cdot R_2(x^\sharp)$$

where $\bar{x} \mapsto (x^\ddagger, x^\sharp)$ is a bijection $S^{(2n)} \rightarrow S^{(2)} \times S^{(2n-2)}$. So putting $\gamma = \gamma_k$ we have

$$\prod_{i=1}^{n^2} T_i = T_{\gamma, \delta}(x^\ddagger) \cdot R(x^\dagger, x^\sharp) \quad (20)$$

with $\gamma, \delta \in \Delta$ and $x \mapsto (x^\ddagger, x^\dagger, x^\sharp)$ a bijection $S^{(n^2)} \rightarrow S^2 \times S^{(2n^2-2n)} \times S^{(2n-2)}$.

In the general case where $D > 1$ we group the $n^2 D$ factors T_i together n^2 at a time, apply (20) to each product of n^2 factors, and then conjugate the resulting terms $T_{\gamma(i), \delta(i)}(x_i^\ddagger)$ by the intervening factors R using Lemma 4.7 to obtain

$$\prod_{i=1}^{n^2 D} T_i = \prod_{i=1}^D T_{\gamma(i), \delta(i)}(\hat{x}) \cdot R(\tilde{x})$$

for a certain bijection $x \mapsto (\tilde{x}, \hat{x}) : S^{(n^2 D)} \rightarrow S^{((2n^2-2)D)} \times S^{(2D)}$.

This completes the proof.

The final reduction step needs the next three lemmas.

Lemma 4.9 *Let $\varepsilon \in (0, 1)$. If $Z \subseteq X \times Y$ satisfies $|Z| \geq (1 - \varepsilon^2) |X \times Y|$ then for at least $(1 - \varepsilon) |X|$ elements $u \in X$ we have $|Z \cap (\{u\} \times Y)| \geq (1 - \varepsilon) |Y|$.*

Proof. Suppose the number of such elements u is $\rho|X|$. Then

$$(1 - \varepsilon^2)|X \times Y| \leq (1 - \rho)|X| \cdot (1 - \varepsilon)|Y| + \rho|X| \cdot |Y|$$

whence $\rho \geq 1 - \varepsilon$. ■

Lemma 4.10

$$\begin{aligned} T_{\alpha, \beta}(x, y) &= [x, \alpha y][y, \beta] \\ z[x, \gamma] &= [x', \gamma]z^\gamma \end{aligned}$$

where $x' = xz^{-1}$.

Recall that for D -tuples x, β , we use the notation $\mathbf{c}(x, \beta) = \prod_{i=1}^D [x_i, \beta_i]$, and $x \cdot \beta = (x_1\beta_1, \dots, x_D\beta_D)$.

Lemma 4.11 *There is a bijection $y \mapsto \bar{y} : S^{(D)} \rightarrow S^{(D)}$, and for each fixed $y \in S^{(D)}$ a bijection $x \mapsto x' : S^{(D)} \rightarrow S^{(D)}$ (depending on y), such that*

$$\mathbf{T}_{\alpha, \beta}(x, y) = \mathbf{c}(x', \bar{y} \cdot \alpha) \cdot h(y),$$

where $h(y)$ depends only on y .

Proof. Using Lemma 4.10 we get

$$\begin{aligned} \mathbf{T}_{\alpha, \beta}(x, y) &= \prod_{i=1}^D [x_i, \alpha_i y_i][y_i, \beta_i] \\ &= \prod_{i=1}^D [x'_i, \alpha_i y_i] \cdot z_D \end{aligned}$$

where $x'_i = x_i z_i^{-1}$ and $z_1 = 1$, $z_i = (z_{i-1}[y_{i-1}, \beta_{i-1}])^{\alpha_i y_i}$ for $1 < i \leq D$. The result follows on setting $\bar{y}_i = y_i^{\alpha_i^{-1}}$. ■

Proposition 4.12 *Let $\alpha, \beta \in \text{Aut}(S)$. Suppose that for each $Y \subseteq S^{(D)}$ with $|Y| \geq (1 - \varepsilon)|S^{(D)}|$ there exists $\mathbf{y} \in Y$ such that*

$$X \subseteq S^{(D)}, |X| \geq (1 - \varepsilon)|S^{(D)}| \implies |\mathbf{c}(X, \mathbf{y} \cdot \alpha)| \geq \lambda|S|. \quad (21)$$

Then $\mathcal{P}(\alpha, \beta; D, \varepsilon^2)$ holds.

Here $\mathbf{c}(X, \mathbf{y} \cdot \alpha) = \{\mathbf{c}(\mathbf{x}, \mathbf{y} \cdot \alpha) \mid \mathbf{x} \in X\}$.

Proof. Suppose that $W \subseteq S^{(2D)}$ satisfies $|W| \geq (1 - \varepsilon^2)|S^{(2D)}|$. Let

$$Y = \left\{ \mathbf{y} \in S^{(D)} \mid |W \cap (S^{(D)} \times \{\mathbf{y}\})| \geq (1 - \varepsilon)|S^{(D)}| \right\}.$$

Lemma 4.9 shows that $|Y| \geq (1 - \varepsilon)|S^{(D)}|$, so we can choose $\mathbf{y} \in Y$ so that (21) holds. There exists $X \subseteq S^{(D)}$ with $|X| \geq (1 - \varepsilon)|S^{(D)}|$ and $X \times \{\mathbf{y}\} \subseteq W$. Let $x \mapsto x'$ be the bijection $S^{(D)} \rightarrow S^{(D)}$ given in Lemma 4.11. Then

$$\mathbf{T}_{\alpha, \beta}(W) \supseteq \mathbf{T}_{\alpha, \beta}(X \times \{\mathbf{y}\}) = \mathbf{c}(X', \bar{\mathbf{y}} \cdot \alpha) \cdot h(\mathbf{y}),$$

a set of size at least $\lambda|S|$. ■

4.1.2 Small groups

Let N^* be an upper bound for the orders of quasisimple groups S such that $l(S) = 2$; that N^* is finite follows from Proposition 1.21 and well-known facts about the alternating groups. We fix a natural number $N_0 \geq N^*$, to be specified later, and denote by $\overline{\mathcal{S}}$ the class of all quasisimple groups of order less than N_0 . Set

$$N_1 = \max_{S \in \overline{\mathcal{S}}} |\text{Out}(S)|.$$

There is a natural number δ_1 such that for each $S \in \overline{\mathcal{S}}$, every element of S is a product of δ_1 commutators (obviously $\delta_1 \leq \delta^*$, given in Proposition 1.18; in fact we can take $\delta_1 \leq 2$).

Define

$$\begin{aligned} \gamma : G \times G &\rightarrow G \\ \gamma(x, y) &= [x, y]. \end{aligned}$$

Lemma 4.13 *Let $\alpha, \beta \in \text{Inn}(S)$. Then there exist a bijection $(x, y) \mapsto (\overline{x}, \overline{y})$ from $S^{(2)}$ to $S^{(2)}$ and an element $t \in S$ such that*

$$T_{\alpha, \beta}(x, y) = [\overline{x}, \overline{y}]t$$

for all $x, y \in S$.

Proof. For simplicity, let α and β denote also elements of S inducing the given inner automorphisms. Now define

$$\begin{aligned} t &= [\alpha^{-1}, \beta], \\ (\overline{x}, \overline{y}) &= (t\beta^{-1}x, t\beta^{-1}\alpha y\beta t^{-1}). \end{aligned}$$

■

Proposition 4.14 *If $S \in \overline{\mathcal{S}}$ then $\mathcal{P}(\text{Aut}(S); D, \varepsilon)$ holds for $\lambda = 1$, with*

$$D = N_1^2 \delta_1, \quad \varepsilon = N_0^{-2\delta_1}.$$

Proof. In view of Proposition 4.4, it will suffice to establish $\mathcal{P}(\text{Inn}(S); \delta_1, \varepsilon)$. Let $X \subseteq S^{(2\delta_1)}$ satisfy $|X| \geq (1 - \varepsilon)|S|^{2\delta_1}$; then $X = S^{(2\delta_1)}$. Let $\alpha, \beta \in \text{Inn}(S)^{(\delta_1)}$. Using Lemma 4.13 we obtain

$$\begin{aligned} \mathbf{T}_{\alpha, \beta}(X) &= \prod_{i=1}^{\delta_1} T_{\alpha_i, \beta_i}(S, S) \\ &= \prod_{i=1}^{\delta_1} \gamma(S \times S) t_i = \gamma(S \times S)^{* \delta_1} \cdot t = S, \end{aligned}$$

where $t = t_1 \dots t_{\delta_1}$. The result follows. ■

4.1.3 Inner automorphisms

The key to this case is a result is due to Garion and Shalev. In [GaSh] they define for each finite group G the invariant

$$\epsilon(G) = (\zeta^G(2) - 1)^{1/4},$$

where $\zeta^G(2) = \sum \chi(1)^{-2}$ summed over irreducible characters χ of G .

Proposition 4.15 ([GaSh], Corollary 1.4(ii)) *For $W \subseteq G \times G$ and $\eta \in (0, 1)$,*

$$|W| \geq (1 - \eta) |G|^2 \implies |\gamma(W)| \geq (1 - \eta - 3\epsilon(G)) |G|.$$

This is useful in combination with Theorem 1.1 of [LiSh2], which implies that $\zeta^G(2) \rightarrow 1$ as $|G| \rightarrow \infty$ when G ranges over quasisimple groups. We may therefore choose $N_2 \in \mathbb{N}$ so that $\epsilon(S) < \frac{1}{24}$ for every quasisimple group S with $|S| \geq N_2$.

Proposition 4.16 *Let S be a quasisimple group with $|S| \geq N_2$. Then $\mathcal{P}(\text{Inn}(S); 1, \frac{1}{8})$ holds with $\lambda = l(S)^{-3/5}$.*

Proof. Let $\alpha, \beta \in \text{Inn}(S)$ and let $X \subseteq S^{(2)}$ satisfy $|X| \geq \frac{7}{8} |S|^2$. According to Lemma 4.13, there exist $t \in S$ and a subset Y of $S^{(2)}$ with $|Y| = |X|$ such that $\mathbf{T}_{\alpha, \beta}(X) = \gamma(Y)t$.

By Proposition 4.15 we have

$$|\gamma(W)| \geq \left(\frac{7}{8} - 3\epsilon(S) \right) |S| \geq \frac{3}{4} |S|.$$

Therefore

$$|\mathbf{T}_{\alpha, \beta}(X)| = |\gamma(Y)| \geq |\gamma(W)| \geq \frac{3}{4} |S| > l(S)^{-3/5} |S|$$

since $l(S) \geq 2$. ■

4.1.4 Diagonal automorphisms

In this subsection and the next, we consider a quasisimple group S of Lie type, of untwisted rank r . This means ([GLS], Section 2.2) that S is the group of fixed points of a Steinberg automorphism σ of order $k \in \{1, 2, 3\}$ of some untwisted Lie type group $S^\diamond \leq \text{GL}_d(q^k)$ of rank r (where $k = 1$ precisely when $S = S^\diamond$ is untwisted). We denote by $D \leq \text{GL}_d(q^k)$ the group of diagonal matrices that induce diagonal automorphisms on S . Thus $S \triangleleft SD$ and the restriction to S of the inner automorphisms of SD is the group $\text{InnDiag}(S)$ of inner-diagonal automorphisms of S . We will use the facts (*loc. cit.* Section 2.5):

$$\begin{aligned} |SD : S| &\leq r + 1, \\ |\mathbf{Z}(SD)| &= |\mathbf{Z}(S)| \leq r + 1. \end{aligned}$$

An abelian subgroup of S consisting of semisimple elements and maximal with this property will be called a *maximal torus* of S (this is the same as the intersection with S of a maximal torus in the underlying algebraic group). The following estimate is easily derived from [C], Proposition 3.3.5:

Lemma 4.17 *The size of a maximal torus of S is at most $(q+1)^r$.*

Proposition 4.18 *There exists $N_3 \in \mathbb{N}$ such that if $|S| \geq N_3$, $r \geq 9$ and $q > 10$ then $\mathcal{P}(\text{InnDiag}(S); 8, 10^{-3})$ holds with $\lambda = l(S)^{-3/5}$.*

This will be deduced from the next two results:

Proposition 4.19 [GL] *If S is a classical group and $h \in D$ then the number of regular semisimple elements in the coset Sh is at least $\left(1 - \frac{3}{q-1} - \frac{1}{(q-1)^2}\right)|S|$, which exceeds $\frac{2}{3}|S|$ if $q > 10$.*

(This follows from the proof of [GL], though it is not explicitly stated there in this form.)

Proposition 4.20 *Assume that $r \geq 9$ and $q > 10$. Let $h_1, h_2, \dots, h_8$ be regular semisimple elements of SD , and let $X \subseteq S^{(8)}$ satisfy $|X| \geq \frac{1}{4}|S|^8$. Then provided $|S|$ is sufficiently large, the number of elements $g \in S$ such that*

$$\mathbf{c}(\mathbf{x}, \mathbf{h}) = \prod_{i=1}^8 [x_i, h_i] = g$$

has a solution $\mathbf{x} = (x_1, \dots, x_8) \in X$ is at least $\frac{1}{6}|S|$.

Before proving this let us deduce Proposition 4.18. Let $\alpha, \beta \in \text{InnDiag}(S)^{(8)}$ and let $Y \subseteq S^{(8)}$ satisfy $|Y| \geq (1 - (\frac{2}{3})^8)|S|^8$. There exist $c_i \in S$ and $h'_i \in D$ such that α_i is induced by $c_i h'_i$ ($i = 1, \dots, 8$). Put $Y' = \{\mathbf{y} \cdot \mathbf{c} \cdot \mathbf{h}' \mid \mathbf{y} \in Y\}$. Then $|Y'| = |Y|$, so Proposition 4.19 ensures that Y' contains at least one element $\mathbf{y} \cdot \mathbf{c} \cdot \mathbf{h}' = (h_1, \dots, h_8)$ with each h_i regular semisimple. Then provided $|S|$ is sufficiently large, Proposition 4.20 gives

$$|\mathbf{c}(X, \mathbf{y} \cdot \alpha)| = |\mathbf{c}(X, \mathbf{h})| \geq \frac{1}{6}|S|$$

whenever $X \subseteq S^{(8)}$ satisfies $|X| \geq \frac{1}{4}|S|^8$. Applying Proposition 4.12 we infer that $\mathcal{P}(\alpha, \beta; 8, (\frac{2}{3})^{16})$ holds with $\lambda = \frac{1}{6}$. Now Proposition 4.18 follows, since $(\frac{2}{3})^{16} > 10^{-3}$ and $l(S) \geq 11^9 - 1 > 6^{5/3}$ by Proposition 1.21.

Proof of Proposition 4.20. Relabelling $(h_1^{-1}, h_2^{-h_1^{-1}}, \dots, h_8^{-(h_1 \dots h_7)^{-1}})$ as $(k_1, k_2, \dots, k_8)$ and $(x_1, x_2^{h_1^{-1}}, \dots, x_8^{(h_1 \dots h_7)^{-1}})$ as $(y_1, \dots, y_8)$, it will suffice to prove that the image of the map

$$f : (y_1, \dots, y_8) \mapsto k_1^{y_1} k_2^{y_2} \dots k_8^{y_8} \cdot (k_1 \dots k_8)^{-1} \in S$$

has size at least $\frac{1}{6}|S|$ when $(y_1, \dots, y_8)$ ranges over a subset of $S^{(8)}$ of proportion $\frac{1}{4}$.

Write $G = SD$. We observe that if g is a semisimple element of S then $C_G(g)$ contains a maximal torus of G and so maps onto G/S . This means that the conjugacy class g^S of g in S is the same as the conjugacy class of g in G . Now we count solutions in conjugacy classes of G :

Lemma 4.21 *Assume that $r \geq 9$, $q > 10$. Let $\delta > 0$ and let $k_1, \dots, k_8$ be regular semisimple elements of G . Put $c_i = |k_i^G|$. There is an integer N_δ such that if $|S| \geq N_\delta$ then the following holds:*

For every $g \in S$ the number of 8-tuples $(a_1, \dots, a_8) \in k_1^G \times \dots \times k_8^G$ such that

$$a_1 \dots a_8 = gk_1 \dots k_8$$

is

$$\frac{c_1 \dots c_8}{|S|} (1 + \gamma_g) \text{ where } |\gamma_g| < \delta.$$

Assuming this for the moment we can finish the proof of Proposition 4.20. Take $\delta = \frac{1}{2}$ and assume that $|S| \geq N_\delta$. Then Lemma 4.21 implies that for each $g \in S$ we have

$$\begin{aligned} |f^{-1}(g)| &= \prod_{i=1}^8 |C_S(k_i)| \cdot \frac{c_1 \dots c_8}{|S|} (1 + \gamma_g) \\ &= |S|^7 (1 + \gamma_g) < \frac{3}{2} |S|^7. \end{aligned}$$

Suppose that $Y \subseteq S^{(8)}$ satisfies $|Y| \geq \frac{1}{4} |S|^8$. Then

$$|f(Y)| > \frac{|Y|}{\frac{3}{2} |S|^7} \geq \frac{1}{6} |S|,$$

as required.

Proof of Lemma 4.21 Let χ be an irreducible character of G . By Clifford theory $\chi \downarrow_S$ is a sum of irreducible characters of S , say $\psi + \phi + \dots$. Then $\chi(1) \geq \psi(1)$. Now if χ is nonlinear then $\psi \in \text{Irr}(S)$ is also nonlinear, and hence $\chi(1) \geq \psi(1) \geq cq^r$ for some absolute constant c (very close to 1), by Proposition 1.21.

Put $p = gk_1 \dots k_8$ and let $s(p)$ denote the number of the number of 8-tuples $(a_1, \dots, a_8) \in k_1^G \times \dots \times k_8^G$ such that $a_1 a_2 \dots a_8 = p$.

A well-known formula (cf. [SGT], 7.2) gives

$$s(p) = \frac{c_1 \dots c_8}{|G|} \sum_{\chi \in \text{Irr}(G)} \frac{\chi(k_1) \dots \chi(k_8) \chi(p^{-1})}{\chi(1)^7}.$$

Since $k_1 \dots k_8 p^{-1} \in S$ and hence lies inside $\ker \chi$ for any linear character χ of G , these contribute precisely $|G/G'| = |G|/|S|$ to the above sum. It therefore suffices to show that

$$|G : S| \sum_{\chi \in \text{Irr}_0(G)} \frac{\chi(k_1) \dots \chi(k_8) \chi(p^{-1})}{\chi(1)^7} \rightarrow 0 \text{ as } |S| \rightarrow \infty,$$

where $\text{Irr}_0(G)$ denotes the set of non-linear irreducible characters of G .

Since $|\chi(p^{-1})|/\chi(1) \leq 1$ it is enough to show that

$$V := (r+1) \sum_{\chi \in \text{Irr}_0(G)} \frac{\chi(k_1) \dots \chi(k_8)}{\chi(1)^6} \rightarrow 0 \text{ as } |S| \rightarrow \infty.$$

Now since k_i is regular semisimple, $C_G(k_i)$ is a torus of $G = SD$, and so $|C_G(k_i)| \leq (q+1)^{r+1}$ by Lemma 4.17.

Hence $|\chi(k_i)| \leq \sqrt{|C_G(k_i)|} \leq (q+1)^{(r+1)/2}$, and we obtain

$$|\chi(k_1) \dots \chi(k_8)| \chi(1)^{-6} \leq \frac{((q+1)^{r+1})^4}{c^6 q^{6r}} = c^{-6} (q+1)^{4+4r} q^{-6r}.$$

By Corollary 1.2 (3) of [FG], $|\text{Irr}(G/\mathbb{Z}(G))| \leq 100q^r$, whence $|\text{Irr}(G)| \leq 100q^r(r+1)$. Moreover $q+1 < q^{1.1}$ when $q > 10$. Consequently

$$V \leq c_5 (r+1)^2 q^{4.4-.6r}$$

for some absolute constant $c_5 > 0$.

As $r \geq 9$ we have $0.6r > 4.4$; consequently $V \rightarrow 0$ as $|S| \rightarrow \infty$, as required.

4.1.5 Field automorphisms

As in the preceding subsection, S denotes a quasisimple group of Lie type, of untwisted rank r . We assume that S is universal, and introduce some more notation (cf. [GLS], Section 2.2). L is a simple simply connected algebraic group L defined over $\mathbb{F}_p$, and $S = L_\sigma \leq L(\mathbb{F}_{q^k})$ is the group of σ -fixed points of a Steinberg automorphism σ acting on L . Here $k \in \{1, 2, 3\}$ and σ^k is the smallest power of σ which is a power of the Frobenius automorphism $[p]$ of L . In fact σ is the product of a graph automorphism of L and some power of $[p]$, so σ commutes with all field automorphisms of L .

We consider L as embedded in some GL_d . Then GL_d contains a torus T that normalizes L and induces the diagonal automorphisms on L . In the same way $D = T_\sigma$ induces the diagonal automorphisms of $S = L_\sigma$.

We consider a field automorphism ϕ of S . Thus ϕ is the restriction to S of $[p]^f$ for some f , and we shall denote $[p]^f$ also by ϕ . Then $\phi^n = \sigma^k$ where n is the order of ϕ as an automorphism of S .

Let q_0 denote the cardinality of the fixed field of ϕ . Thus $q_0 = p^f$, while $\mathbb{F}_{q^k}$ is the fixed field of ϕ^n , so $q^k = p^{nf}$ and

$$q^k = q_0^n. \tag{22}$$

We remark that $k \leq 2$ unless S is of type 3D_4 , with $r = 4$; and q might be the square root of a non-square integer if S is a Suzuki or Ree group.

For an algebraic subgroup M of $\mathrm{GL}_d(\overline{\mathbb{F}_p})$ we denote by M_ϕ the fixed-point set of ϕ in M . Later, we shall need to consider the groups

$$G = L_\phi, \quad H = T_\phi$$

Thus G is an untwisted quasisimple group of Lie type, say X , over $\mathbb{F}_{q_0}$ of rank r equal to the rank of L . The group H induces the diagonal automorphisms on G . Since σ commutes with ϕ it preserves G and acts on it as an automorphism of order k (since $\sigma^k = \phi^n$).

We shall consider automorphisms

$$\alpha = ch\phi^{-1}$$

where

- ϕ is a field automorphism of S having order $n > 50$,
- h is a diagonal automorphism of S (we identify h with an element of D),
- c is an inner automorphism of S (we will identify c with an element of S).

Proposition 4.22 *With α as above, $\mathcal{P}(\alpha, \beta; 1, \frac{3}{5})$ and $\mathcal{P}(\beta, \alpha^{-1}; 1, \frac{3}{5})$ hold for every $\beta \in \mathrm{Aut}(S)$, with $\lambda = l(S)^{-3/5}$.*

This will follow from

Proposition 4.23 *Let*

$$W = \left\{ x \in S \mid |C_S(xh\phi^{-1})| < l(S)^{1/2} \right\}. \quad (23)$$

Then $|W| > \frac{4}{5}|S|$.

To deduce Proposition 4.22, suppose $Y \subseteq S$ satisfies $|Y| \geq \frac{1}{5}|S|$. Then $Yc \cap W$ is non-empty; choose $y \in Y$ with $yc \in W$. Then $|C_S(y\alpha)| < l(S)^{1/2}$, so for any subset X of S with $|X| \geq \frac{1}{5}|S|$ we have

$$\begin{aligned} |\mathbf{c}(X, y\alpha)| &= |\{[x, y\alpha] \mid x \in X\}| \\ &\geq |X|l(S)^{-1/2} \geq \frac{1}{5}l(S)^{-1/2}|S|. \end{aligned}$$

With Proposition 4.12, this shows that $\mathcal{P}(\alpha, \beta; 1, \frac{3}{5})$ holds with $\lambda = \frac{1}{5}l(S)^{-1/2}$ (as $\frac{3}{5} < (\frac{4}{5})^2$). Since

$$T_{\beta^{-1}, \alpha^{-1}}(x, y) = T_{\alpha, \beta}(y^{\alpha^{-1}}, x^{\beta^{-1}})^{-1},$$

this implies also that $\mathcal{P}(\beta^{-1}, \alpha^{-1}; 1, \frac{3}{5})$ holds with the same value of λ .

Suppose that $k \leq 2$. Then (22) implies that $q > 2^{25}$. Proposition 1.21 then implies that $l(S) \geq (q-1)/2 \geq 2^{24}$. If $k = 3$ then $S = {}^3D_4(q)$ and

Proposition 1.21 gives $l(S) \geq (q^4 - 1) > 2^{4 \cdot 50/3} - 1 > 2^{66}$. In any case, then, $l(S)^{-1/10} \leq 2^{-2.4} < \frac{1}{5}$, whence

$$\frac{1}{5}l(S)^{-1/2} > l(S)^{-3/5}.$$

Proposition 4.22 follows.

We proceed to the proof of Proposition 4.23. We are given $h \in D = T_\sigma$. By Lang's theorem ([GLS], Theorem 2.1.1) we may choose $\kappa \in T$ with $h = \kappa^{-1}\kappa^\phi$. Put $h' = \kappa\kappa^{-\sigma}$. Note that

$$\begin{aligned} (\kappa^{-1}\kappa^\phi)^\sigma &= h^\sigma = h = \kappa^{-1}\kappa^\phi, \\ h'^\phi &= (\kappa\kappa^{-\sigma})^\phi = \kappa\kappa^{-\sigma} = h', \end{aligned}$$

so $h' \in H$. Define

$$\begin{aligned} \mu, \nu &: L \rightarrow LT \\ \mu(x) &= [x\kappa, \phi], \quad \nu(x) = [(x\kappa)^{-1}, \sigma]. \end{aligned}$$

Lemma 4.24 (i) $\mu^{-1}(Sh) = \nu^{-1}(Gh')$;
(ii) if $g \in Sh$ then $|\mu^{-1}(g)| = |G|$;
(iii) if $z \in Gh'$ then $|\nu^{-1}(z)| = |S|$.

Proof. (i).

$$\begin{aligned} \mu(x) \in Sh &\iff \mu(x)^\sigma = \mu(x) \\ &\iff \kappa^{-1}x^{-1}x^\phi\kappa^\phi = \kappa^{-\sigma}x^{-\sigma}x^{\sigma\phi}\kappa^{\sigma\phi} \\ &\iff x^\sigma\kappa^\sigma\kappa^{-1}x^{-1} = x^{\sigma\phi}\kappa^{\sigma\phi}\kappa^{-\phi}x^{-\phi} \\ &\iff \nu(x) = v(x)^\phi \iff \nu(x) \in Gh'. \end{aligned}$$

(ii), (iii). Let $g \in Sh$. Then $g = \kappa^{-1}g'\kappa^\phi$ with $g' \in L$, and by Lang's theorem again we have $g' = [x, \phi]$ for some $x \in L$. Then $\mu(x) = g$, and we see that $\mu^{-1}(g) = xL_\phi = xG$. Similarly we find that $\nu^{-1}(z) = y\kappa L_\sigma \kappa^{-1} = yS\kappa^{-1}$ where $y = \kappa y_1 \kappa^{-1}$ and $z = \kappa \cdot y_1 y_1^{-\sigma} \cdot \kappa^\sigma$. ■

Now consider the semi-direct product $G_1 = GH \rtimes \langle \sigma \rangle$. We define a permutation action of G on G_1 as follows: for $x \in G$ and $a \in G_1$,

$$a^{\widehat{x}} = x^{-1}ax^\sigma.$$

We will call this the *twisted action*. For $a \in G_1$ we denote the stabilizer of a in G under this action by $C(a)$, i.e.

$$C(a) = \{x \in G \mid ax^\sigma = xa\}.$$

Set $Y = \mu^{-1}(Sh) = \nu^{-1}(Gh')$.

Lemma 4.25 *Let $y \in Y$ and put $g = \mu(y)$, $z = \nu(y)$. Then*

$$|C_S(g\phi^{-1})| = |C(z)|.$$

Proof. Let $a \in L$ and put $b = y\kappa a\kappa^{-1}y^{-1}$. The condition $a^{g\phi^{-1}} = a$ is equivalent to $b^\phi = b$, i.e. $b \in L_\phi = G$. The condition $a \in S = L_\sigma$ is equivalent to $(b^{y\kappa})^\sigma = b^{y\kappa}$, i.e. $zb^\sigma = bz$. So

$$C_S(g\phi^{-1}) = (y\kappa)^{-1}C(z)y\kappa.$$

■

If we put

$$Z = \left\{ z \in Gh' \mid |C(z)| < l(S)^{1/2} \right\}$$

$$Y^* = \nu^{-1}(Z),$$

the two preceding lemmas give

$$|W| = |G|^{-1} |Y^*| = |S| |G|^{-1} |Z|. \quad (24)$$

Lemma 4.26 (i) *If $S \neq {}^3D_4(q)$ then $l(S)^{1/2} > q_0^{11r}$;*
(ii) *If $S = {}^3D_4(q)$ then $l(S)^{1/2} > |G|$.*

Proof. Proposition 1.21 says that $l(S)$ is at least $q^r - 1$ unless $S = \text{SL}_2(q)$ when $l(S) = (q - 1)/2$. Also $q = q_0^{n/k} \geq q_0^{51/k}$.

In case (i) we have $k \leq 2$. Then $l(S) > \frac{1}{2}(q_0^{25r} - 1)$, whence $l(S) \geq q_0^{24r}$ and the result follows.

In case (ii), $k = 3$ and $G = D_4(q_0)$. In this case, we have

$$l(S) \geq (q^4 - 1) \geq q_0^{68} - 1,$$

$$|G| < q_0^{28} < l(S)^{1/2}.$$

■

Since $C(z) \leq G$ for each $z \in Gh'$, it follows in case (ii) that $Z = Gh'$ and hence that $|W| = |S|$.

Henceforth, **we assume that $S \neq {}^3D_4(q)$.**

Let $c(G)$ denote the number of conjugacy classes of G .

Lemma 4.27 *The coset $Gh' \subseteq G_1$ is a union of at most $|G_1 : G| c(G)$ orbits of G with the twisted action.*

Proof. For $z \in Gh'$ and $x \in G$ we have

$$(z \cdot \sigma^{-1})^x = x^{-1}zx^\sigma \cdot \sigma^{-1} = z^{\hat{x}} \cdot \sigma^{-1}$$

in G_1 . This shows that the twisted action on G on the coset Gh' is equivalent to the conjugation action of G on $Gh'\sigma^{-1} \subseteq G_1$. The number of orbits of G acting by conjugation on G_1 is

$$\begin{aligned} |G|^{-1} \sum_{g \in G} |C_{G_1}(g)| &\leq |G|^{-1} |G_1 : G| \sum_{g \in G} |C_G(g)| \\ &= |G|^{-1} |G_1 : G| |G| c(G) = |G_1 : G| c(G). \end{aligned}$$

The result follows. ■

Since $G = L_\phi$ is a quasisimple group of untwisted Lie type,

$$|GH : G| \leq |\text{Outdiag}(G)| \leq \min\{r+1, q_0-1\} < q_0.$$

The automorphism σ has order 1 or 2. Thus $|G_1 : G| \leq 2q_0$. Now Theorem 1.1 (1) in [FG] shows that $c(G) \leq 30q_0^r$. Applying Lemma 4.26, we deduce that if $y \in Gh' \setminus Z$ then

$$|y^{\widehat{G}}| = \frac{|G|}{|C(y)|} < \frac{|G|}{q_0^{11r}};$$

Hence by Lemma 4.27 $Gh' \setminus Z$ is the union of at most $60q_0^{r+1}$ orbits of this size, whence

$$|Gh' \setminus Z| < 60q_0^{-10r+1} |G|.$$

Therefore $|Z| \geq \eta |G|$ where $\eta = 1 - 60/2^9 > \frac{4}{5}$.

Now Proposition 4.23 follows from (24).

4.1.6 Proof of Theorem 4.1

As explained in Subsection 4.1.1, we have to find $D \in \mathbb{N}$ and $\varepsilon > 0$ such that $\mathcal{P}(\text{Aut}(S); D, \varepsilon)$ holds with λ given by (19) for every quasisimple group S : i.e. $\lambda = l(S)^{-3/5}$ if $l(S) \geq 3$, $\lambda = 1$ if $l(S) = 2$. Henceforth, when we say that $\mathcal{P}(\dots)$ holds for some group S , we will mean that it holds with λ given by (19).

Set $N_0 = \max\{N_2, N_3, 1 + |M|\}$ where N_i are the bounds introduced above and M denotes the largest sporadic (quasi)simple group (it happens to be simple).

Now let S be a quasisimple group. We consider several cases.

Case 1. Where $|S| < N_0$. Proposition 4.14 shows that $\mathcal{P}(\text{Aut}(S); D_1, \varepsilon_1)$ holds for some D_1 and ε_1 .

We assume henceforth that $|S| \geq N_0$. Putting $\Gamma_0 = \text{Inn}(S)$, Proposition 4.16 shows that $\mathcal{P}(\Gamma_0; 1, \frac{1}{8})$ holds.

Case 2. Where $S/Z(S)$ is an alternating group. Then $|\text{Aut}(S) : \Gamma_0| = 2$, and Proposition 4.4 gives $\mathcal{P}(\text{Aut}(S); 4, \frac{1}{8})$.

From now on, S is a group of Lie type, of rank r over $\mathbb{F}_q$. We denote by Φ the group of field automorphisms of S . Then $\text{Aut}(S)$ has normal subgroups

$$\text{Aut}(S) \geq \Gamma \geq \Gamma_1 \geq \Gamma_2 \geq \Gamma_0 = \text{Inn}(S)$$

where $\Gamma_2 = \text{InnDiag}(S)$, $\Gamma = \Gamma_2\Phi$, and $\Gamma_1 = \Gamma_2\Phi_1$ where Φ_1 is the subgroup of Φ generated by all elements of order at most 50.

Put $n_0 = \text{lcm}[50]$, and define $n_1 = \min\{q+1, r+1\}$ if S has type A_r or 2A_r , $n_1 = 4$ otherwise. We have

$$\begin{aligned} |\text{Aut}(S) : \Gamma| &\leq 6, \\ |\Gamma : \Gamma_2| &\leq \log_p(q^3) \leq 3\log_2(q), \\ |\Gamma_1 : \Gamma_2| &\leq n_0, \\ |\Gamma_2 : \Gamma_0| &\leq n_1 \end{aligned}$$

where $p = \text{char}(\mathbb{F}_q)$ (see [GLS], Section 2.5).

Case 3. Where $q \leq 10$. In this case, $|\text{Aut}(S) : \Gamma_0| \leq 600$. As in Case 2, we may deduce that $\mathcal{P}(\text{Aut}(S); D_2, \frac{1}{8})$ holds where $D_2 = 360,000$.

Case 4. Where $q > 10$. If $r < 9$ we have $|\Gamma_2 : \Gamma_0| \leq n_1 \leq 9$; we deduce as before that $\mathcal{P}(\Gamma_2; 81, \frac{1}{8})$ holds. If $r \geq 9$, Proposition 4.18 gives $\mathcal{P}(\Gamma_2; 8, \frac{1}{2})$. Taking $D_3 = 81n_0^2$, we infer in any case that $\mathcal{P}(\Gamma_1; D_3, \frac{1}{8})$ holds, whatever the rank r .

Now let $\alpha, \beta \in \Gamma^{(D_3)}$. If α_i and β_i lie in Γ_1 for every i then we have $\mathcal{P}(\alpha, \beta; D_3, \frac{1}{8})$. If not, let us suppose for convenience that $\alpha_1 \notin \Gamma_1$. Then $\alpha_1 = ch\phi$ where $c \in \Gamma_0$, h is diagonal, and $\phi \in \Phi$ has order exceeding 50. Proposition 4.22 now shows that $\mathcal{P}(\alpha_1, \beta_1; 1, 3/5)$ holds. As in the proof of Proposition 4.3, this in turn implies $\mathcal{P}(\alpha, \beta; D_3, 3/5)$.

Thus $\mathcal{P}(\Gamma; D_3, \frac{1}{8})$ holds in either case. Since $|\text{Aut}(S) : \Gamma| \leq 6$, a final application of Proposition 4.4 gives $\mathcal{P}(\text{Aut}(S); D_4, \frac{1}{8})$ where $D_4 = 36D_3$.

Conclusion. Take $D = \max\{4, D_1, D_2, D_4\}$ and $\varepsilon = \min\{\varepsilon_1, \frac{1}{8}\}$. Then $\mathcal{P}(\text{Aut}(S); D, \varepsilon)$ holds in all cases, by Proposition 4.3.

4.2 Commutators in semisimple groups

In this subsection, D and ε are the constants introduced in subsection 4.1. We will say that a multiset Y has the (k, η) -fpp on a $\langle Y \rangle$ -set Ω if at least k elements of Y have the η -fpp on Ω .

Theorem 4.28 *Let N be a finite quasisemisimple group with at least 3 non-abelian composition factors. Let $\mathbf{y}_1, \dots, \mathbf{y}_{10}$ be m -tuples of automorphisms of*

N . Assume that for each i , the group $\langle \mathbf{y}_i \rangle$ permutes the set Ω of quasisimple factors of N transitively and that $\mathbf{y}_i$ has the (k, η) -fpp on Ω , where $k\eta \geq 4 + 2D$. For each i let $W(i) \subseteq N^{(m)}$ be a subset with $|W(i)| \geq (1 - \varepsilon/6) |N|^m$. Then

$$\prod_{i=1}^{10} W(i) \phi(i) = N$$

where $\phi(i) : N^{(m)} \rightarrow N$ is given by

$$(x_1, \dots, x_m) \phi(i) = \prod_{j=1}^m [x_i, y_{ij}].$$

The action of $\text{Aut}(N)$ lifts to an action on the universal cover $\tilde{N}$ of N , and $\tilde{N} = S_1 \times \dots \times S_n$ where the S_i are quasisimple groups. Replacing N by $\tilde{N}$ and each $W(i)$ by its inverse image in $\tilde{N}^{(m)}$, we may suppose that in fact $N = S_1 \times \dots \times S_n$. Since $\langle \mathbf{y}_1 \rangle$ permutes $\Omega = \{S_1, \dots, S_n\}$ transitively, the groups S_i are all isomorphic to a quasisimple group S .

Now let $G = \langle g_1, \dots, g_m \rangle \leq \text{Aut}(N)$ and denote by e_i the number of cycles (including fixed points) of g_i in its action on Ω . Define $\phi : N^{(m)} \rightarrow N$ by

$$\mathbf{x} \phi = \mathbf{c}(\mathbf{x}, \mathbf{g}) = \prod_{i=1}^m [x_i, g_i].$$

We shall prove

Proposition 4.29 *Suppose that that*

$$(m-2)n - \sum_{i=1}^m e_i \geq 2D. \quad (25)$$

(1) *Let $W \subseteq N^{(m)}$ satisfy $|W| \geq (1 - \varepsilon/6) |N|^m$. Then*

$$|W \phi| \geq l(S)^{-4/5} |N|.$$

(2) *If D is replaced by $D_1 = 5D$, then ϕ is surjective, and each fibre of ϕ has size at least $|N|^{-2D_1/n} |N|^{m-1}$.*

Part (2) is a sharper version of [NS], Proposition 9.1; we will not be needing it, and include it in a revisionist spirit, to show how the main results of [NS] can be reproduced using these methods.

To deduce Theorem 4.28 from (1), note that for each $i = 1, \dots, 10$, the total number of cycles for $y_{i1}, \dots, y_{im}$ on Ω is at most

$$(m-k)n + k(1 - \eta/2)n \leq (m-2)n - nD,$$

which implies condition (25) since $n \geq 3$. So taking $\mathbf{g} = \mathbf{y}_i$ and writing $\phi(i)$ for the corresponding map ϕ , we may infer that

$$|W(i) \phi(i)| \geq l(S)^{-4/5} |N|.$$

Now $l(S) = l(N) = l$, say, and we have

$$\prod_{i=1}^{10} |W(i)\phi(i)| \geq \frac{|N|^{10}}{l^8}.$$

It follows by the ‘Gowers trick’ that $\prod_{i=1}^{10} W(i)\phi(i) = N$, and this is the statement of Theorem 4.28 since $W(i)\phi(i) = \prod_{j=1}^m [W(i), y_{ij}]$.

4.2.1 Proof of Proposition 4.29

Lemma 4.30 *Suppose that $G = \langle g_1, \dots, g_m \rangle$ acts transitively on a set J of size n . Let $t_1 \in J$. Then there exist $i(2), \dots, i(n) \in [m]$ and $\varepsilon_2, \dots, \varepsilon_n \in \{\pm 1\}$ such that $J = \{t_1, \dots, t_n\}$ where $t_j = t_{j-1}g_{i(j)}^{\varepsilon_j}$ for $j = 2, \dots, n$.*

Proof. Just take a Schreier transversal $\{1\} \cup \{g_{i(2)}^{\varepsilon_2} g_{i(3)}^{\varepsilon_3} \cdots g_{i(l)}^{\varepsilon_l} \mid 2 \leq l \leq n\}$ to the cosets of $\text{stab}_G(t_1)$. ■

Keeping G and J as above, re-label the elements of J as $\{1, 2, \dots, n\}$ with $t_l = l$ for each l . Say g_i has cycles Δ_{il} , $l = 1, \dots, e_i$ (including cycles of length 1); if $i = i(j)$ choose Δ_{i1} to be the cycle containing j . In general, we also write

$$\Delta_{il} = \Delta_i(j) \text{ if } j \in \Delta_{il}.$$

Let δ_{il} denote the first member of Δ_{il} . Note that if $i = i(j)$ then $j > j-1 \geq \delta_{i1} = \hat{j}$, say.

Put

$$\begin{aligned} \Delta'_{il} &= \Delta_{il} \setminus \{\delta_{il}\}, \\ J'_i &= \bigcup_{l=1}^{e_i} \Delta'_{il}. \end{aligned}$$

Each Δ_{il} is ordered as a g_i -cycle starting with δ_{il} (not with the induced order from J); for $j \in \Delta'_{il}$ let $j^- \in \Delta_{il}$ denote the immediate predecessor of j (this also depends on i but its meaning will be clear from the context).

Let S be a finite group, $N = S^J$, and suppose that G acts on N , permuting the factors according to the action of G on J . Write elements of N as $x = (x(j))_{j \in J}$.

For any subset T of $[m] \times J$ write $\pi_T : N^{(m)} \rightarrow S^T$ for the projection map

$$(x_1, \dots, x_m)\pi_T = (x_i(j))_{(i,j) \in T}.$$

For $x \in S$ an expression x^* will mean x^α where α is some fixed automorphism of S , depending on the context but not on x , and $x^{-*} = (x^*)^{-1}$.

We write

$$[\mathbf{x}, \mathbf{g}] = ([x_1, g_1], \dots, [x_m, g_m]).$$

Lemma 4.31 *Let $x, y \in N$. Then $[x, g_i] = y$ if and only if*

$$y(\delta_{il}) = x(\delta_{il})^{-1} x(\delta_{il})^* \prod_{j \in \Delta'_{il}} y(j)^{-*} \quad (26)$$

$$x(j) = x(j^-)^* y(j)^{-1} \quad (j \in \Delta'_{il}) \quad (27)$$

for $1 \leq l \leq e_i$.

Proof. Compare the j -components of $u = [x, g_i]$ and of y as j runs over a given cycle Δ_{il} . To simplify notation let's suppose that $\Delta_{il} = (1, 2, \dots, s)$, with $\delta_{il} = 1$. For $1 \leq j \leq s$ we have

$$u(j) = x(j)^{-1} x(j-1)^{\alpha_j}$$

(writing $x(0) = x(s)$) where $\alpha_j \in \text{Aut}(S)$ depends on j and g_i . Using these to eliminate $x(2), \dots, x(s)$ in turn we get

$$x(1)^{-1} x(1)^\beta = u(1) u(s)^{\alpha_s} u(s-1)^{\alpha_{s-1} \alpha_s} \dots u(2)^{\alpha_2 \dots \alpha_s},$$

where $\beta = \alpha_1 \dots \alpha_s$ is the automorphism induced by g_i^s on the first component of $S^{\Delta_{il}}$. Thus (26) and (27) hold with u in place of y . The lemma follows since these equations determine y uniquely, given x . ■

Put

$$\begin{aligned} C &= \{(i, \delta_{il}) \mid 1 \leq i \leq m, 1 \leq l \leq e_i\} \\ K &= \{(i, j) \mid 1 \leq i \leq m, j \in J'_i\} \\ K' &= K \setminus \{(i(j), j) \mid j = 2, \dots, n\}. \end{aligned}$$

Define $\Theta : N^{(m)} \rightarrow S^C \times S^K = S^{(mn)}$ by

$$\mathbf{x}\Theta = (\mathbf{x}\pi_C, [\mathbf{x}, \mathbf{g}]\pi_K).$$

Lemma 4.31 shows that Θ is bijective.

Now define $\phi : N^{(m)} \rightarrow N = S^{(n)}$ by

$$\begin{aligned} \mathbf{x}\phi &= \prod_{i=1}^m [x_i, g_i] \\ &= (\mathbf{x}\phi_1, \dots, \mathbf{x}\phi_n). \end{aligned}$$

Define $\Psi : N^{(m)} \rightarrow S^C \times S^{K'} \times S^{(n-1)} = S^{(mn)}$ by

$$\mathbf{x}\Psi = (\mathbf{x}\pi_C, [\mathbf{x}, \mathbf{g}]\pi_{K'}, (\mathbf{x}\phi_2, \dots, \mathbf{x}\phi_n)).$$

Lemma 4.32 *The mapping $\Psi : S^{(mn)} \rightarrow S^{(mn)}$ is bijective.*

Proof. Let $(\mathbf{u}, \mathbf{v}, z_2, \dots, z_n) \in S^C \times S^{K'} \times S^{(n-1)}$. We have to show that there exists a unique $\mathbf{x} \in N^{(m)}$ such that $\mathbf{x}\pi_C = \mathbf{u}$, $[\mathbf{x}, \mathbf{g}]\pi_{K'} = \mathbf{v}$ and $\mathbf{x}\phi_j = z_j$ for $j = 2, \dots, n$.

Since Θ is bijective, for each tuple $\eta = (\eta_2, \dots, \eta_n) \in S^{(n-1)}$ there exists a unique $\mathbf{x} \in N^{(m)}$ with

$$\begin{aligned} \mathbf{x}\pi_C &= \mathbf{u}, [\mathbf{x}, \mathbf{g}]\pi_{K'} = \mathbf{v}, \\ [x_{i(j)}, g_{i(j)}]\pi_j &= \eta_j \quad (j = 2, \dots, n). \end{aligned}$$

Write $y_i = [x_i, g_i]$. Then

$$\mathbf{x}\phi_j = y_1(j)y_2(j) \dots y_m(j).$$

If $(i, j) \in K'$ then $y_i(j)$ is the (i, j) -component of $[\mathbf{x}, \mathbf{g}]\pi_{K'} = \mathbf{v}$. If $(i, j) \in C$ then $y_i(j)$ is determined by equation (26); this involves $x_i(j)$, a component of $\mathbf{x}\pi_C = \mathbf{u}$, and further factors $y_i(r)$ where $r > j$.

If $(i, j) \notin C \cup K'$ then $i = i(j)$ and $y_i(j) = \eta_j$. Now we can solve the equations

$$\eta_j = y_{i-1}(j)^{-1} \dots y_1(j)^{-1} z_j y_m(j)^{-1} \dots y_{i+1}(j)^{-1} \quad (28)$$

successively for $j = n, n-1, \dots, 2$, uniquely for η . The result follows. ■

Lemma 4.33 *Let $K'' \subseteq K'$ be a set of size $|K'| - 2D \geq 0$. Suppose that $W \subseteq N^{(m)}$ satisfies $|W| \geq (1 - \varepsilon/q) |N^{(m)}|$. Let $\mathcal{P}$ be the set of elements $\mathbf{z} \in S^{(n-1)}$ for which there exist $\mathbf{u} \in S^C$, $\mathbf{v} \in S^{K''}$ such that*

$$|\{w \in W \mid w\Psi\pi_{C \cup K'' \cup [n-1]} = (\mathbf{u}, \mathbf{v}, \mathbf{z})\}| \geq (1 - \varepsilon) |S|^{2D}.$$

Then $|\mathcal{P}| \geq (1 - \frac{1}{q}) |S|^{n-1}$.

Proof. Put $\sigma = |S|$. Recall that $|C| + |K''| = mn - (n-1) - 2D$, and that Ψ is bijective. Suppose that $|\mathcal{P}| = \lambda\sigma^{n-1}$. Then

$$|W| \leq \lambda\sigma^{n-1}\sigma^{mn-n+1-2D} \cdot \sigma^{2D} + (1 - \lambda)\sigma^{n-1}\sigma^{mn-n+1-2D} \cdot (1 - \varepsilon)\sigma^{2D}.$$

It follows that

$$1 - \varepsilon/q \leq \lambda + (1 - \lambda)(1 - \varepsilon),$$

which implies that $\lambda \geq 1 - \frac{1}{q}$. ■

Observe now that $\mathbf{x}\phi = (z_1, \dots, z_n)$ if and only if

$$\mathbf{x}\phi_1 = z_1 \quad (29)$$

and

$$\mathbf{x}\Psi = (\mathbf{u}, \mathbf{v}, z_2, \dots, z_n) \quad (30)$$

for some $(\mathbf{u}, \mathbf{v}) \in S^C \times S^{K'}$.

Putting $y_i = [x_i, g_i]$ as above we have

$$\mathbf{x}\phi_1 = y_1(1)y_2(1)\dots y_m(1). \quad (31)$$

Now the following hold:

If $(i, j) \in C$ then $j = \delta_{il}$ for some $l \leq e_i$, and

$$y_i(j) = x_i(j)^{-1}x_i(j)^* \prod_{k \in \Delta'_{il}} y_i(k)^{-*}; \quad (\text{S}(i, j))$$

note that for each factor $y_i(k)$ occurring on the right we have $(i, k) \notin C$ and $k > j$.

If $i = i(j)$ then

$$y_i(j)^{-1} = y_{i+1}(j)\dots y_m(j)z_j^{-1}y_1(j)\dots y_{i-1}(j); \quad (\text{S}(j))$$

note that for each factor $y_r(j)$ occurring on the right we have $r \neq i(j)$.

Hence we may transform the right-hand member of (31) by successively eliminating factors $y_i(j)$ with $(i, j) \in C$, using $(\text{S}(i, j))$, and factors $y_i(j)^{-1}$ with $i = i(j)$, using $(\text{S}(j))$.

To analyse this process, for the time being we consider the $y_i(j)$, $y_i(j)^{-1}$, $x_i(j)$, $x_i(j)^{-1}$ and z_j^{-1} as abstract symbols (but allowing the automorphisms denoted by $*$ to distribute over the factors in the usual way). If U is a product of such symbols, possibly decorated with $*$ s, the *support* $\text{sup}(U)$ is the multiset of symbols that occur in U , with their multiplicities. For $(i, j) \in C$ let Y_{ij} denote the right-hand side of $(\text{S}(i, j))$, and for $(i, j) \notin C$ set $Y_{ij} = y_i(j)$. For $j = 2, \dots, n$ put

$$Z_j = Y_{i(j)+1,j}\dots Y_{mj}z_j^{-1}Y_{1j}\dots Y_{i(j)-1,j}.$$

Then

$$\begin{aligned} \text{sup}(Y_{ij}) &= \{x_i(j)^{-1}, x_i(j), y_i(k)^{-1} \mid k \in \Delta'_i(j)\} \text{ if } (i, j) \in C, \\ \text{sup}(Y_{ij}) &= \{y_i(j)\} \text{ if } (i, j) \notin C \end{aligned}$$

and

$$\text{sup}(Z_j) = \{z_j^{-1}\} \cup \bigcup_{i \neq i(j)} \text{sup}(Y_{ij})$$

(disjoint union).

Now set

$$U_1 = \prod_{i=1}^m Y_{i1}.$$

Then

$$\text{sup}(U_1) = \bigcup_i \text{sup}(Y_{i1}) \ni y_{i(2)}(2)^{-1},$$

because $(i, 1) \in C$ for every i , and $2 \in \Delta'_{i(2)}(1)$. Let U_2 be the expression obtained from U_1 on replacing $y_{i(2)}(2)^{-1}$ by Z_2 . Then

$$\begin{aligned} \sup(U_2) &= \sup(U_1) \cup \sup(Z_2) \setminus \{y_{i(2)}(2)^{-1}\} \\ &= \bigcup_i \sup(Y_{i1}) \cup \{z_2^{-1}\} \cup \bigcup_{i \neq i(2)} \sup(Y_{i2}) \setminus \{y_{i(2)}(2)^{-1}\}. \end{aligned}$$

If $(i(3), 2) \in C$ then $i(3) \neq i(2)$ and $y_{i(3)}(3)^{-1} \in \sup(Y_{i(3),2})$ because $3 \in \Delta'_{i(3)}(2)$; if $(i(3), 2) \notin C$ then $3 \in \Delta'_{i(3)}(1)$ and $y_{i(3)}(3)^{-1} \in \sup(Y_{i(3),1})$. So in any case $y_{i(3)}(3)^{-1} \in \sup(U_2)$.

Let U_3 be obtained from U_2 on replacing $y_{i(3)}(3)^{-1}$ by Z_3 . Iterating this process, suppose that after $j-1 < n-1$ steps we obtain U_j , where $\sup(U_j)$ contains

$$\bigcup_{r=1}^j \bigcup_{i \neq i(r)} (\sup(Y_{ir}) \setminus \{y_{i(r)}(r)^{-1}\}).$$

Say $\widehat{j+1} = r$, so $r \leq j$ and $j+1 \in \Delta'_{i(j+1)}(r)$. Then $(i(j+1), r) \in C$ and (if $r > 1$) $i(j+1) \neq i(r)$, so $y_{i(j+1)}(j+1)^{-1} \in \sup(Y_{i(j+1),r}) \subseteq \sup(U_j)$. Now replace $y_{i(j+1)}(j+1)^{-1}$ in U_j by Z_{j+1} to obtain U_{j+1} .

After $n-1$ such steps we obtain an expression U_n with

$$\sup(U_n) = \mathcal{X} \cup \mathcal{Y} \cup \mathcal{Z}$$

where

$$\begin{aligned} \mathcal{X} &= \{x_i(j), x_i(j)^{-1} \mid (i, j) \in C\}, \quad \mathcal{Y} = \{y_i(j), y_i(j)^{-1} \mid (i, j) \in K'\}, \\ \mathcal{Z} &= \{z_2^{-1}, \dots, z_n^{-1}\}. \end{aligned}$$

To any formal product V of factors $x_i(j)^{\pm*}$, $y_i(j)^{\pm*}$, z_j^{-*} we assign a numerical sequence $\tau(V)$ as follows: reading V from left to right, ignore all factors $x_i(j)^{\pm*}$ and z_j^{-*} ; to each factor $y_i(j)^*$ assign the label i , and to each maximal product of consecutive terms of the form $y_i(k)^{-*}$ (fixed i , varying k) assign the label i .

Claim 1: For each $j = 1, \dots, n$, $\tau(U_j)$ is a subsequence of

$$\mathcal{S}(j) = (1, \dots, m, 1, \dots, m, \dots, 1, \dots, m)$$

where $1, \dots, m$ is repeated j times.

Proof. This is clear for $j = 1$. Let $j \geq 1$ and suppose inductively that $\tau(U_j)$ is a subsequence of $\mathcal{S}(j)$. Put $i = i(j+1)$; then $y_i(j+1)^{-*}$ is a factor in U_j , and we obtained U_{j+1} by replacing it with Z_{j+1}^* .

Thus

$$\tau(U_j) = (I_1, P, i, Q, I_2)$$

where (P, i, Q) is a subsequence of $(1, \dots, m)$, I_1 is a subsequence of $\mathcal{S}(p)$ and I_2 is a subsequence of $\mathcal{S}(q)$ and $p+1+q = j$ (here p or q could be 0, with $\mathcal{S}(0) = \emptyset$); the displayed i is due to $y_i(j+1)^{-*}$. Substituting Z_{j+1}^* for $y_i(j+1)^{-*}$ has the effect of replacing i by $(\underline{i}, i+1, \dots, m, 1, \dots, i-1, \underline{i})$, where the underlined i s may or may not be present (depending on whether $y_i(j+1)^{-*}$ appears in the middle or at either end of a product of consecutive terms of the form $y_i(k)^{-*}$). In any case,

$$(P, \underline{i}, i+1, \dots, m, 1, \dots, i-1, \underline{i}, Q)$$

is a subsequence of $\mathcal{S}(2)$, and so $\tau(U_{j+1})$ is a subsequence of

$$(\mathcal{S}(p), \mathcal{S}(2), \mathcal{S}(q)) = \mathcal{S}(p+2+q) = \mathcal{S}(j+1).$$

Claim 2: There exist $2D$ distinct elements $\xi_1, \eta_1, \dots, \xi_D, \eta_D$ of $\mathcal{Y}$ such that the following holds. There exist R, A_i, B_i, C_i, D_i ($i = 1, \dots, D$), each of which is a product of factors t^* with $t \in \mathcal{X} \cup \mathcal{Z} \cup \mathcal{Y} \setminus \{\xi_1, \eta_1, \dots, \xi_D, \eta_D\}$, such that

$$U_n \simeq \prod_{i=1}^D (A_i \xi_i B_i)^{-*} (C_i \eta_i D_i)^{-*} (A_i \xi_i B_i)^* (C_i \eta_i D_i)^* \cdot R, \quad (32)$$

meaning that the two sides represent the same element in the free group on all the occurring symbols t^* , $t \in \mathcal{X} \cup \mathcal{Y} \cup \mathcal{Z}$.

Proof. This follows from hypothesis (25) and Claim 1 by (the proof of) [NS], Prop. 8.4.

Now we can complete the proof of Proposition 4.29. For some subset $L \subseteq K'$ of size $2D$ we have

$$\{\xi_1, \eta_1, \dots, \xi_D, \eta_D\} = \{y_i(j) \mid (i, j) \in L\}.$$

Put $K'' = K' \setminus L$. Recall that $W \subseteq N^{(m)}$ satisfies $|W| \geq (1 - \varepsilon/6) |N|^m$. Let $\mathcal{P} \subseteq S^{(n-1)}$ be the set defined in Lemma 4.33; thus $|\mathcal{P}| \geq \frac{5}{6} |S|^{n-1}$.

By definition, for each $\mathbf{z} \in \mathcal{P}$ there exist $\mathbf{u}_{\mathbf{z}} \in S^C$, $\mathbf{v}_{\mathbf{z}} \in S^{K''}$ and $W_{\mathbf{z}} \subseteq W$ with $|W_{\mathbf{z}}| \geq (1 - \varepsilon) |S|^{2D}$ such that

$$W_{\mathbf{z}} \Psi \pi_{C \cup K'' \cup [n-1]} = \{(\mathbf{u}_{\mathbf{z}}, \mathbf{v}_{\mathbf{z}}, \mathbf{z})\}.$$

As Ψ is a bijection this implies that $|W_{\mathbf{z}} \Psi \pi_L| = |W_{\mathbf{z}}| \geq (1 - \varepsilon) |S|^{2D}$.

Now let $\mathbf{x} \in W_{\mathbf{z}}$. Then

$$\begin{aligned} \mathbf{x}\phi &= (\mathbf{x}\phi_1, \dots, \mathbf{x}\phi_n) \\ &= (\mathbf{x}\phi_1, \mathbf{z}) \end{aligned}$$

and

$$\mathbf{x}\phi_1 = U(X, Y, Z).$$

In the expression (32) for $U(X, Y, Z)$, each of the factors R, A_i, B_i, C_i, D_i is a product of terms $t^{\pm*}$ where t is a component of $\mathbf{x}\Psi\pi_{C\cup K''\cup[n-1]} = (\mathbf{u}_z, \mathbf{v}_z, \mathbf{z})$. Therefore

$$\begin{aligned} \mathbf{x}\phi_1 &= \prod_{i=1}^D (a_i \xi_i b_i)^{-\alpha_i} (c_i \eta_i d_i)^{-\beta_i} (a_i \xi_i b_i)^{\gamma_i} (c_i \eta_i d_i)^{\delta_i} \cdot r \\ &= \prod_{i=1}^D T_{\sigma_i, \tau_i}(\bar{\xi}_i, \bar{\eta}_i) \cdot r \end{aligned} \quad (33)$$

where a_i, b_i, c_i, d_i and r depend only on $\mathbf{z}$ and $\alpha_i, \beta_i, \gamma_i, \delta_i$ are certain automorphisms of S , independent of everything else, and

$$\begin{aligned} \bar{\xi}_i &= (a_i \xi_i b_i)^{-\alpha_i}, \quad \bar{\eta}_i = (c_i \eta_i d_i)^{-\beta_i} \\ \sigma_i &= \alpha_i^{-1} \gamma_i, \quad \tau_i = \beta_i^{-1} \delta_i. \end{aligned}$$

Now $(\xi_1, \eta_1, \dots, \xi_D, \eta_D) = \mathbf{x}\Psi\pi_L$ takes $|W_{\mathbf{z}}\Psi\pi_L| \geq (1 - \varepsilon) |S|^{2D}$ values as $\mathbf{x}$ ranges over $W_{\mathbf{z}}$; hence so does the tuple $(\bar{\xi}_1, \bar{\eta}_1, \dots, \bar{\xi}_D, \bar{\eta}_D)$. According to Theorem 4.1 this implies that $\prod_{i=1}^D T_{\sigma_i, \tau_i}(\bar{\xi}_i, \bar{\eta}_i)$ takes at least $\lambda |S|$ values, where $\lambda = l(S)^{-3/5}$ if $l(S) \geq 3$, $\lambda = 1$ if $l(S) = 2$; therefore so does $\mathbf{x}\phi_1$, by (33). It follows that

$$\begin{aligned} |W\phi| &\geq \sum_{\mathbf{z} \in \mathcal{P}} |W_{\mathbf{z}}\phi_1| \geq \frac{5}{6} |S|^{n-1} \cdot \lambda |S| \\ &\geq l(S)^{-4/5} |N| \end{aligned}$$

since $|S|^n = |N|$ and $(\frac{6}{5})^5 < 3$. This completes the proof of (1).

To prove (2), we replace D by D_1 in the above. Let $(\mathbf{u}, \mathbf{v}_0, \mathbf{z})$ be an arbitrary element of $S^C \times S^{K''} \times S^{(n-1)}$, and let $z \in S$. For each $\xi = (\xi_1, \eta_1, \dots, \xi_{D_1}, \eta_{D_1}) \in S^L$ there exists $\mathbf{x} \in N^{(m)}$ such that

$$\begin{aligned} \mathbf{x}\Psi\pi_{C\cup K''\cup[n-1]} &= (\mathbf{u}, \mathbf{v}_0, \mathbf{z}) \\ \mathbf{x}\Psi\pi_L &= \xi. \end{aligned}$$

Take $\mathbf{u}_z = \mathbf{u}$ and $\mathbf{v}_z = \mathbf{v}_0$ in the above discussion. Then $\mathbf{x}\phi_1$ is given by (33). Now Corollary 4.2 says that

$$S = \prod_{i=1}^{D_1} T_{\sigma_i, \tau_i}(S, S).$$

We may therefore choose ξ so that

$$\prod_{i=1}^{D_1} T_{\sigma_i, \tau_i}(\bar{\xi}_i, \bar{\eta}_i) = zr^{-1},$$

and so ensure that $\mathbf{x}\phi = (z, \mathbf{z})$. It follows that

$$\begin{aligned} |(z, \mathbf{z})\phi^{-1}| &\geq |S^C \times S^{K''}| = |S|^{mn-(n-1)-2D_1} \\ &> |N|^{-2D_1/n} |N|^{m-1}. \end{aligned}$$

5 Applications

5.1 Finite to profinite

Here we recall some standard compactness arguments. We refer to subsection 1.2.1 for the statements of the following theorems, concerning a finitely generated profinite group G with closed normal subgroups K and H .

Proof of Theorem 1.5. Write $I = \{(i, j) \mid 1 \leq i \leq r, 1 \leq j \leq f_0\}$. For each open normal subgroup N of G let

$$X(N) = \left\{ \mathbf{x} = (x_{ij}) \in K^{(rf_0)} \mid G = N \langle y_i^{x_{ij}} \mid (i, j) \in I \rangle \right\}.$$

Theorem 1.1, applied to the finite group G/N , shows that each set $X(N)$ is non-empty. Also $X(N)$ is closed in $K^{(rf_0)}$, being a union of cosets of $(N \cap K)^{(rf_0)}$, and if $N > M$ then $X(N) \supseteq X(M)$. It follows by compactness that $\bigcap_N X(N)$ is non-empty, taking the intersection over all open normal subgroups N of G . Let $\mathbf{x}$ be in this intersection. Then

$$G = \bigcap_N N \langle y_i^{x_{ij}} \mid (i, j) \in I \rangle = \overline{\langle y_i^{x_{ij}} \mid (i, j) \in I \rangle}.$$

Proof of Theorems 1.6 and 1.7. Let R denote the right-hand side of equation (†) or equation (‡) (see Subsection 1.2.1). Then R is a closed subset of G . Now let N be an open normal subgroup of G . Then Theorem 1.2, respectively Theorem 1.3, applied to the finite group G/N shows that $[H, G]N = RN$. As R is closed, intersecting over all open normal subgroups N of G we get

$$R = \bigcap_N RN \supseteq [H, G],$$

and the results follow since $R \subseteq [H, G]$.

5.2 Verbal subgroups

Here we show how the main results of [NSP] may be quickly derived from Theorems 1.1 and 1.2.

Let w be a group word in k variables, and G a group. The corresponding *verbal subgroup* is $w(G) = \langle G_w \rangle$, where

$$G_w = \left\{ w(\mathbf{g})^{\pm 1} \mid \mathbf{g} \in G^{(k)} \right\}$$

denotes the (symmetrized) set of w -values in G . We say that w has *width* m in G if

$$w(G) = G_w^{*m};$$

if this holds for some finite m we denote the least such m by $m_w(G)$, and say that w has finite width in G .

The following elementary result is Proposition 2.1.2 of [S2]:

Lemma 5.1 *If G is abelian-by-finite then $m_w(G)$ is finite.*

Now define

$$\beta(w, G) = |G : w(G)|.$$

Let us call the word w *d-bounded* if there exists $\beta_w = \beta_w(d) \in \mathbb{N}$ such that $\beta(w, G) \leq \beta_w(d)$ whenever G is a d -generator finite group. The positive solution of the Restricted Burnside Problem [Z] asserts that the word $w = x^q$ is d -bounded, for all natural numbers d and q . This implies that every *non-commutator* word w is d -bounded, since for any group G we have $w(G) \geq G^q$ where $q = |\mathbb{Z}/w(\mathbb{Z})|$. (In fact it is easy to see that, conversely, every d -bounded word is a non-commutator word.)

Proposition 5.2 *Suppose that w is d -bounded (for some $d \geq 1$). Then there exists $m_0 = m_0(w)$ such that w has width m_0 in every finite semisimple group.*

Proof. It suffices to prove this for a simple group G . Take $q = |\mathbb{Z}/w(\mathbb{Z})|$. We consider three cases.

- (i) Where $w(G) = 1$. Then $m_w(G) = 1$.
- (ii) Where $w(G) \neq 1$ but $G^q = 1$. There are only finitely many possibilities for G in this case [J]. Since G_w generates G it follows that $G = G_w^{*n}$ where $n = n(q)$ is the maximal order of any such group G .
- (iii) Where $G^q \neq 1$. In this case, a theorem of Martinez-Zelmanov [MZ] and Saxl-Wilson [SW] shows that every element of G is a product of $h(q)$ q th powers; as each q th power is a w -value it follows that $m_w(G) \leq h(q)$. ■

The main result is now

Theorem 5.3 *Let w be a d -bounded word and G a finite d -generator group. Then $m_w(G) \leq f(w, d)$ where $f(w, d)$ depends only on w and d .*

Proof. Put $W = w(G)$ and set $\beta = |G : W|$, so that $\beta \leq \beta_w(d)$. By Schreier's formula we then have

$$d(W) \leq d_0 := 1 + \beta(d - 1).$$

Let F be the free group on d generators, and let K be the intersection of the kernels of all homomorphisms $F \rightarrow \text{Sym}(\beta)$. Lemma 5.1 shows that w has finite width $m_1 = m_1(d, \beta)$ in the group F/K' . As G/W' is an image of F/K' it follows that $m_w(G/W') \leq m_1$, so

$$W = W' \cdot G_w^{*m_1}. \quad (34)$$

Hence there exists $Y_1 \subseteq G_w$ such that

$$W = W' \langle Y_1 \rangle, \quad |Y_1| \leq m_1 d_0.$$

Now recall Theorem 1.1: this associates to W a characteristic subgroup W_0 such that $W^{(3)}W_0/W_0$ is semisimple. Applying Theorem 1.1 to the soluble group $W/W^{(3)}$, we find a set $Y_2 \subseteq Y_1^W \subseteq G_w$ such that

$$W = W^{(3)} \langle Y_2 \rangle, \quad |Y_2| \leq h_1 = m_1 d_0 \cdot f_0(m_1 d_0, d_0).$$

Applying Proposition 5.2 to the semisimple group $W^{(3)}W_0/W_0$, we can write each of the d_0 generators of W as an element of W_0 times a product of $m_0(w)$ w -values times an element of $\langle Y_2 \rangle$; thus there exists $Y_3 \subseteq G_w \cup Y_2 = G_w$ such that $Y_3 \supseteq Y_2$ and

$$W = W_0 \langle Y_3 \rangle, \quad |Y_3| \leq h_2 = h_1 + m_0(w) d_0.$$

Now Applying Theorem 1.1 to W we may infer that $W = \langle Y \rangle$ where $Y \subseteq Y_3^W \subseteq G_w$ and $|Y| \leq h_3 = h_2 \cdot f_0(h_2, d_0)$.

Put $\tilde{Y} = Y \cup Y^{-1}$. It then follows by Theorem 1.2 that

$$\begin{aligned} W' &= \left(\prod_{y \in \tilde{Y}} [W, y] \right)^{*f_1(2h_3, h_3)} \\ &\subseteq G_w^{*4h_3 f_1(2h_3, h_3)}, \end{aligned}$$

since $[a, y^{\pm 1}]$ is a product of two conjugates of $y^{\pm 1}$ for for any $y \in Y$ and $a \in W$. With (34) this shows that w has width $f(w, d) = 4h_3 f_1(2h_3, h_3) + m_1$ in G . ■

Now let G be a d -generator profinite group. Suppose that $m_w(Q) \leq m < \infty$ for every continuous finite quotient Q of G . Then

$$w(G)N = G_w^{*m} \cdot N$$

for every open normal subgroup N of G . But G_w^{*m} is a closed subset of G , because $w : G^{(k)} \rightarrow G$ is continuous; therefore

$$w(G) \subseteq \bigcap_N G_w^{*m} \cdot N = G_w^{*m} \subseteq G_w,$$

so $w(G) = G_w^{*m}$ is a closed subgroup of G .

If also w is d -bounded, then $\beta(w, Q) \leq \beta_w(d) = \beta$, say, for every continuous finite quotient Q of G . Thus

$$|G : w(G)N| \leq \beta$$

for each open normal subgroup N of G . Choosing an open normal subgroup M for which $|G : w(G)M|$ is maximal we infer (given that $w(G)$ is closed) that

$$w(G) = \bigcap_N w(G)N = w(G)M.$$

Thus $w(G)$ is an open subgroup of G .

Theorem 5.3 now gives

Theorem 5.4 *Let G be a d -generator profinite group and w a d -bounded word. Then the verbal subgroup $w(G)$ is open in G .*

This shows that $w(G)$ is open whenever G is a finitely generated profinite group and w is any non-commutator word, the main result of [NSP]. The point of our clumsier formulation is that the theorem as stated is independent of the Restricted Burnside Problem. The following elementary observation is proved on page 173 of [NS]:

Lemma 5.5 *Let Q be a finite group and let $d \in \mathbb{N}$. Then there exists a d -bounded word w such that $w(Q) = 1$.*

It is now easy to deduce the main result of [NS]:

Theorem 5.6 *If G is a finitely generated profinite group then every subgroup of finite index in G is open.*

Proof. Let H be a subgroup of finite index in the d -generator profinite group G . Then H contains a normal subgroup N of finite index in G . Lemma 5.5 provides a d -bounded word w such that $w(G/N) = 1$. Then

$$H \geq N \geq w(G),$$

and $w(G)$ is open by Theorem 5.4. Therefore H is open. ■

5.3 Verbal subgroups in compact groups

Throughout this subsection, we suppose that G is a compact group and that *the profinite quotient G/G^0 is finitely generated*. Several of the preceding results can be generalized.

Corollary 5.7 *If w is a non-commutator word then $w(G)$ is open in G .*

Proof. Set $q = |\mathbb{Z}/w(\mathbb{Z})|$. Every element of G^0 is a q th power ([HM], Theorem 9.35), so $G^0 \leq w(G)$, and so $w(G/G^0) = w(G)/G^0$. The result follows by the remark following Theorem 5.4. ■

The next corollary follows likewise from Theorem 5.6:

Corollary 5.8 *Every subgroup of finite index in G is open.*

Lemma 5.9 *Suppose that $A \leq Z(G^0)$ is closed and normal in G . Then $[A, G]$ is closed in G .*

Proof. Let us consider A as an additively-written $\Gamma = G/G^0$ -module. By hypothesis, Γ has a dense finitely generated (abstract) subgroup $X = \langle x_1, \dots, x_d \rangle$. Now

$$[A, X] = A(x_1 - 1) + \dots + A(x_d - 1);$$

this is an X -submodule of A , and it is closed in A because A is compact. Therefore $[A, X]$ is a Γ -submodule, because X is dense in Γ . Therefore $C := C_\Gamma(A/[A, X])$ is closed in Γ , and as $X \leq C$ it follows that $C = \Gamma$. Hence $[A, G] = [A, \Gamma] = [A, X]$ is closed. ■

Corollary 5.10 *The derived group G' is closed in G .*

Proof. Let $P = (G^0)'$ denote the derived group of G^0 . Then P is closed, by [HM], Theorem 9.2. So replacing G by G/P we may suppose that G^0 is abelian. Then $[G^0, G]$ is closed by the preceding Lemma, so we may factor it out and reduce to the case where G^0 is central in G . Now according to [HM], theorem 9.41, we have $G = G^0 D$ for some closed profinite subgroup D . Since $D/(D \cap G^0) \cong G/G^0$ is finitely generated, $D = (D \cap G^0)H$ for some finitely generated profinite group H . Then $G' = H'$ is closed by the remark following Theorem 1.6. ■

Remark. It would be interesting to know if $[H, G]$ is closed for every closed normal subgroup H of G . This is true if $G^0 = 1$, by Theorem 1.6, and if $G = G^0$, from the known structure of connected compact groups.

5.4 Quotients of semisimple compact groups

In this subsection we consider a topological group

$$G = \prod_{i \in I} S_i, \tag{35}$$

where I is an index set, and *either*:

- (a) each S_i is a nonabelian finite simple group, *or*
- (b) each S_i is a compact connected simple Lie group

(here, by a ‘simple Lie group’ we mean the analogue of a *quasisimple* finite group: i.e. it may have a non-trivial centre, but is simple modulo the centre and perfect).

Remark: Hofmann and Morris [HM] call a compact connected group G ‘semisimple’ if it is perfect, i.e. if $G = \overline{G'}$, equivalently if $G = G'$ (*loc. cit.* Theorem 9.2). However, this holds if and only if $G = \tilde{G}/C$ where $\tilde{G}$ is a product of compact connected simply-connected simple Lie groups and C is a totally disconnected normal subgroup (*loc. cit.* Theorem 9.19); thus any quotient of G is also a quotient of a group of the form (35).

Theorem 5.11 *Let Q be an infinite quotient of (the underlying abstract group) G . Then $|Q| \geq 2^{\aleph_0}$.*

This depends on the following technical device:

Proposition 5.12 *Let L be (a) a nonabelian finite simple group or (b) a compact connected simple Lie group. In Case (b), let T be a maximal torus of L , in Case (a) let $T = L$. There is a function $\lambda = \lambda_L : T \rightarrow [0, 1]$ with the following properties:*

- (i) $\lambda(s) = 0 \iff s \in Z(L)$;
- (ii) $\lambda(s^{-1}) = \lambda(s^t) = \lambda(s)$ and $\lambda(st) \leq \lambda(s) + \lambda(t)$ for all $s, t \in T$;
- (iii) if $t \in T$ and $\lambda(t) \geq \varepsilon > 0$ then

$$L = (t^L \cup t^{-L})^{*f(\varepsilon)}$$

where $f(\varepsilon) \in \mathbb{N}$ depends only on ε ;

- (iv) in Case (a), $1 \neq s \in L$ implies $\lambda_L(s) \geq \varepsilon(r)$ where $\varepsilon(r) > 0$ depends only on $r = \text{rank}(L)$;
- (va) in Case (a): given $\beta, \varepsilon \in (0, 1)$, there exists $s \in L$ with

$$|\lambda_L(s) - \beta| < \varepsilon,$$

provided that $\text{rank}(L) \geq n(\varepsilon)$, where $n(\varepsilon)$ depends only on ε ;

- (vb) in Case (b): for each $\beta \in [0, 1]$ there exists $s \in T$ with $\lambda_L(s) = \beta$.

Recall that $\text{rank}(L)$ means the (untwisted) Lie rank of L if L is of Lie type, n if $L \cong \text{Alt}(n)$, and 0 otherwise. The proof is postponed to the following subsections.

Given an ultrafilter $\mathcal{U}$ on I , one defines the *ultralimit* of a bounded family $(a_i)_{i \in I}$ of real numbers to be the unique number $\alpha = \lim_{\mathcal{U}} a_i$ such that

$$\epsilon > 0 \implies \{i \in I \mid |a_i - \alpha| < \epsilon\} \in \mathcal{U}$$

(cf. [KL], Section 3.1). We remark that if $\mathcal{U}$ is the principal ultrafilter $\mathcal{U}(j)$ over some element $j \in I$, then $\lim_{\mathcal{U}} a_i = a_j$.

In Case (a), set $T_i = S_i$ for each i ; in Case (b), we choose a maximal torus T_i in S_i . In either case, let $G_{\bullet} = \prod_{i \in I} T_i$. Now define a function $h_{\mathcal{U}} : G_{\bullet} \rightarrow [0, 1]$ by

$$h_{\mathcal{U}}(\mathbf{g}) = \lim_{\mathcal{U}} \lambda_{S_i}(g_i) \text{ for } \mathbf{g} = (g_i)_i.$$

The analogue of property (ii) obviously holds for the function $h_{\mathcal{U}}$. This implies that the set

$$K_{\mathcal{U}} := h_{\mathcal{U}}^{-1}(0) \tag{36}$$

is a normal subgroup of $G_{\bullet}$, and that $h_{\mathcal{U}}$ is constant on the cosets of $K_{\mathcal{U}}$.

For a subset J of I we set

$$N(J) = \prod_{i \in J} Z_j \times \prod_{i \in I \setminus J} S_i,$$

the kernel of the projection $G \rightarrow \prod_{j \in J} S_j/Z_j$, where $Z_j = Z(S_j)$. Each $N(J)$ is a closed normal subgroup of G .

Now we can prove Theorem 5.11. Let $Q = G/H$ where H is a normal subgroup of infinite index in G . Suppose we are in Case (b) (Lie groups); if $H \leq N(j)$ then Q maps onto S_j/Z_j and the result is clear. Suppose we are in Case (a), and let J be the set of indices j such that $H \leq N(j)$. Then $|Q| = |G/N(J)| |G_1/H_1|$ where $G_1 = \prod_{i \in I \setminus J} S_i$ and H_1 denotes the projection of H into G_1 . If J is infinite, then $G/N(J)$ is an infinite profinite group and again the result is clear. If J is finite, then H_1 has infinite index in G_1 , and we can replace G by G_1 .

Thus in any case, we may assume that $H \not\leq N(j)$ for every $j \in I$. We shall show that in this case,

(*) *There exists a non-principal ultrafilter $\mathcal{U}$ on I such that $H_\bullet := H \cap G_\bullet \leq K_\mathcal{U}$;*

(**) $|G_\bullet/K_\mathcal{U}| \geq 2^{\aleph_0}$.

(Recall that $G_\bullet = G$ in Case (a).)

Proof of ().**

Case 1. The S_i are finite simple groups, and for some $m \in \mathbb{N}$, the set

$$D = D(m) = \{i \mid \text{rank}(S_i) \leq m\}$$

belongs to $\mathcal{U}$. Then $N(D) \leq K_\mathcal{U}$, so $G/K_\mathcal{U} \cong G_1/K_{\mathcal{U}_1}$ where $G_1 = \prod_{i \in D} S_i$ and $\mathcal{U}_1$ is the restriction of $\mathcal{U}$ to D . Now property (iv) of the functions λ_{S_i} implies that $\mathbf{g} \in K_{\mathcal{U}_1}$ precisely when the set $\{i \in D \mid g_i = 1\}$ belongs to $\mathcal{U}_1$. Therefore the quotient $G_1/K_{\mathcal{U}_1}$ coincides with the ultraproduct $\prod_{i \in D} S_i/\mathcal{U}_1$. But an ultraproduct of finite sets is either finite or has cardinality at least $2^{\aleph_0}$ ([FMS], Theorem 1.31), and (**) follows.

Case 2. The S_i are finite simple groups and $D(m) \notin \mathcal{U}$ for each $m \in \mathbb{N}$. Let $\beta \in (0, 1)$. For each $i \in I$ we choose $g_i \in S_i$ so as to minimize

$$|\lambda_{S_i}(g_i) - \beta| = \varepsilon_i,$$

say. Property (va) ensures that for any $\varepsilon > 0$, we have $\varepsilon_i < \varepsilon$ whenever $\text{rank}(S_i) \geq n(\varepsilon)$. We claim that $h_\mathcal{U}(\mathbf{g}) = \beta$. Indeed, suppose that $h_\mathcal{U}(\mathbf{g}) = \beta' \neq \beta$, and put $\varepsilon = |\beta' - \beta|$. Then

$$|\lambda_{S_i}(g_i) - \beta'| < \varepsilon/2 \implies \varepsilon_i = |\lambda_{S_i}(g_i) - \beta| > \varepsilon/2 \implies i \in D(m)$$

where $m = n(\varepsilon/2)$; thus $D(m)$ contains a member of $\mathcal{U}$ and so $D(m) \in \mathcal{U}$, a contradiction.

It follows that $h_{\mathcal{U}}(G) = [0, 1]$. Since $h_{\mathcal{U}}$ is constant on cosets of $K_{\mathcal{U}}$ this now implies that $G/K_{\mathcal{U}}$ has the cardinality of $[0, 1]$, and $(**)$ follows.

Case 3. The S_i are connected simple Lie groups. Let $\beta \in (0, 1)$. Using Property (vb), choose $g_i \in T_i$ with $\lambda_{S_i}(g_i) = \beta$ for each i . Then $\mathbf{g} = (g_i) \in G_{\bullet}$ and $h_{\mathcal{U}}(\mathbf{g}) = \beta$; and $(**)$ follows as in the preceding case.

Proof of (*).

H is a normal subgroup of infinite index in G , and $H \not\leq N(j)$ for any $j \in I$. For $\mathbf{t} = (t_i)_i \in H_{\bullet}$ and $\epsilon > 0$ put

$$A(\mathbf{t}, \epsilon) = \{i \in I \mid \lambda_{S_i}(t_i) < \epsilon\},$$

and let U be the collection of all subsets $A(\mathbf{t}, \epsilon)$ with $\mathbf{t} \in H_{\bullet}$ and $\epsilon > 0$.

We claim that *every finite subset of U has nonempty intersection*. Indeed, suppose that

$$A(\mathbf{t}_1, \epsilon_1) \cap A(\mathbf{t}_2, \epsilon_2) \cap \dots \cap A(\mathbf{t}_k, \epsilon_k) = \emptyset.$$

Put $\epsilon = \min_i \{\epsilon_i\}$ and suppose that $\mathbf{t}_i = (t_{i,j})_j$ with $t_{i,j} \in T_j$.

Then for each index $j \in I$ there is some $i \leq k$ such that $j \notin A(\mathbf{t}_i, \epsilon)$, so $\lambda_{S_j}(t_{i,j}) \geq \epsilon$. Now (iii) gives

$$S_j = \left(t_{i,j}^{S_j} \cup t_{i,j}^{-S_j} \right)^{*n},$$

where $n = f(\epsilon)$. Considering independently each coordinate $j \in I$ we see that

$$G = \prod_{i=1}^k (\mathbf{t}_i^G \cup \mathbf{t}_i^{-G})^{*n} \subseteq H,$$

a contradiction.

On the other hand, *the intersection of the collection U is empty*. Let T_j^* denote the projection of $H_{\bullet}$ into S_j . If j belongs to every member of U then $\lambda_{S_j}(t) = 0$ for every $t \in T_j^*$, whence $T_j^* \leq Z(S_j)$ by property (i). Since the conjugates of T_j^* generate the projection of H into S_j , this implies that $H \leq N(j)$, contrary to hypothesis.

Now a standard application of Zorn's lemma establishes the existence of a non-principal ultrafilter $\mathcal{U}$ on I containing U . From the definition of U it follows that $h_{\mathcal{U}}(\mathbf{t}) = 0$ for all $\mathbf{t} \in H_{\bullet}$, and $(*)$ follows.

5.4.1 The profinite case

In Case (a) we can say rather more:

Theorem 5.13 *Suppose that $G = \prod_{i \in I} S_i$ where each S_i is a finite (non-abelian) simple group. Then*

- every proper normal subgroup of G is contained in a maximal one;
- the maximal proper normal subgroups of G are precisely the subsets $K_{\mathcal{U}}$ for ultrafilters $\mathcal{U}$ on I ;
- the normal subgroup $K_{\mathcal{U}}$ is closed in G if and only if $\mathcal{U}$ is principal.

Proof. If $\mathcal{U} = \mathcal{U}(j)$ is principal then $K_{\mathcal{U}} = N(j)$ is a closed maximal normal subgroup. If $\mathcal{U}$ is non-principal, then $K_{\mathcal{U}}$ has infinite index in G , by (**). We claim that in this case too, $K_{\mathcal{U}}$ is a maximal normal subgroup. Suppose that $\mathbf{g} = (g_i)_i \in G$ is not in $K = K_{\mathcal{U}}$. This means that $h_{\mathcal{U}}(\mathbf{g}) > 0$, which in turn implies that for some $\alpha > 0$ the set

$$A = \{i \in I \mid \lambda_{S_i}(g_i) > \alpha\}$$

belongs to $\mathcal{U}$.

Now if $i \in A$, we see from (iii) in Proposition 5.12 that

$$S_i = (g_i^{S_i} \cup g_i^{-S_i})^{*n}$$

where $n = f(\alpha)$. It follows that

$$G = N(A) \cdot (\mathbf{g}^G \cup \mathbf{g}^{-G})^{*n}.$$

As $\mathcal{U}$ is a filter and $A \in \mathcal{U}$ it is easy to see that $N(A) \leq K$, and so

$$G = K (\mathbf{g}^G \cup \mathbf{g}^{-G})^{*n} \subseteq K \langle \mathbf{g}^G \cup \mathbf{g}^{-G} \rangle.$$

Since $\mathbf{g}$ was an arbitrary element of $G \setminus K$ it follows that G/K is simple.

Now suppose that H is any proper normal subgroup of G . Then either $H \leq N(j) = K_{\mathcal{U}(j)}$ for some $j \in I$, or (*) provides a non-principal ultrafilter $\mathcal{U}$ such that $H \leq K_{\mathcal{U}}$.

It remains only to observe that if $\mathcal{U}$ is a non-principal ultrafilter then $K_{\mathcal{U}}$ contains the restricted direct product of the S_i , which is dense in G , and so $K_{\mathcal{U}}$ cannot be closed. ■

5.4.2 The connected case: automorphisms

The material in this subsection will only be needed for the proof of Theorem 5.27 in Subsection 5.6. We consider $G = \prod_{i \in I} S_i$ where I is an infinite set and each S_i is a compact connected simple Lie group. In this case, our functions $h_{\mathcal{U}}$ were only defined on $G_{\bullet} = \prod_{i \in I} T_i$, which depends on a choice of maximal torus T_i in each S_i . Suppose that in each S_i we choose maximal tori $T_i^{(l)}$, $l = 1, \dots, d$. Let $\lambda_{S_i}^{(l)} : T_i^{(l)} \rightarrow [0, 1]$ be as in Proposition 5.12, put $\mathbf{T}^{(l)} = \prod_{i \in I} T_i^{(l)}$, and define $h_{\mathcal{U}}^{(l)} : \mathbf{T}^{(l)} \rightarrow [0, 1]$ and $K_{\mathcal{U}}^{(l)} := h_{\mathcal{U}}^{-1}(0) \leq \mathbf{T}^{(l)}$ as before, using the maps $\lambda_{S_i}^{(l)}$. A subgroup of the form $\mathbf{T}^{(l)}$ will be called a ‘maximal pro-torus’ of G (cf. [HM]). We will write λ_i for λ_{S_i} where the meaning is clear.

Lemma 5.14 *Let H be a proper normal subgroup of G with $H \not\subseteq N(j)$ for all $j \in I$. Then there exists a non-principal ultrafilter $\mathcal{U}$ on I such that $H_{(l)} := H \cap \mathbf{T}^{(l)} \leq K_{\mathcal{U}}^{(l)}$ for $l = 1, \dots, d$.*

Proof. For $\mathbf{t} \in H_{(l)}$ and $\epsilon > 0$ define $A^{(l)}(\mathbf{t}, \epsilon)$ as in the proof of (*), above, using $\lambda^{(l)}$ in place of λ . Let $U^{(l)}$ be the collection of all subsets $A^{(l)}(\mathbf{t}, \epsilon)$ with $\mathbf{t} \in H_{(l)}$ and $\epsilon > 0$. As above, it will suffice to show that every finite subcollection of $U^{(1)} \cup \dots \cup U^{(d)}$ has non-empty intersection. Arguing as before, we see that if

$$\bigcap_{l=1}^d \left(A^{(l)}(\mathbf{t}_1^{(l)}, \epsilon_{l1}) \cap A^{(l)}(\mathbf{t}_2^{(l)}, \epsilon_{l2}) \cap \dots \cap A^{(l)}(\mathbf{t}_k^{(l)}, \epsilon_{lk}) \right) = \emptyset,$$

then for each $j \in I$ there exist $l \leq d$ and $i \leq k$ such that $\lambda_{S_j}^{(l)}(t_{i,j}^{(l)}) \geq \epsilon$ where $\epsilon = \min \epsilon_{l'j'}$. As before this yields the contradiction

$$G = \prod_{l=1}^d \prod_{i=1}^k \left(\mathbf{t}_i^{(l)G} \cup \mathbf{t}_i^{(l)-G} \right)^{*n} \subseteq H.$$

■

Now let y be a continuous automorphism of G . The action of y induces a permutation $y \cdot$ on the index set I , so that $S_i^y = S_{iy \cdot}$ for each i . Let $\mathcal{C}$ denote the set of orbits of $\langle y \cdot \rangle$ on I , and for each $J \in \mathcal{C}$ pick $i(J) \in J$. Then

$$\prod_{i \in J} S_i = \begin{cases} \prod_{n \in \mathbb{Z}} S^{y^n} & (J \text{ infinite}) \\ \prod_{n=0}^{e-1} S^{y^n} & (|J| = e < \infty) \end{cases}$$

where $S = S_{i(J)}$. Choose a maximal torus $T_{i(J)}$ in $S_{i(J)}$, and for $i = i(J)y^n$ (where $0 \leq n < e$ if $|J| = e < \infty$) set $T_i = T_{i(J)}^{y^n}$. Thus $\mathbf{T} = \prod_{i \in I} T_i$ becomes a maximal pro-torus in G , and $\mathbf{T}$ is ‘almost’ y -invariant, in the following sense. For each $J \in \mathcal{C}$ with $|J| < \infty$ put $l(J) = i(J)y^{-1}$, and set

$$Z = \{l(J) \mid J \in \mathcal{C}, |J| < \infty\},$$

$$\mathbf{T}(Z) = \{\mathbf{t} = (t_i) \in \mathbf{T} \mid t_i = 1 \ \forall i \in Z\};$$

for $i \notin Z$ we may identify $S_{iy \cdot}$ with S_i via the action of y , and then for $\mathbf{t} = (t_i) \in \mathbf{T}(Z)$ we have

$$(\mathbf{t}^y)_{iy \cdot} = t_i \ \forall i \in I, \tag{37}$$

so $\mathbf{T}(Z)^y \leq \mathbf{T}$.

Set $Z^c = I \setminus Z$. For $\alpha \in [0, 1]$ and $\epsilon > 0$ define

$$A(\mathbf{t}, \alpha, \epsilon) = \{i \in I \mid |\lambda_{S_i}(t_i) - \alpha| < \epsilon\}.$$

Lemma 5.15 *Let $\mathcal{U}$ be a non-principal ultrafilter on I with $Z^c \in \mathcal{U}$, and put $\mathcal{U}' = \mathcal{U}|_{Z^c}$. Then*

$$\mathcal{U}' = \{A(\mathbf{t}, 1/2, 1/4) \mid \mathbf{t} \in \mathbf{T}(Z), h_{\mathcal{U}}(\mathbf{t}) = 1/2\}.$$

Proof. Let $\mathcal{V}$ denote the family of sets on the right-hand side of the equation. Then $\mathcal{V} \subseteq \mathcal{U}'$ by the definition of $h_{\mathcal{U}}(\mathbf{t})$.

Now suppose that $Y \subseteq Z^c$ and $Y \in \mathcal{U}$. Choose $t_i \in T_i$ so that

$$\begin{aligned} t_i &= 1 \text{ for } i \in Z \\ \lambda_i(t_i) &= 1/2 \text{ for } i \in Y \\ \lambda_i(t_i) &= 1 \text{ for } i \notin Y \cup Z. \end{aligned}$$

Then $\mathbf{t} = (t_i) \in \mathbf{T}(Z)$ and $A(\mathbf{t}, \frac{1}{2}, \epsilon) = Y$ for every $\epsilon \in (0, \frac{1}{2}]$, so $h_{\mathcal{U}}(\mathbf{t}) = \frac{1}{2}$. Therefore $Y \in \mathcal{V}$. Thus $\mathcal{U}' \subseteq \mathcal{V}$. ■

Lemma 5.16 *Suppose that $Z^c \in \mathcal{U}$ and that $\mathbf{t}^{-1}\mathbf{t}^y \in K_{\mathcal{U}}$ for all $\mathbf{t} \in \mathbf{T}(Z)$. Then $\mathcal{U}^{y\cdot} = \mathcal{U}$.*

Proof. Let $X \in \mathcal{U}$. Then $X \supseteq X \cap Z^c = A(\mathbf{t}, \frac{1}{2}, \frac{1}{4})$ for some $\mathbf{t} \in \mathbf{T}(Z)$ with $h_{\mathcal{U}}(\mathbf{t}) = \frac{1}{2}$. Now

$$\begin{aligned} h_{\mathcal{U}}(\mathbf{t}^y) &= h_{\mathcal{U}}(\mathbf{t} \cdot \mathbf{t}^{-1}\mathbf{t}^y) \leq h_{\mathcal{U}}(\mathbf{t}) + h_{\mathcal{U}}(\mathbf{t}^{-1}\mathbf{t}^y) = h_{\mathcal{U}}(\mathbf{t}), \\ h_{\mathcal{U}}(\mathbf{t}) &= h_{\mathcal{U}}(\mathbf{t}^{-1}) = h_{\mathcal{U}}(\mathbf{t}^{-1}\mathbf{t}^y \cdot \mathbf{t}^{-y}) \leq h_{\mathcal{U}}(\mathbf{t}^{-1}\mathbf{t}^y) + h_{\mathcal{U}}(\mathbf{t}^{-y}) = h_{\mathcal{U}}(\mathbf{t}^y), \end{aligned}$$

so $h_{\mathcal{U}}(\mathbf{t}^y) = \frac{1}{2}$. Now it follows from (37) that

$$A(\mathbf{t}, 1/2, 1/4)^{y\cdot} = A(\mathbf{t}^y, 1/2, 1/4) = B,$$

say, and $B \in \mathcal{U}$ since $h_{\mathcal{U}}(\mathbf{t}^y) = \frac{1}{2}$. Therefore $X^{y\cdot} \supseteq B \in \mathcal{U}$ and so $X^{y\cdot} \in \mathcal{U}$. Thus $\mathcal{U}^{y\cdot} \subseteq \mathcal{U}$, and the result follows since $\mathcal{U}^{y\cdot}$ is an ultrafilter. ■

Lemma 5.17 *If $\mathcal{U}^{y\cdot} = \mathcal{U}$ then $\text{fix}(y\cdot) \in \mathcal{U}$.*

Proof. Here $\text{fix}(y\cdot)$ denotes the set of fixed points of $y\cdot$. We can partition I as

$$I = A_1 \dot{\cup} A_2 \dot{\cup} A_3 \dot{\cup} \text{fix}(y\cdot)$$

where $A_i^{y\cdot} \cap A_i = \emptyset$ for $i = 1, 2, 3$. To see this, it suffices to partition each $\langle y\cdot \rangle$ -orbit J of length at least 2 into three pieces J_i such that $J_i^{y\cdot} \cap J_i = \emptyset$. Identifying J with $\mathbb{Z}$ or with $(1, 2, \dots, e)$ where $y\cdot$ takes i to $i + 1 \pmod{e}$, let

$$\begin{aligned} J_1 &= 2\mathbb{Z}, \quad J_2 = 2\mathbb{Z} + 1, \quad J_3 = \emptyset \text{ if } |J| = \infty; \\ J_1 &= 2\mathbb{Z} \cap J, \quad J_2 = (2\mathbb{Z} + 1) \cap J, \quad J_3 = \emptyset \text{ if } |J| \text{ is even}; \\ J_1 &= \{2, \dots, 2n\}, \quad J_2 = \{1, \dots, 2n - 1\}, \quad J_3 = \{2n + 1\} \text{ if } |J| = 2n + 1. \end{aligned}$$

Then set $A_i = \cup_{J \in \mathcal{C}} J_i$ for $i = 1, 2, 3$.

If $\mathcal{U}^{y\cdot} = \mathcal{U}$ then $A_i \notin \mathcal{U}$ for each i , since $\emptyset \notin \mathcal{U}$. Therefore $A_i^c \in \mathcal{U}$ for each i , whence

$$\text{fix}(y\cdot) = A_1^c \cap A_2^c \cap A_3^c \in \mathcal{U}.$$

(We are grateful to Martin Kassabov for pointing us to this lemma, which suggested the possibility of Proposition 5.19, below.) ■

Lemma 5.18 *Suppose that $Z \in \mathcal{U}$ and that $\mathbf{t}^{-1}\mathbf{t}^y \in K_{\mathcal{U}}$ for all $\mathbf{t} \in \mathbf{T}(Z)$. Then $\text{fix}(y) \in \mathcal{U}$.*

Proof. If J is an orbit of $\langle y \rangle$ of length at least 2, choose $t_J \in T_{i(J)}$ with $\lambda_{i(J)}(t_J) = 1$. Then set

$$\begin{aligned} t_{i(J)y^n} &= t_J^{y^n} \quad \forall n \in \mathbb{Z} \text{ if } J \text{ is infinite,} \\ t_{i(J)y^n} &= t_J^{y^n} \quad (0 \leq n \leq e-2), \quad t_{i(J)} = 1 \text{ if } |J| = e < \infty; \end{aligned}$$

and set $t_i = 1$ for each $i \in \text{fix}(y)$ (recall that $l(J) = i(J)y^{(e-1)}$). Then $\mathbf{t} = (t_i) \in \mathbf{T}(Z)$, and whenever $\infty > |J| \geq 2$ we have

$$(\mathbf{t}^{-1}\mathbf{t}^y)_{l(J)} = t_J.$$

Now $\mathbf{t}^{-1}\mathbf{t}^y \in K_{\mathcal{U}}$ implies that $A(\mathbf{t}^{-1}\mathbf{t}^y, 0, \frac{1}{2}) \in \mathcal{U}$; consequently $A(\mathbf{t}^{-1}\mathbf{t}^y, 0, \frac{1}{2}) \cap Z \in \mathcal{U}$. As $Z = \{l(J) \mid 2 \leq |J| < \infty\} \cup \text{fix}(y)$, we see that $A(\mathbf{t}^{-1}\mathbf{t}^y, 0, \frac{1}{2}) \cap Z = \text{fix}(y)$. ■

Proposition 5.19 *Let $y_1, \dots, y_d$ be continuous automorphisms of G and let H be a proper normal subgroup of G with $[G, y_l] \subseteq H$ for each l . Suppose that $H \not\subseteq N(j)$ for all $j \in I$. Then there exists a non-principal ultrafilter $\mathcal{U}$ on I such that*

$$\bigcap_{l=1}^d \text{fix}(y_l) \in \mathcal{U}.$$

Hence $\bigcap_{l=1}^d \text{fix}(y_l)$ is infinite.

Proof. For each l choose a maximal pro-torus $\mathbf{T}^{(l)}$ corresponding to y_l as above, and apply Lemma 5.14 to find a non-principal ultrafilter $\mathcal{U}$ such that $H \cap \mathbf{T}^{(l)} \leq K_{\mathcal{U}}^{(l)}$ for $l = 1, \dots, d$. Now the last three lemmas show that $\text{fix}(y_l) \in \mathcal{U}$ for each l , and the result follows. ■

5.4.3 Proposition 5.12, finite case

Now L is a finite simple group. We define

$$\lambda(s) = \frac{\log |s^L|}{\log |L|}.$$

Properties (i) and (ii) are clear, and (iii) follows from Proposition 1.22. (iv) follows from Proposition 1.23.

It remains to establish property (v). Given $\beta, \varepsilon \in (0, 1)$, we have to show that provided $\text{rank}(L)$ is sufficiently large, there exists $g \in L$ such that

$$\frac{\log |C_L(g)|}{\log |L|} \in (\alpha - \varepsilon, \alpha + \varepsilon)$$

where $\alpha = 1 - \beta$. As we only need to consider groups of large rank, we may suppose that L is either alternating or a classical group.

If $L = \text{Alt}(n)$, take g to be an even cycle of length $l \sim \beta n$ in $\text{Alt}(n)$. Note that $|C_L(g)|$ is roughly $l \cdot \tilde{l}!/2$ where $\tilde{l} \sim \alpha n$. By Stirling's formula, $\log(n!) \sim n \log n$ and hence $\log(l \cdot \tilde{l}!/2) \sim \alpha \log(n!/2)$ as $n \rightarrow \infty$.

If L is a simple classical group, consider the corresponding universal quasi-simple classical group $\tilde{L}$ acting on its natural module V over a finite field of size q equipped with a bilinear form f (symmetric, sesquilinear, alternating or just equal to 0 in case L has type PSL_n). Note that $\dim(V) \rightarrow \infty$ as $\text{rank}(L) \rightarrow \infty$. We have $L = \tilde{L}/Z$ where Z is the centre of $\tilde{L}$; and if $g = \tilde{g}Z \in L$ with $\tilde{g} \in \tilde{L}$ then

$$|g^L| \leq |\tilde{g}^{\tilde{L}}| \leq |Z||g^L|.$$

Since Z has asymptotically negligible size compared to L it is enough to find an element $\tilde{g} \in \tilde{L}$ with $\log |C_{\tilde{L}}(\tilde{g})| \sim \alpha \log |\tilde{L}|$.

We can decompose V as $V_0 \oplus V_1 \oplus V_2$ so that:

- $\dim V_0$ is about $\sqrt{\alpha} \dim V$, and $\dim V_1 = \dim V_2$,
- $V_1 \oplus V_2$ is orthogonal to V_0 , and
- The form f is nondegenerate on both V_0 and $V_1 \oplus V_2$ and is isotropic on V_1 and on V_2

Let $\tilde{g} \in \tilde{L}$ be equal to the identity on V_0 and act on each of V_1 and V_2 as a cyclic transformation without fixed vectors. In other words there is a vector $v_i \in V_i$, ($i = 1, 2$) such that $v_i, \tilde{g}v_i, \tilde{g}^2v_i, \dots$ is a basis for V_i .

Now $C_{\tilde{L}}(\tilde{g})$ contains the classical group H on V_0 preserving f , and by the choice of $\dim V_0$ we have $\log |H| / \log |\tilde{L}| \sim (\dim V_0 / \dim V)^2$ which tends to α as $\dim V \rightarrow \infty$.

On the other hand if $s \in \tilde{L}$ commutes with $\tilde{g}$ then s must stabilize V_0 , the fixed space of $\tilde{g}$. Since V_1 and V_2 are cyclic modules for $\tilde{g}$, the action of s on V_1 and V_2 is determined by $s \cdot v_1$ and $s \cdot v_2$. Hence s is completely known from its restriction to V_0 and from the two vectors $sv_1, sv_2 \in V$. Denote by $\text{Gf}(V_0)$ the subgroup of $\text{GL}(V_0)$ which preserves f . We have $|\text{Gf}(V_0)| \leq q |H|$.

Therefore

$$|H| \leq C_{\tilde{L}}(\tilde{g}) \leq |\text{Gf}(V_0)||V|^2 \leq q^{1+2 \dim V} |H|$$

which gives

$$\log |C_{\tilde{L}}(\tilde{g})| / \log |\tilde{L}| \sim \log |H| / \log |\tilde{L}| \rightarrow \alpha$$

as $\dim V$ tends to infinity.

5.4.4 Proposition 5.12, connected case

We shall need some information about the tori and roots of compact simple Lie groups; see for example [Bu], Chapter 19, [HM], Chapter 6. By S^1 we shall denote the group of complex numbers of absolute value 1 under multiplication. It is a compact torus of dimension 1.

Let L be a compact simple Lie group with centre Z (possibly nontrivial). Let T be a maximal torus of L (this is unique up to conjugacy). Every element of L is conjugate to an element of T . Let Φ be a set of roots with respect to T . We choose and fix a set of fundamental roots $\Pi = \{\beta_1, \dots, \beta_r\}$; r is the rank of L . Every root $\alpha \in \Phi$ corresponds to a character $T \rightarrow S^1$ which we will also denote by α . We have

$$\bigcap_{i=1}^r \ker \beta_i = Z.$$

There is also a cocharacter $h_\alpha : S^1 \rightarrow T$ such that $\alpha(h_\alpha(\mu)) = \mu^2$ for all $\mu \in S^1$. For every pair $(\pm\alpha)$ of opposite roots of Φ there is a homomorphism $f_\alpha : \text{SU}(2) \rightarrow L$ such that h_α is the restriction of f_α to the diagonal subgroup $\text{diag}(\mu, \mu^{-1})$ of $\text{SU}(2)$ (and $h_{-\alpha} = h_\alpha^{-1}$). Let $S_\alpha = S_{-\alpha}$ be the image of $\text{SU}(2)$ in L under f_α . Then S_α is either $\text{SU}(2)$ or $\text{PSU}(2) \cong \text{SO}(3)$. Moreover S_α commutes elementwise with the closed subgroup $T_\alpha := \{g \in T \mid \alpha(g) = 1\}$ of T , and the central product $S_\alpha T_\alpha$ contains T .

Now we have to define $\lambda : T \rightarrow [0, 1]$ so that properties (i) – (iii) and (v) of Proposition 5.12 hold.

We can write a complex number $\mu \in S^1$ in a unique way as $\mu = e^{i\theta}$ with $\theta \in (-\pi, \pi]$. Set $l(\mu) := |\theta|$. We shall refer to $l(\mu)$ as the *angle* of μ .

Definition. For an element $g \in T$ define

$$\lambda(g) = \frac{1}{\pi r} \sum_{i=1}^r l(\beta_i(g))$$

Clearly $\lambda(g)$ is the same as $\lambda(\bar{g})$ for $\bar{g} = gZ$, if λ is defined taken with respect to the torus T/Z of L/Z .

It is also clear that (i) $\lambda(g) = 0$ if and only if $g \in Z$, and (ii) $\lambda(h_1) = \lambda(h_1^{-1})$ and $\lambda(h_1 h_2) \leq \lambda(h_1) + \lambda(h_2)$ for any $h_1, h_2 \in T$. Since $l(\mu)$ takes all values in $[0, \pi]$ and T is a torus, we see that $\lambda(T) = [0, 1]$, which is property (v).

Example: If $L = \text{SU}(2)$ and g is an element of the diagonal subgroup of L with eigenvalues μ and μ^{-1} then $l(g)$ is the angle of μ^2 . From here and the isomorphism $\text{PSU}(2) \cong \text{SO}(3)$ we see that if $g \in \text{SU}(2)$ then $\lambda(g)$ is $|\theta|/\pi$ where θ is the angle of the image $\bar{g} \in \text{PSU}(2) = \text{SO}(3)$ considered as a rotation of $\mathbb{R}^3$.

Property (iii) follows from

Lemma 5.20 *There is an absolute constant $C > 0$ such that if $g \in T$ and $C/(\lambda(g))^2 < M \in \mathbb{N}$ then $K^{*M} = L$, where $K = g^L \cup g^{-L}$.*

First we consider a special case:

Lemma 5.21 *If $L = \mathrm{SU}(2)$ and $g \in L$ with $\lambda(g) = \epsilon > 0$ then every element of L is a product of $N = \lceil 2/\epsilon \rceil$ conjugates of g . Moreover $L = [L, g]^{*N}$.*

Proof. Consider the realization of $\mathrm{SU}(2) < \mathrm{GL}_2(\mathbb{C})$ by unitary matrices:

$$\mathrm{SU}(2) = \left\{ \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} \mid |a|^2 + |b|^2 = 1 \right\}.$$

The conjugacy class of an element $h \in \mathrm{SU}(2)$ is uniquely determined by its trace $\mathrm{tr}(h) \in [-2, 2]$. Write $\mathrm{tr}(g) = 2 \cos \gamma$ with $\gamma \in [0, \pi]$; then for fundamental roots α we have $\alpha(g) = e^{\pm 2i\gamma}$, and so $\lambda(g) = 2\gamma/\pi$ if $\gamma \in [0, \pi/2]$, $\lambda(g) = 2(\pi - \gamma)/\pi$ otherwise. Of course $\lambda(g) = \lambda(-g)$ and $(g^L)^{*N} = L$ is equivalent to $((-g)^L)^{*N} = L$. So by replacing g with $-g$ if necessary we may assume that $\lambda(g) = 2\gamma/\pi = \epsilon > 0$ and $\gamma = \pi\epsilon/2 \in (0, \pi/2]$. Now a direct computation shows that if $h \in L$ is a diagonal element with $\mathrm{tr}(h) = 2 \cos \theta$ then for any $\theta_1 \in [\theta - \gamma, \theta + \gamma]$ we may find a matrix $g' \in g^L$ (i.e. such that $\mathrm{tr}(g') = 2 \cos \gamma$) with $\mathrm{tr}(hg') = 2 \cos \theta_1$. This shows that for any integer $m > 1$, any element of L with trace $2 \cos \theta_2$ with $\theta_2 \in [0, m\gamma]$ is a product of m conjugates of g . Taking $N = \lceil 2/\epsilon \rceil$ we have $N\gamma \geq \pi$ and so $(g^L)^{*N} = L$.

This proves the first claim of the Lemma. The second claim follows since $[L, g] = (g^{-1})^L g = g^L \cdot g$ and

$$[L, g]^{*N} = (g^L \cdot g)^{*N} = (g^L)^{*N} g^N = L g^N = L.$$

■

We now consider the general case of Lemma 5.20. It is enough to prove it when L is simply connected, since the definition of λ was the same for L and $L/Z(L)$. We shall assume this from now on.

Let us write H_α for the one-parameter torus $\{h_\alpha(t) \mid t \in S_1\}$ given by the image of the cocharacter h_α . Thus we have $T = H_{\beta_1} \times \cdots \times H_{\beta_r}$. Take an element $g \in L$ with $\lambda(g) = \epsilon > 0$. Then for at least one fundamental root β_j we have $l(\beta_j(g)) \geq \epsilon\pi$. Fix $N = \lceil 2/\epsilon \rceil$ as above.

Case 1: Assume that the rank r of L satisfies $r \leq \max\{10, 4/\epsilon\}$.

In the central product $S_\beta T_\beta$ we can write g as $g = g_1 g_2$ where $g_1 \in S_\beta$ and $g_2 \in T_\beta$. Now S_β is a copy of $\mathrm{SU}(2)$ (not $\mathrm{PSU}(2)$ since L is simply connected), and by Lemma 5.21 we can express any $h \in S_\beta$ as $h = \prod_{i=1}^N [s_i, g_1]$ for some $s_i \in S_\beta$. Then

$$h = \prod_{i=1}^N [s_i, g]$$

and in particular the subgroup $H_\beta \leq S_\beta$ is contained in K^{*2N} . Recall that the Weyl group W acts on H . For a pair of roots γ_1, γ_2 of Φ of the same length there is some element $v \in W$ such that $\gamma_1^v = \gamma_2$ and consequently $H_{\gamma_1}^v = H_{\gamma_2}$.

Moreover, if γ, δ are two roots of different lengths in Φ then γ is in the linear span of roots δ_1 and δ_2 in the orbit of δ under W , and then

$$H_\gamma \leq H_{\delta_1} H_{\delta_2} = H_\delta^{u_1} H_\delta^{u_2} \quad \text{for some } u_1, u_2 \in W.$$

Therefore each of the groups H_{β_i} is contained in K^{*4N} . But T is a product of all the H_{β_i} for $i = 1, \dots, r$ and hence

$$T \subseteq K^{*4rN}.$$

Now the right-hand side is a union of conjugacy classes of L ; since every conjugacy class intersects T we have $L = K^{*4rN} = K^{*M}$ as long as $M \geq 4rN = O(\epsilon^{-2})$, since $r \leq \max\{10, 4/\epsilon\}$.

Case 2: The Lie rank of L exceeds both 10 and $4/\epsilon$. This means that L is a classical Lie group of type A_r, B_r, C_r or D_r . In all these cases we can label the fundamental roots in Π so that $\beta_1, \dots, \beta_{r-1}$ span a root system of type A_{r-1} and the angle between β_i and β_{i+1} is $2\pi/3$ for $i = 1, \dots, r-2$. (This is the labelling on the vertices of the Dynkin diagram of L where we number the vertices on the A_{r-1} part of the diagram consecutively.) The last root β_r may have different length from the others.

Put $\eta = \epsilon/8$. It is immediate that for a subset $\Delta \subseteq \Pi$ of size at least $4\eta r$ we must have $l(\beta_i(g)) \geq \epsilon\pi/2$ for all $i \in \Delta$: otherwise, the average on Π could not be $\epsilon\pi$ since each $l(\beta_i(g)) \leq \pi$. Define

$$\Pi_1 = \{\beta_i \mid 1 \leq i \leq r-1 \text{ and } i \text{ even}\}, \quad \Pi_2 = \Pi \setminus (\Pi_1 \cup \{\beta_r\}).$$

Then each Π_i consists of pairwise orthogonal roots and their union is $\Pi \setminus \{\beta_r\}$.

Observe that $|\Delta| \geq \epsilon r/2 \geq 2$ since $r \geq 4/\epsilon$. Put $\Delta_i = \Pi_i \cap \Delta$. Since $|\Delta_1| + |\Delta_2| \geq |\Delta| - 1 \geq |\Delta|/2 \geq 2\eta r$ we have either $|\Delta_1| \geq \eta r$ or $|\Delta_2| \geq \eta r$. Without loss of generality assume that $|\Delta_1| \geq \eta r$.

The roots in Δ_1 are pairwise orthogonal. The group $Q := \langle T, S_\beta \mid \beta \in \Delta_1 \rangle$ is therefore isomorphic to the central product

$$\left(\prod_{\beta \in \Delta_1} S_\beta \right) \circ T_{\Delta_1},$$

where $T_{\Delta_1} = \{h \in T \mid \beta(h) = 1 \ \forall \beta \in \Delta_1\}$ and $\prod_{\beta \in \Delta_1} S_\beta$ is the direct product of the S_β .

Now if $\beta \in \Delta_1$ we have $l(\beta(g)) \geq \epsilon\pi/2$. Just as in Case 1, working independently in each S_β and using Lemma 5.21 we deduce that

$$\prod_{\beta \in \Delta_1} H_\beta \subseteq [Q, g]^{*N_1} \subseteq K^{2N_1} \tag{38}$$

where $N_1 = \lceil 4/\epsilon \rceil$. We now refer to the following straightforward

Lemma 5.22 *Let Ψ be the set of roots in the root system of type A_n . For an integer $m \leq n/2$ let $X, Y \in \Psi^{(m)}$ be two m -tuples of elements of Ψ each consisting of pairwise orthogonal roots. Then $X = Y^w$ for an element w in the Weyl group of Ψ .*

Proof. This can be done directly from the realization of Ψ and the fact that $W = \text{Sym}(n+1)$. Alternatively it follows by induction on m and using that for any root $\alpha \in \Psi$, the orthogonal complement $\Psi \cap \alpha^\perp$ is a root system of type A_{n-2} . ■

Now the set Π_1 is a union of at most $|\Pi_1|/\eta r + 1$ subsets of size $|\Delta_1|$ and the same holds for Π_2 . Altogether $\Pi_1 \cup \Pi_2$ is a union of at most $r/\eta r + 2 = 1/\eta + 2$ subsets of size $|\Delta_1|$. Using Lemma 5.22 and (38) we see that

$$\prod_{i=1}^{r-1} H_{\beta_i} \subseteq K^{*N_2}$$

where $N_2 = 2 \lceil 1/\eta + 2 \rceil N_1$. Finally $H_{\beta_r} \subseteq K^{*4N}$, and hence $T \subseteq K^{N_2+4N}$. Again, it follows that $L = K^{*M}$ as long as $M \geq N_2 + 4N = O(\epsilon^{-2})$.

5.5 Countable quotients of compact groups

In this subsection, by a *quotient* of a topological group G we mean a quotient of the underlying abstract group, unless stated otherwise. We will be interested in countable quotients: in this subsection, one can always replace ‘countable’ with ‘of cardinality strictly less than $2^{\aleph_0}$ ’.

Until further notice, we assume that G is a compact group such that the profinite quotient G/G^0 is finitely generated (topologically). Recall (Corollary 5.10) that the derived group G' of G is closed; this applies likewise if G is replaced by any open subgroup of G .

The following observation is an immediate consequence of Corollary 5.8:

Corollary 5.23 *If M is a normal subgroup of G and G/M is residually finite then M is closed.*

Indeed, M is an intersection of normal subgroups of finite index, each of which is open.

Suppose to begin with that G is abelian. If G/G^0 is infinite, then G has $\mathbb{Z}_p$ as a quotient for some prime p ; as observed in the introduction, this yields a homomorphism

$$G \rightarrow \mathbb{Z}_p \rightarrow \mathbb{Q}_p \rightarrow \mathbb{Q}$$

with countably infinite image. If $G^0 \neq 1$, then G^0 maps onto a torus T . Let D be the torsion subgroup of T . Then T/D is a divisible torsion-free abelian

group, so a vector space over $\mathbb{Q}$; choosing an epimorphism $T/D \rightarrow \mathbb{Q}$ we obtain an epimorphism (of abstract groups) $G^0 \rightarrow \mathbb{Q}$.

Now suppose that G has an open normal subgroup K such that K/K' is infinite. The preceding remarks shows that K , and therefore also G , has a countably infinite quotient.

A group Q is said to be *FAb* if every virtually-abelian quotient of Q is finite; when Q is a topological group, this refers to *continuous* quotients.

Theorem 5.24 *Let G be a compact group such that G/G^0 is (topologically) finitely generated. Then every countable FAb quotient of G is finite.*

Before giving the proof, let us deduce

Corollary 5.25 *G has a countably infinite quotient if and only if G is not FAb.*

We remark that many familiar compact groups are FAb: among connected groups, these are just the semisimple ones; among profinite groups, examples include $\mathfrak{S}(\mathbb{Z}_p)$ for Chevalley groups $\mathfrak{S}$.

Proof. The remarks above show that if G is not FAb then G has a countably infinite quotient. Suppose conversely that G has a countably infinite quotient G/N . By Theorem 5.24, we may suppose that G/N is virtually abelian, so G has a normal subgroup K of finite index with $K' \leq N \leq K$. Now K is open by Corollary 5.8 and so K' is closed. Thus G/K' is an infinite virtually-abelian continuous quotient of G , so G is not FAb. ■

Proof of Theorem 5.24. Let H be a normal subgroup of G such that G/H is countable and FAb, and suppose that G/H is infinite.

Set $P = (G^0)'$. Then P is closed in G and P is a semisimple connected compact group, hence has no proper countable quotient, by Theorem 5.11 (and the *remark* preceding it). So $H \geq P$, and replacing G by G/P we may suppose that G^0 is abelian.

Since G^0H/H is abelian, G/G^0H must be infinite. Replacing G by G/G^0 and H by G^0H/G^0 , we may suppose that G is a finitely generated profinite group. Put $K = \overline{H}$; then K is open in G , so K is again a finitely generated profinite group. Now $G/K'H$ is virtually abelian and therefore finite. Thus $K'H$ is open by Theorem 5.6, and so $K'H = K$.

Now recall the definition of K_0 (see the Introduction). This is a characteristic closed subgroup of K such that $K^{(3)}K_0/K_0$ is semisimple, where $K/K^{(3)}$ is soluble of derived length at most 3. Since any soluble FAb group is finite, we infer that $G/K^{(3)}K_0H$ is finite, and as before conclude that $K^{(3)}K_0H = K$. Thus K/HK_0 is a countable image of the semisimple group $K^{(3)}K_0/K_0$; so K/HK_0 is finite by Theorem 5.11, and as above it follows that $HK_0 = K$.

Now Corollary 1.8 shows that $H = K$. Hence G/H is finite, a contradiction.

■

Now we consider arbitrary compact groups:

Theorem 5.26 *Let G be a compact group and N a normal subgroup of (the underlying abstract group) G . If G/N is finitely generated then G/N is finite.*

Proof. Suppose that G/N is finitely generated and infinite. Then $G = N \langle X \rangle$ for some finite subset X . Let $K = \overline{\langle X \rangle}$ be the subgroup topologically generated by X . Then $G/N \cong K/(K \cap N)$, so replacing G by K we may suppose that G is topologically finitely generated. Now G/N is countable, hence by Theorem 5.24 there exists $M \triangleleft G$ with $M \geq N$ such that G/M is infinite and virtually abelian. But a finitely generated virtually abelian group is residually finite; hence M is closed in G , by Corollary 5.23. Thus G/M is both countably infinite and compact, a contradiction. ■

5.6 Dense normal subgroups

Let G be a compact group such that G/G^0 is (topologically) finitely generated. If $N \triangleleft G$ and G/N is countable then the closure $\overline{N}$ of N is open in G ; in this case, we say that N is *virtually dense*. Generalizing the preceding subsection, we can ask: under what conditions does G have a virtually dense normal subgroup N of infinite index? Note that N has infinite index if and only if N is not closed, in view of Corollary 5.8.

Suppose that G is abelian. If G/G^0 is infinite, then G/G^0 contains a dense (abstractly) finitely generated subgroup. If G^0 is infinite, then G^0 has a dense proper subgroup (necessarily of infinite index), because it maps onto a torus.

A group of the form $\prod_{i \in I} S_i$ is said to be *strictly infinite semisimple* if the index set I is infinite and *either* each S_i is a finite (non-abelian) simple group *or* each S_i is a connected compact simple Lie group. Such a group has a *characteristic* dense subgroup of infinite index, namely the restricted direct product N of the S_i . Note that N is *countable* if the S_i are finite groups.

It turns out that these examples essentially account for all possibilities:

Theorem 5.27 *Let G be a compact group such that G/G^0 is (topologically) finitely generated. Then G has a virtually dense normal subgroup of infinite index if and only if G has an open normal subgroup H and a closed normal subgroup $K < H$ such that H/K is either infinite and abelian or strictly infinite semisimple.*

In one direction, this follows quickly from the preceding observations. Supposing that H and K exist as indicated, we may as well assume that $K = 1$. Necessarily $H \geq G^0$. If H is strictly infinite semisimple, then H has a characteristic dense subgroup N of infinite index, and then N is normal in G .

Now suppose that H is abelian. If G/G^0 is infinite then H/G^0 has a countable dense subgroup M/G^0 . Then $N := \langle M^G \rangle = M^{g_1} \dots M^{g_n}$ is virtually dense and normal in G , where $\{g_1, \dots, g_n\}$ is a set of coset representatives for G/H , and N/G^0 is countable, so N has infinite index in G . Suppose finally that G/G^0 is finite. As G^0 is a compact connected abelian group, it has a subgroup T such that G^0/T is a one-dimensional torus. Put $S = T^{g_1} \cap \dots \cap T^{g_n}$ where $\{g_1, \dots, g_n\}$ is a set of coset representatives for G/G^0 . Then G^0/S is a torus, so

has a countable dense subgroup M/S (in fact we can choose M/S to be cyclic). Now take $N = \langle M^G \rangle = M^{g_1} \dots M^{g_n}$ as before.

For the converse, let N be a normal subgroup of infinite index in (the abstract group) G such that $L = \overline{N}$ is open in G . Note that $L \geq G^0$ and that L/G^0 is a finitely generated profinite group. It will suffice to find an open normal subgroup H of G and a closed normal subgroup K of H such that H/K is either infinite and abelian or strictly infinite semisimple; for if $\{g_1, \dots, g_n\}$ is a set of coset representatives for G/H then $K_* = K^{g_1} \cap \dots \cap K^{g_n}$ is closed and normal in G , and H/K_* is a subdirect product of copies of H/K , hence shares the given property of H/K .

Now we separate cases.

Case 1: where $G^0 = 1$, i.e. G is profinite.

Recall that L' is closed, by Corollary 5.10. Suppose that both L/L' and L/L_0 are finite. Then both L' and L_0 are open in L , so $NL' = NL_0 = L$. It follows by Corollary 1.8 (applied to the finitely generated profinite group L) that $N = L$, a contradiction. Therefore at least one of L/L' , L/L_0 is infinite.

If L/L' is infinite we set $H = L$ and $K = L'$. Suppose finally that L/L_0 is infinite, and put $T = L^{(3)}L_0$; recall that T/L_0 is semisimple (a consequence of Proposition 1.17). If L/T is finite, then T/L_0 is infinite; in this case, set $H = T$ and $K = L_0$. If L/T is infinite, then some term S of the derived series of L must satisfy: L/S is finite and S/S' is infinite. In this case, we take $H = S$ and $K = S'$.

Case 2: where G is connected.

In this case, N is dense in G . According to [HM], Theorem 9.24, G is a quotient $(A \times P)/Z$ where A is a connected compact abelian group, $P = \prod_{i \in I} S_i$ is a connected compact semisimple group, and $Z \leq Z(P)$. If we assume that G has no infinite abelian image, it follows that $G \cong P/(P \cap Z)$. If G has a proper dense normal subgroup, then so does P . Now the claim (*) in Subsection 5.4, above, shows that there exists a non-principal ultrafilter on the index set I : but this implies that I is infinite. Thus $G \cong P/(P \cap Z)$ has a strictly infinite semisimple quotient G/K isomorphic to the product $\prod_{i \in I} S_i/Z(S_i)$.

The General Case.

If $NG^0 < L$ the result follows by Case 1 applied to G/G^0 . So we may assume that $NG^0 = L$. Let $Z = Z(G^0)$. If L/ZN is finite then $H := ZN$ is open in G and $K := H'$ is closed (Corollary 5.10); and K has infinite index in H because $K \leq N$.

So replacing G by G/Z and N by ZN/N we may assume that $Z(G^0) = 1$. In this case, $G^0 = \prod_{i \in I} S_i$ where each S_i is a connected (and centreless) simple Lie group ([HM], *loc. cit.*). Put $D = G^0 \cap N$. Then $[G^0, N] \leq D$. It follows that

$$G^0 = G^{0'} \leq [G^0, L] = [G^0, \overline{N}] \leq \overline{D},$$

so D is dense in G^0 . In particular, in view of Case 2 above, the index set I must be infinite.

Since G/G^0 is finitely generated, so is L/G^0 ; thus $L = G^0 \overline{\langle y_1, \dots, y_d \rangle}$ for some $y_l \in N$. Then $[G^0, y_l] \subseteq D$ for each l . Applying Proposition 5.19 we deduce that there exists an infinite subset J of I such that each y_l normalizes S_i for every $i \in J$. As $N_L(S_i)$ is closed and contains G^0 , it follows that S_i is normal in L for every $i \in J$. Put $C_i = C_L(S_i)$. Then $L/C_i S_i$ embeds in the outer automorphism group of S_i , which embeds in $\text{Sym}(3)$ (cf. [HM], page 256). As the finitely generated profinite group L/G^0 admits only finitely many homomorphisms into $\text{Sym}(3)$ and $C_i S_i \geq G^0$, it follows that L has a characteristic open subgroup $H \geq G^0$ such that $C_i S_i \geq H$ for all $i \in J$.

Thus putting $X = \prod_{i \in J} S_i$ we have $H = C_H(X) \times X$; indeed, if $h \in H$ then $h = c_i s_i$ ($c_i \in C_i$, $s_i \in S_i$) for each $i \in J$, and if $x = (s_i)_{i \in J}$ then $[hx^{-1}, s_j] = 1$ for every $j \in J$, so $hx^{-1} \in C_H(X)$. To complete the proof we may therefore take $K = C_H(X)$.

Remark. It might be more natural to ask: when does G have a *virtually normal* virtually dense subgroup? (N is *virtually normal* if the normalizer $N_G(N)$ has finite index in G).

Corollary 5.28 *G has a virtually normal virtually dense subgroup of infinite index if and only if G has a normal virtually dense subgroup of infinite index.*

This follows from the theorem: suppose that R is a subgroup of finite index in G , that H is open and normal in R , and that $K < H$ is a closed normal subgroup of R . Then as above we can replace K by a closed normal subgroup K_* of G such that H/K_* is a subdirect product of $|G : R|$ copies of H/K , and replace H by $H_* \triangleleft G$, where H_* is normal of finite index in G . Then H_* is open by Corollary 5.8, whence H_*/K_* is again an infinite abelian or semisimple group of the same type as H/K .

The question: *under what circumstances can a finitely generated profinite group G have a proper dense normal subgroup?* turns out to be more delicate. The answer requires a

Definition. Let S be a finite simple group. Then $Q(S)$ denotes the following subgroup of $\text{Aut}(S)$:

$$\begin{aligned} & \text{InnDiag}(S) \langle \tau \rangle \text{ if } S = D_n(q), \ n \geq 5 \\ & \text{InnDiag}(S) \langle [q] \rangle \text{ if } S = {}^2D_n(q) \\ & \text{InnDiag}(S) \text{ if } S \text{ is of another Lie type} \\ & \text{Aut}(S) \text{ in all other cases} \end{aligned}$$

where τ is the non-trivial graph automorphism of $D_n(q)$ and $[q]$ denotes the field automorphism of order 2 of ${}^2D_n(q)$.

A finite group H is *Q -almost-simple* if $S \triangleleft H \leq Q(S)$ for some finite simple groups S .

If H is Q -almost-simple as above, the *rank* of H is then the rank of S , namely the (untwisted) Lie rank if S is of Lie type, n if $S \cong \text{Alt}(n)$, and zero otherwise.

Theorem 5.29 *Let G be a finitely generated profinite group. Then G has no proper dense normal subgroup if and only if*

- G^{ab} is finite, and
- the simple quotients of G have bounded orders, and
- the Q -almost-simple quotients of G have bounded ranks.

The proof, which will appear elsewhere, depends on Corollary 1.8 and further arguments in the spirit of Subsection 4.1.

References

- [AG] M. Aschbacher and R. M. Guralnick, Some applications of the first cohomology group, *J. Algebra* **90** (1984), 446-460.
- [As] M. Aschbacher, *Finite group theory*, Cambridge Univ. Press, Cambridge, 1988.
- [B] H. Blau, A fixed-point theorem for central elements in quasisimple groups, *Proc. AMS* **122** (1994), 79-84.
- [BCP] L. Babai, P. J. Cameron and P. Pálffy, On the orders of primitive groups with restricted non-abelian composition factors, *J. Algebra* **79** (1982), 161-168.
- [BNP] L. Babai, N. Nikolov and L. Pyber, Product Growth and Mixing in Finite Groups, *19th ACM-SIAM Symposium on Discrete Algorithms*, SIAM, 2008, Pages 248-257.
- [Bu] D. Bump, *Lie groups*, Springer-Verlag, New York, 2004.
- [C] R. W. Carter, *Finite groups of Lie type: conjugacy classes and complex characters*, Wiley and Sons, London, 1985.
- [DM] J. D. Dixon and B. Mortimer, *Permutation groups*, Springer-Verlag, New York, 1996.
- [FG] J. Fulman, R. Guralnick, Bounds on the number and sizes of conjugacy classes in finite Chevalley groups with applications to derangements, <http://arxiv.org/abs/0902.2238>
- [FMS] T. Frayne, A. Morel and D. Scott, Reduced direct products, *Fund. Math.* **51** (1962), 195-228.
- [FJ] M. Fried and M. Jarden, *Field arithmetic*, Springer-Verlag, Berlin – Heidelberg, 1986.

- [GaSh] S. Garion and A. Shalev, Commutator maps, measure preservation, and T -systems, *Trans. Amer. Math. Soc.* **361** (2009), 4631–4651.
- [Gch] W. Gaschütz, Zu einem von B. H. und H. Neumann gestellten Problem, *Math. Nachrichten* **14** (1955), 249–252.
- [GL] R. M. Guralnick and F. Lübeck, On p -singular elements in Chevalley groups in characteristic p , in *Groups and computation III*, 169–182, Ohio State Univ. Math. Res. Inst. Publ. **8**, de Gruyter, Berlin, 2001.
- [GLS] D. Gorenstein, R. Lyons and R. Solomon, *The classification of the finite simple groups, no.3*, American Math. Soc., Providence, Rhode Island, 1998.
- [Go] D. Gorenstein, *Finite Groups*, 2nd ed., Chelsea, New York, 1980.
- [Gt] M. Goto, A theorem on compact semisimple groups. *J. Math. Soc. Japan* **1** (1949), 270–272.
- [GFSG] D. Gorenstein, *Finite simple groups*, Plenum Press, New York and London, 1982.
- [GSS] D. Gluck, A. Seress and A. Shalev, Bases for primitive permutation groups and a conjecture of Babai, *J. Algebra* **199** (1998), 367–378.
- [HM] K. H. Hofmann and S. A. Morris, *The structure of compact groups*. 2nd edn., de Gruyter Studies in Mathematics, **25**. Walter de Gruyter & Co., Berlin, 2006.
- [J] G. A. Jones, Varieties and simple groups, *J. Austral. Math. Soc.* **17** (1974), 163–173.
- [KL] M. Kapovich and B. Leeb, On asymptotic cones and quasi-isometry of fundamental groups of 3-manifolds, *GAF* **5** (1995), 582–603.
- [LaS] V. Landazuri and G. M. Seitz, On the minimal degrees of projective representations of the finite Chevalley groups, *J. Algebra* **32** (1974), 418–443.
- [LiSh] M. W. Liebeck and A. Shalev, Diameters of finite simple groups: sharp bounds and applications, *Annals of Math.* **154** (2001), 383–406.
- [LiSh2] M. W. Liebeck and A. Shalev, Fuchsian groups, finite simple groups and representation varieties. *Invent. Math.* **159** (2005), 317–367.
- [LOST] M. Liebeck, E. O’Brien, A. Shalev and P. Tiep, The Ore conjecture, *J. European Math. Soc.* **12** (2010), 939–1008.
- [LOST2] M. Liebeck, E. O’Brien, A. Shalev and P. Tiep, Commutators in finite quasisimple groups, *to appear*

- [MZ] C. Martinez and E. Zelmanov, Products of powers in finite simple groups, *Israel J. Math.* **96** (1996), 469–479.
- [NS] N. Nikolov and D. Segal, On finitely generated profinite groups, I: strong completeness and uniform bounds, *Annals of Math.* **165** (2007), 171–238.
- [NS2] N. Nikolov and D. Segal, On finitely generated profinite groups, II: products in quasisimple groups, *Annals of Math.* **165** (2007), 239–273.
- [NSP] N. Nikolov and D. Segal, Powers in finite groups, *Groups, Geometry and Dynamics*, to appear; arXiv:0909.6439
- [SW] J. Saxl and J. S. Wilson, A note on powers in simple groups, *Math. Proc. Cambridge Philos. Soc.* **122** (1997), 91–94.
- [S1] D. Segal, Closed subgroups of profinite groups, *Proc. London Math. Soc.* **81** (2000), 29–54.
- [S2] D. Segal, *Words: notes on verbal width in groups*, London Math. Soc. Lecture Notes Series **361**, Cambridge Univ. Press, Cambridge, 2009.
- [SGT] J.-P. Serre, *Topics in Galois Theory*, Res. Notes Math. **1**, Jones and Bartlett, Boston – London, 1992.
- [W] J. S. Wilson, On simple pseudofinite groups, *J. London Math. Soc.* **51** (1995), 471–490.
- [Z] E. I. Zelmanov, On the restricted Burnside problem, *Proc. Internl. Congress Math. Kyoto 1990*, Math. Soc. Japan, Tokyo, 1991, pp. 395–402.